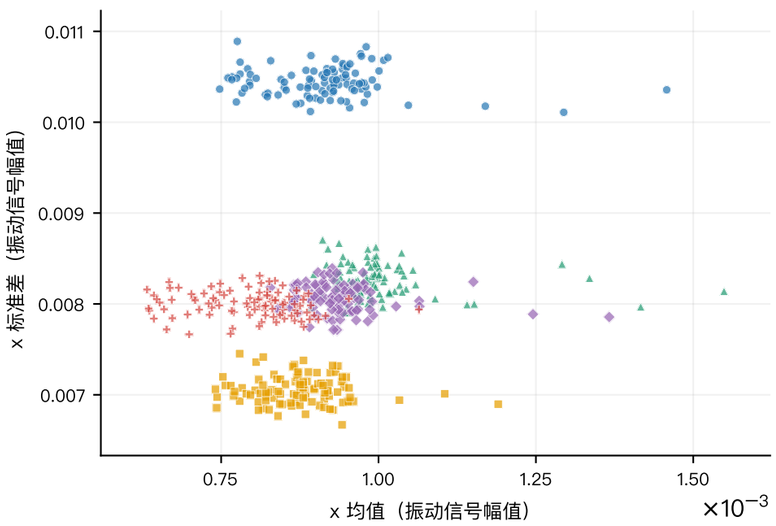
30_2 | 全部双特征组合

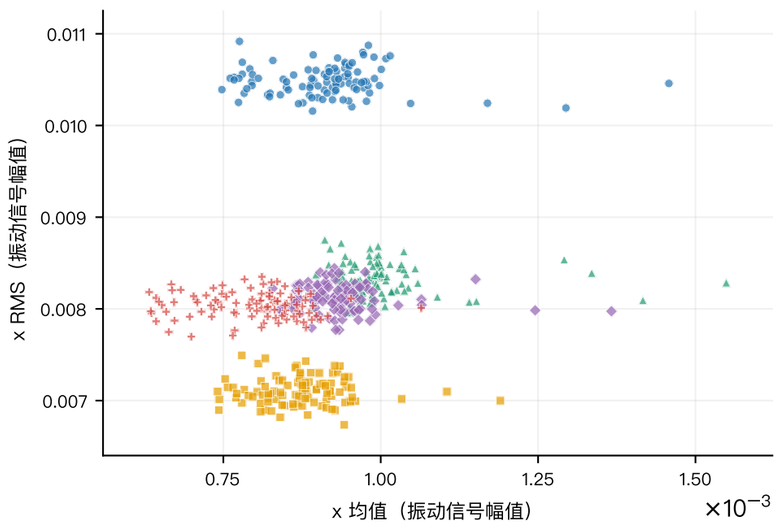
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

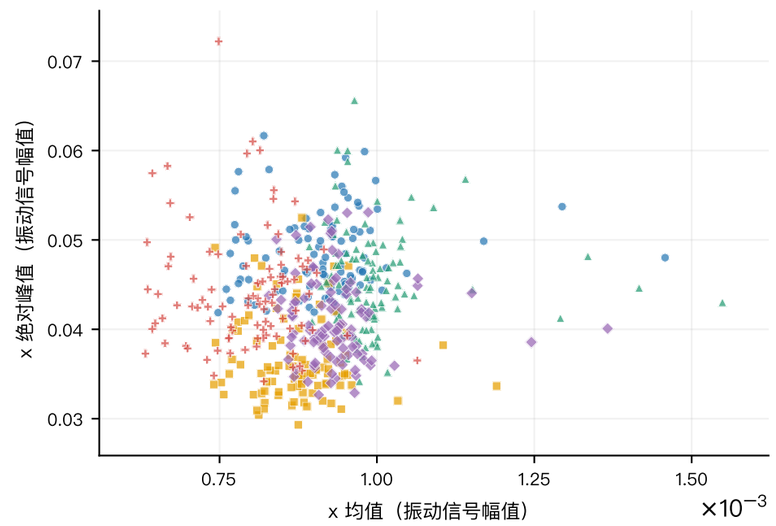
001 x 均值 × x 标准差



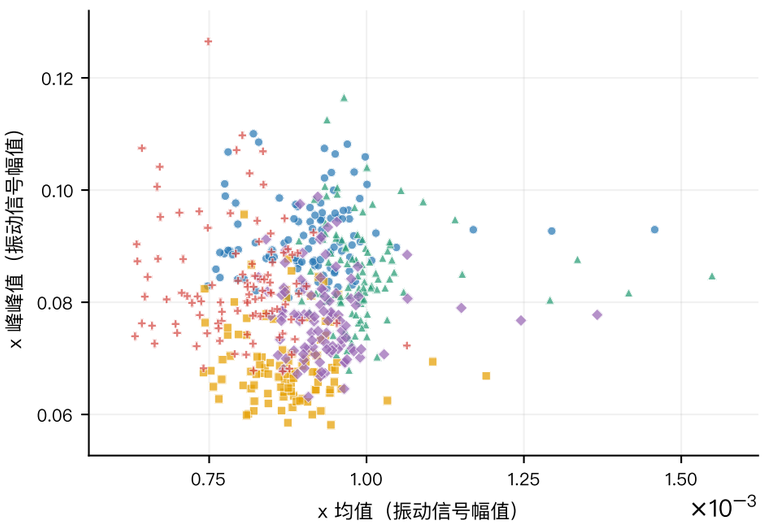
002 x 均值 × x RMS



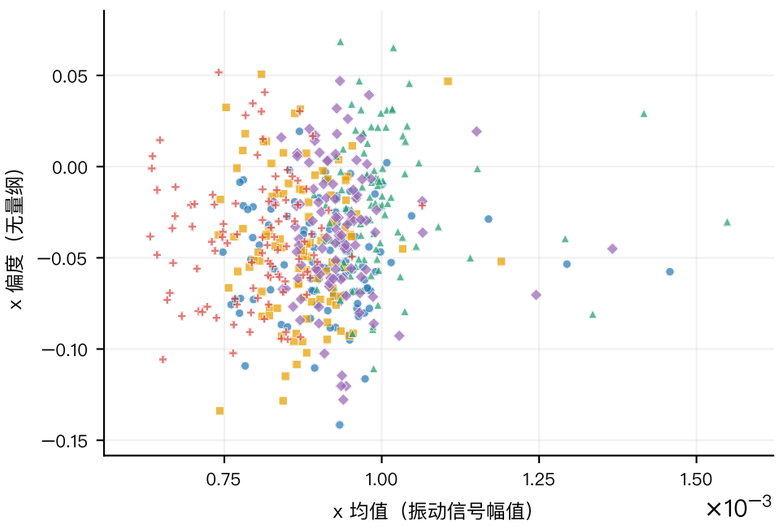
003 x 均值 × x 绝对峰值



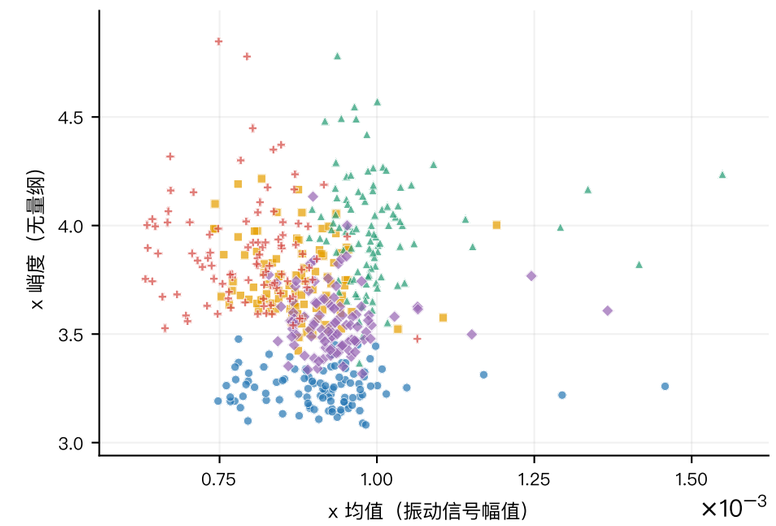
004 x 均值 × x 峰峰值



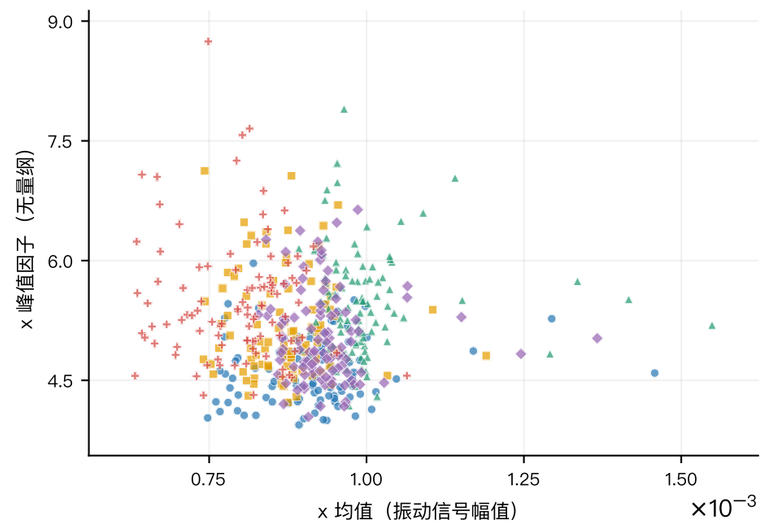
005 x 均值 × x 偏度



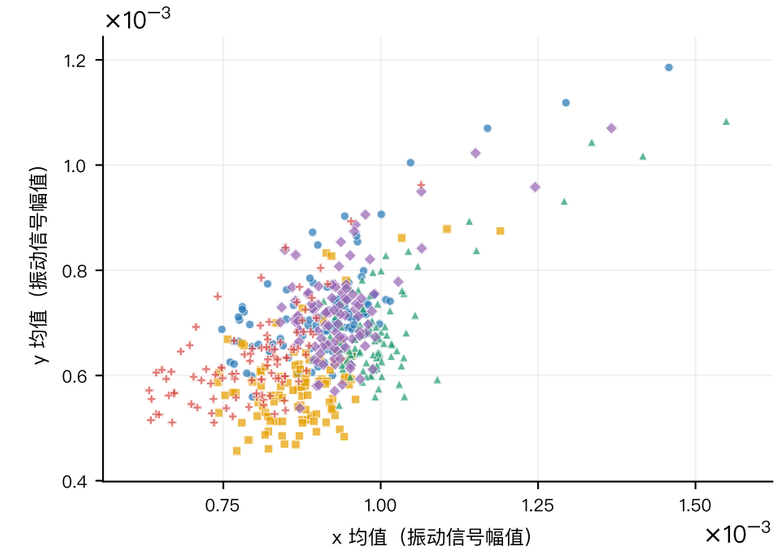
006 x 均值 × x 峭度



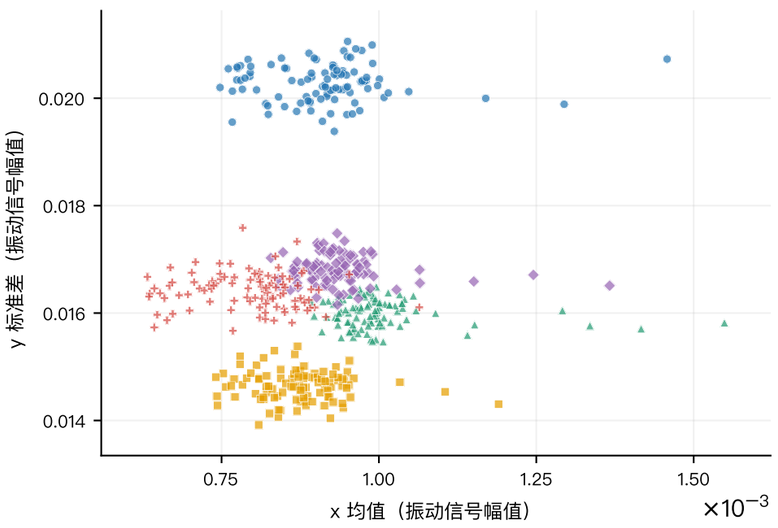
007 x 均值 × x 峰值因子



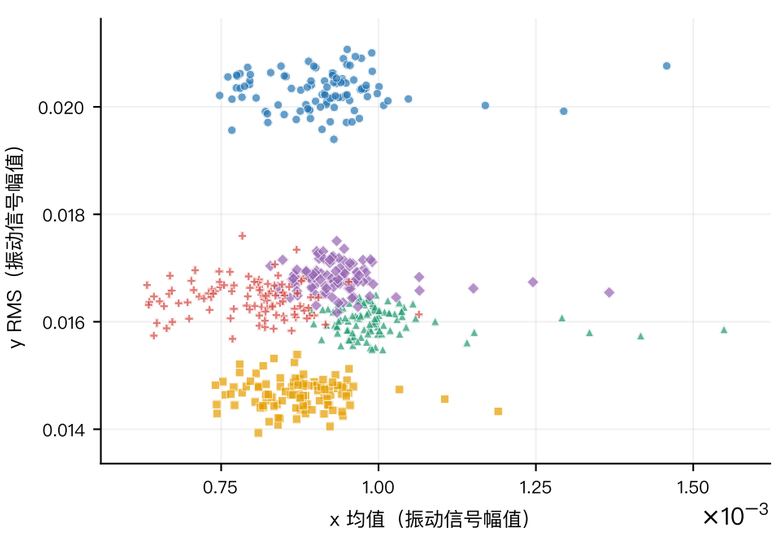
008 x 均值 × y 均值



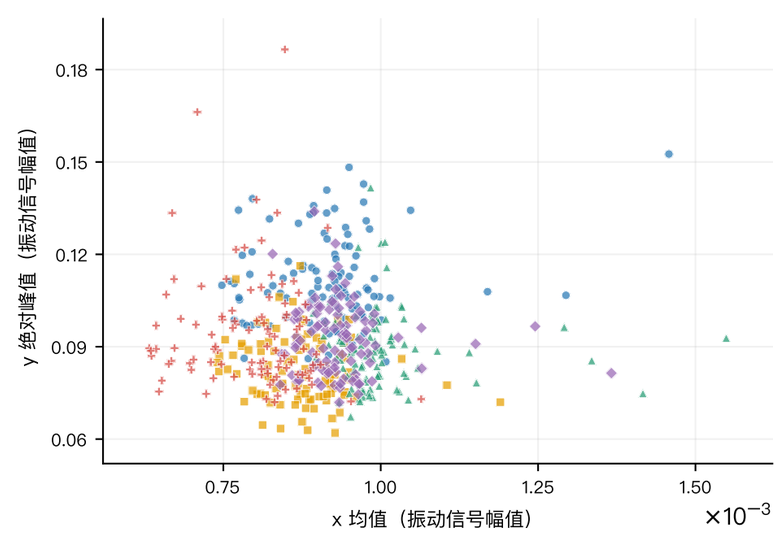
009 x 均值 × y 标准差



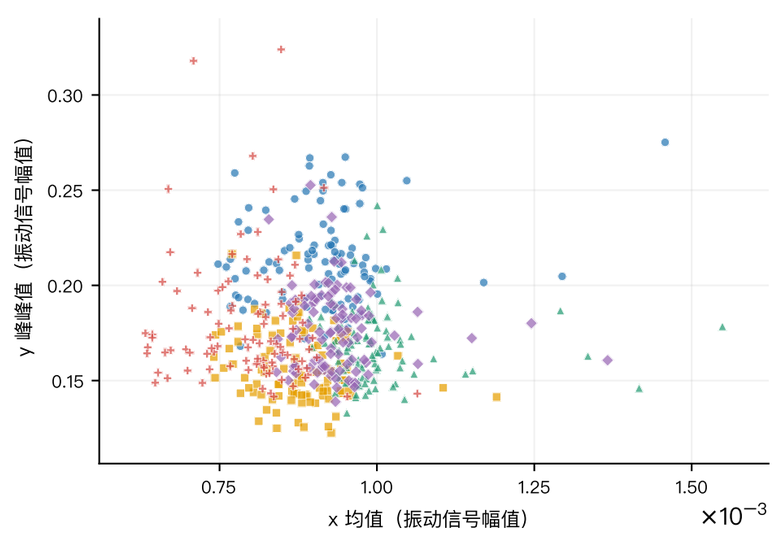
010 x 均值 × y RMS



011 x 均值 × y 绝对峰值



012 x 均值 × y 峰峰值



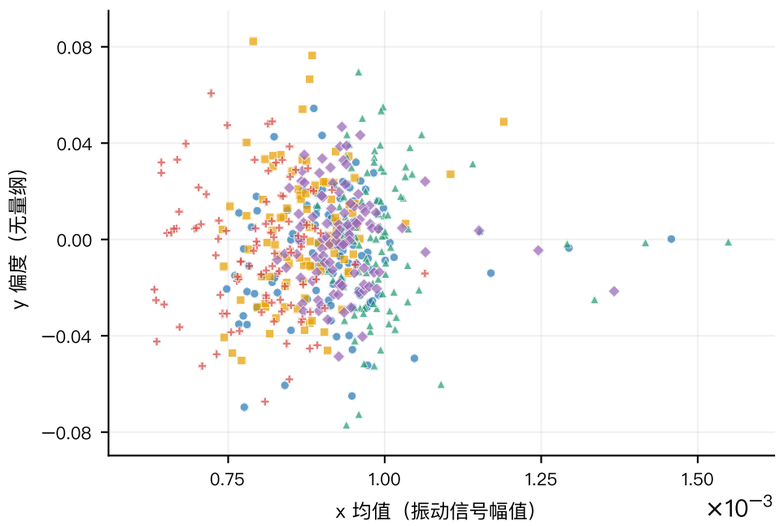
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

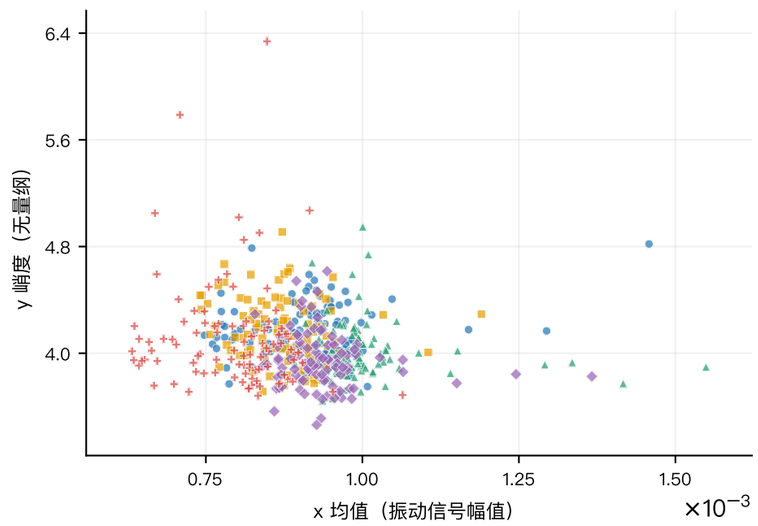
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

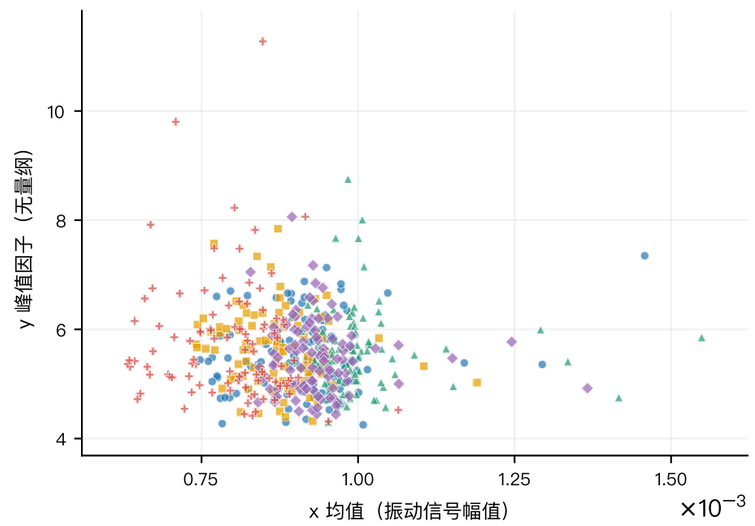
013 x 均值 × y 偏度



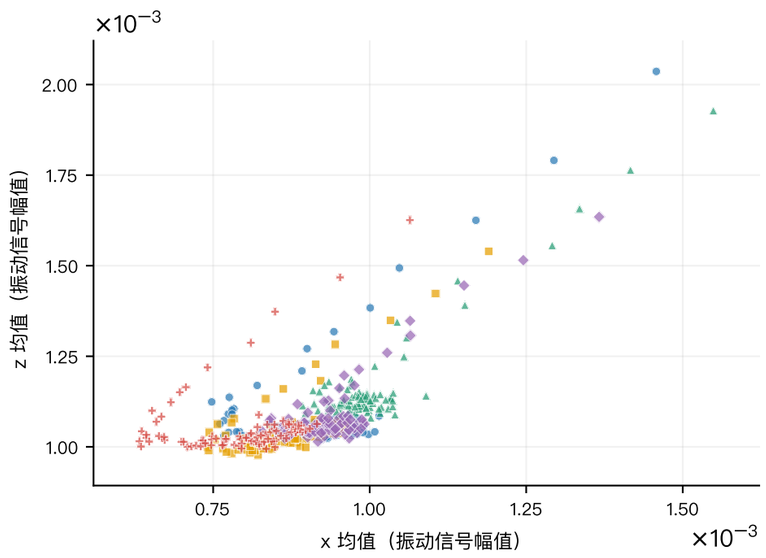
014 x 均值 × y 峭度



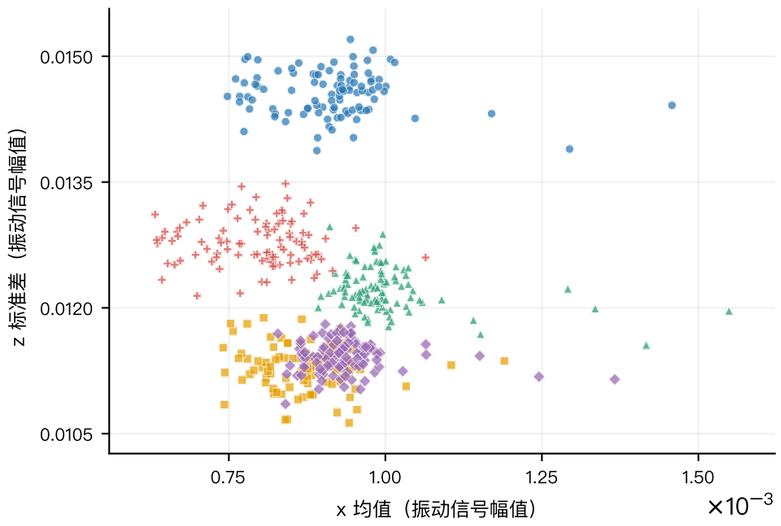
015 x 均值 × y 峰值因子



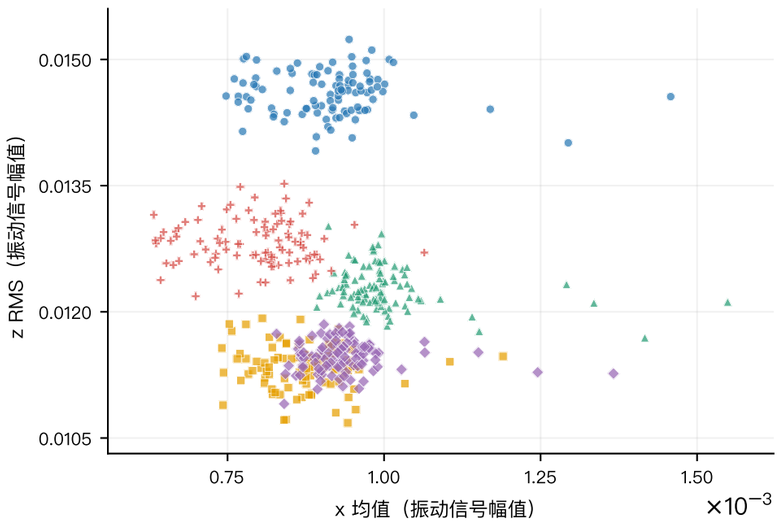
016 x 均值 × z 均值



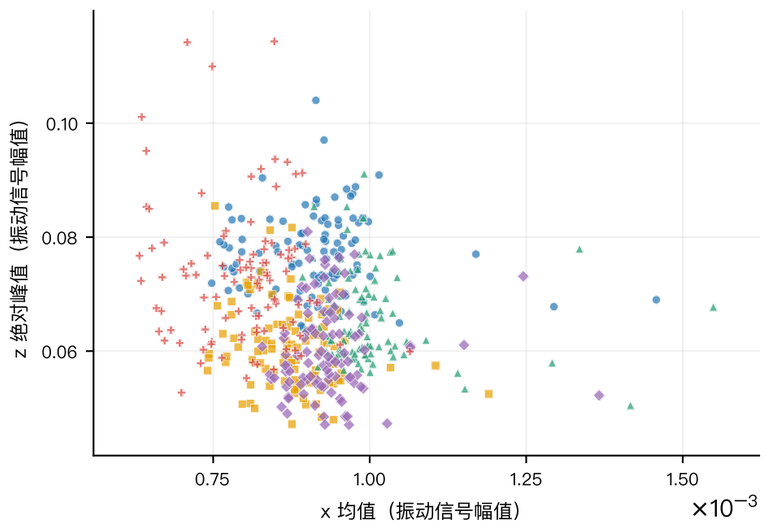
017 x 均值 × z 标准差



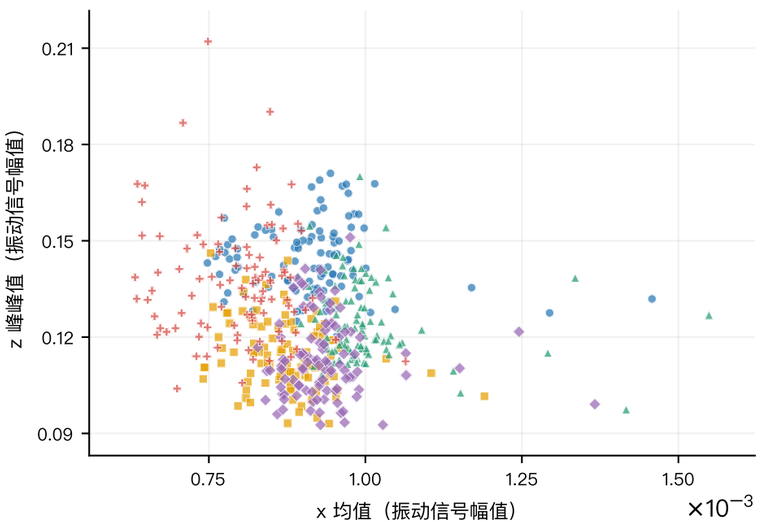
018 x 均值 × z RMS



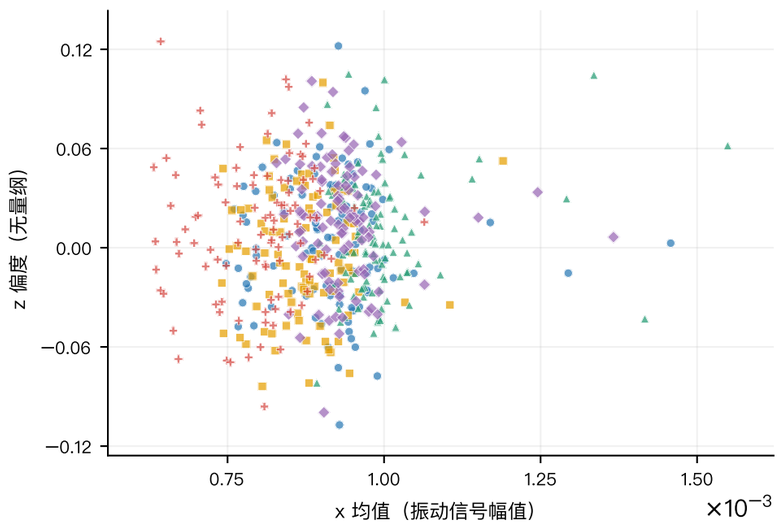
019 x 均值 × z 绝对峰值



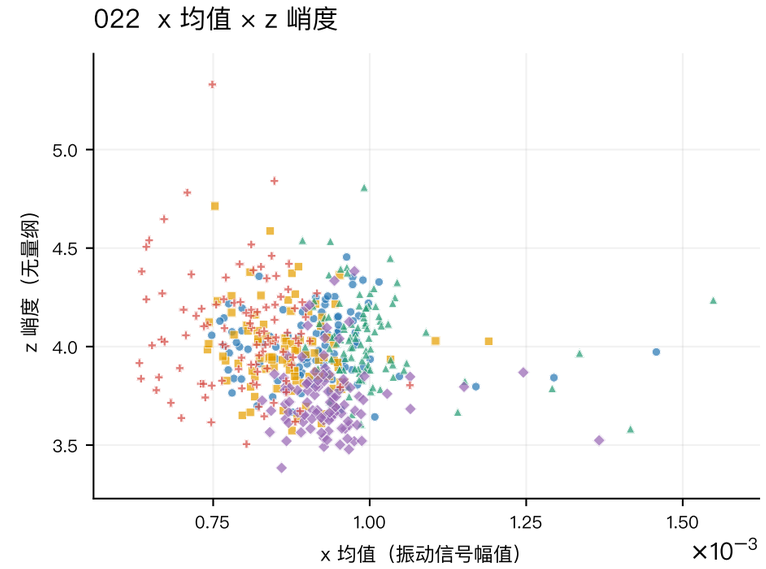
020 x 均值 × z 峰峰值



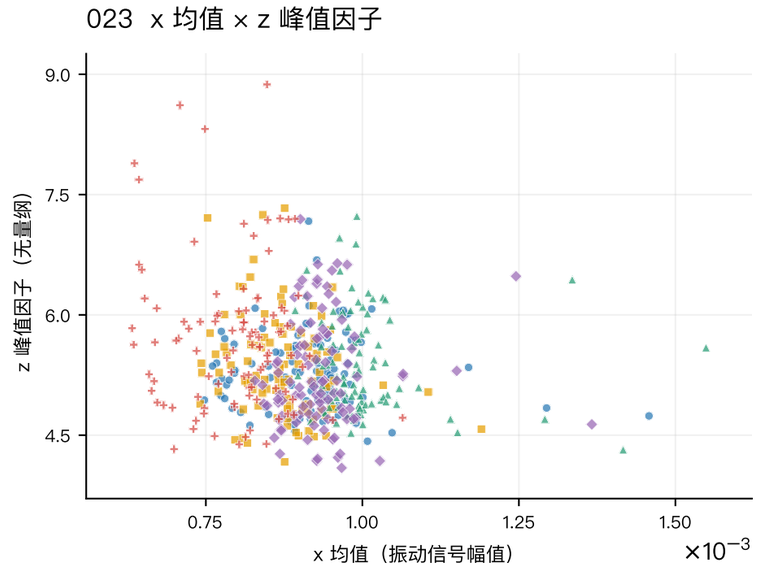
021 x 均值 × z 偏度



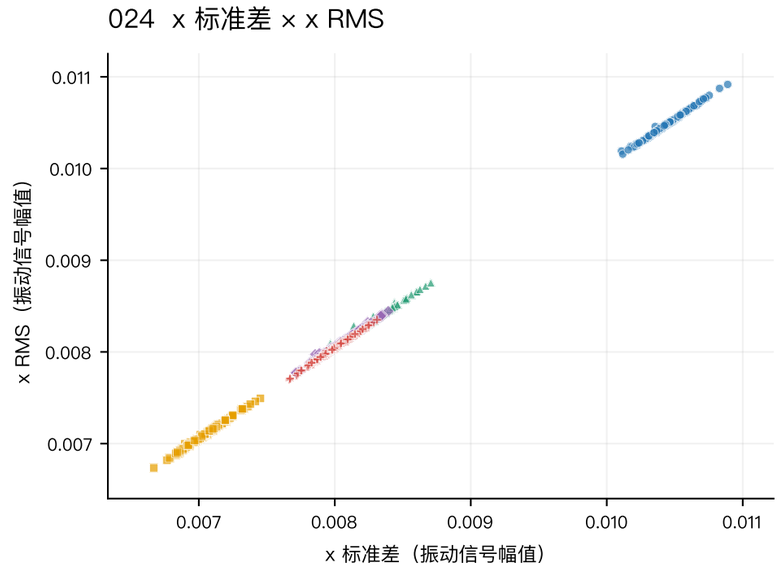
022 x 均值 × z 峭度



023 x 均值 × z 峰值因子



024 x 标准差 × x RMS



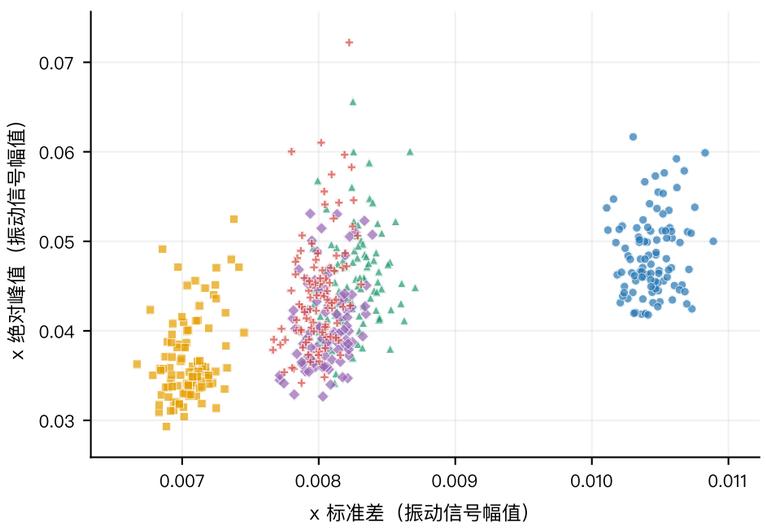
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

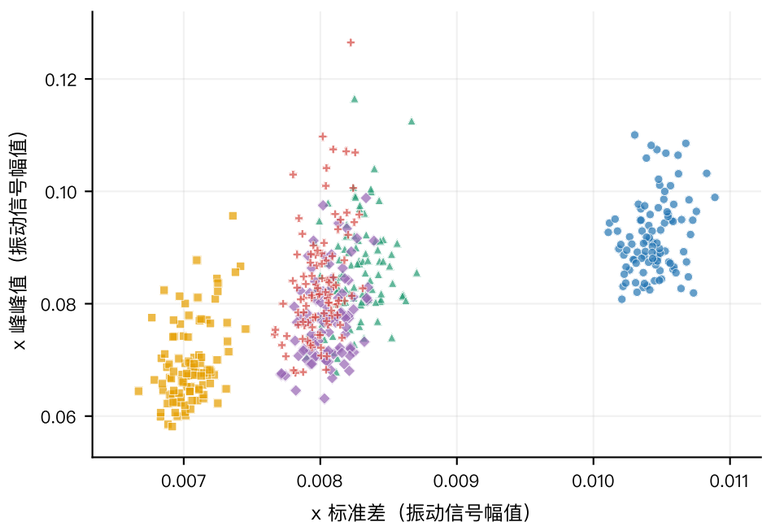
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

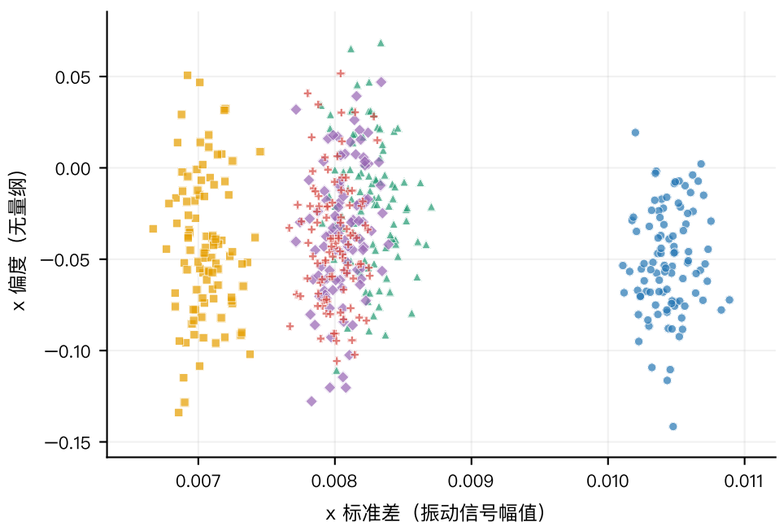
025 x 标准差 × x 绝对峰值



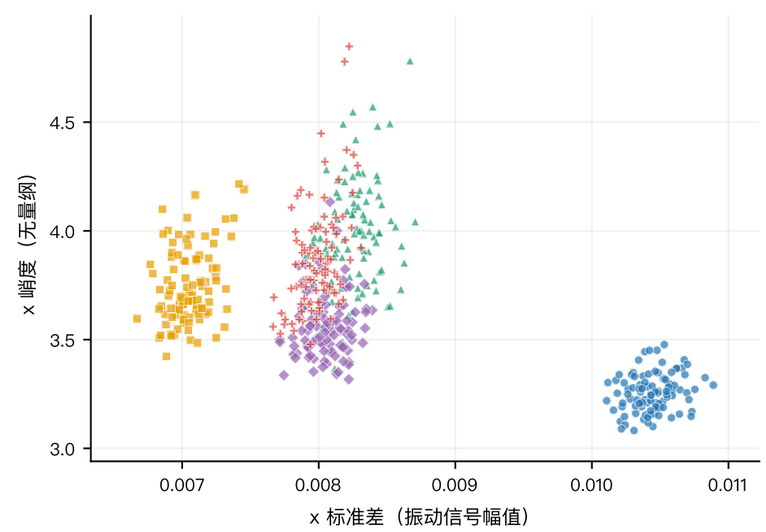
026 x 标准差 × x 峰峰值



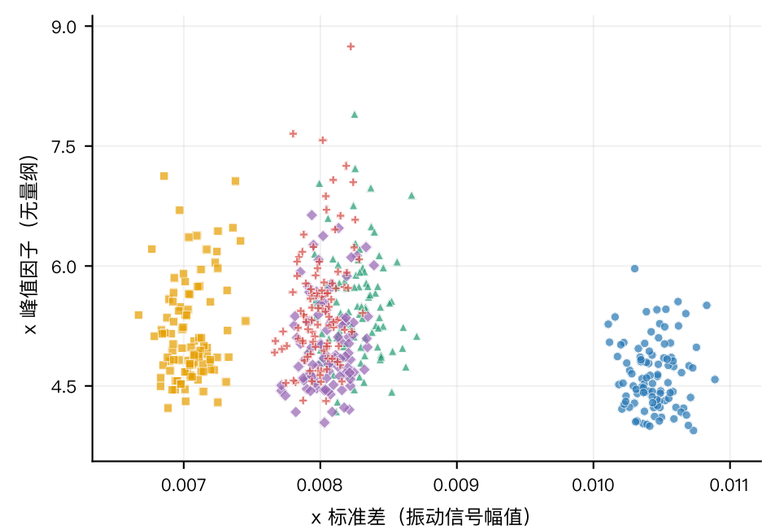
027 x 标准差 × x 偏度



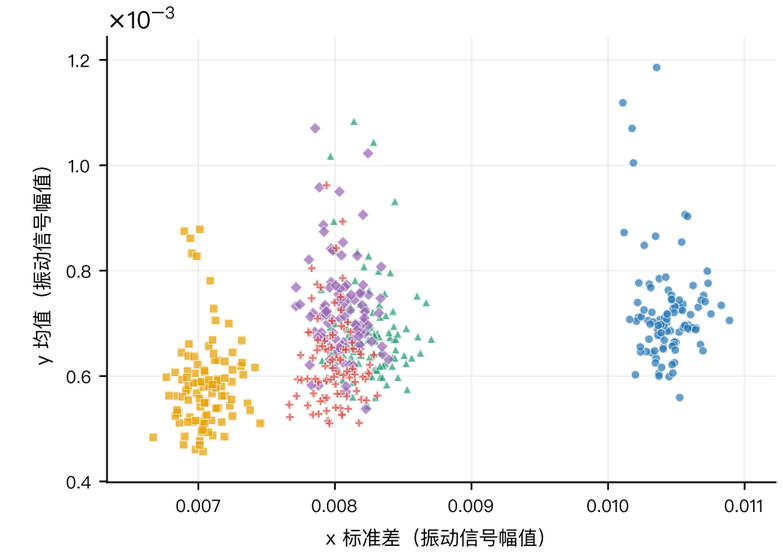
028 x 标准差 × x 峭度



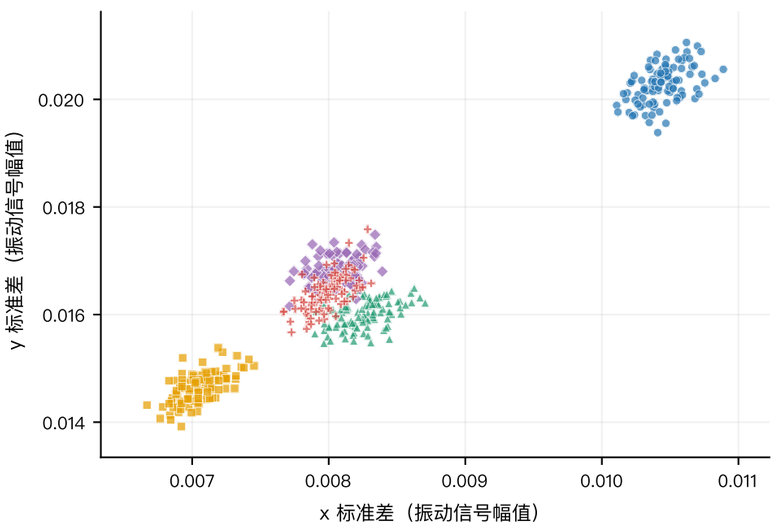
029 x 标准差 × x 峰值因子



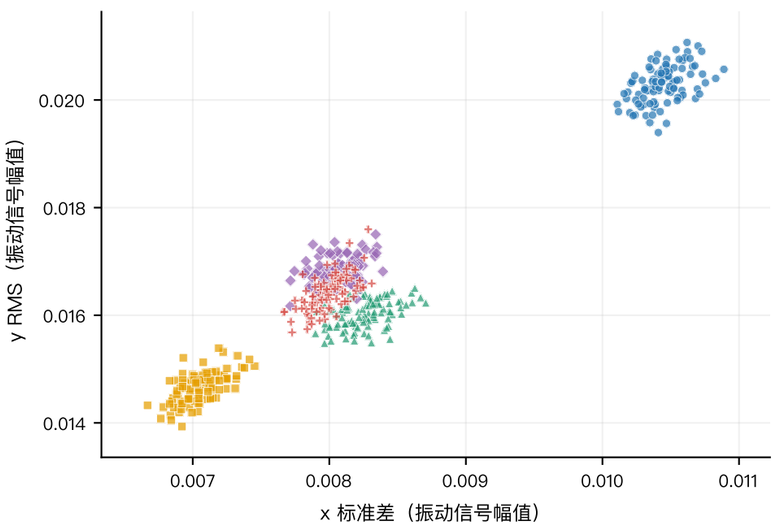
030 x 标准差 × y 均值



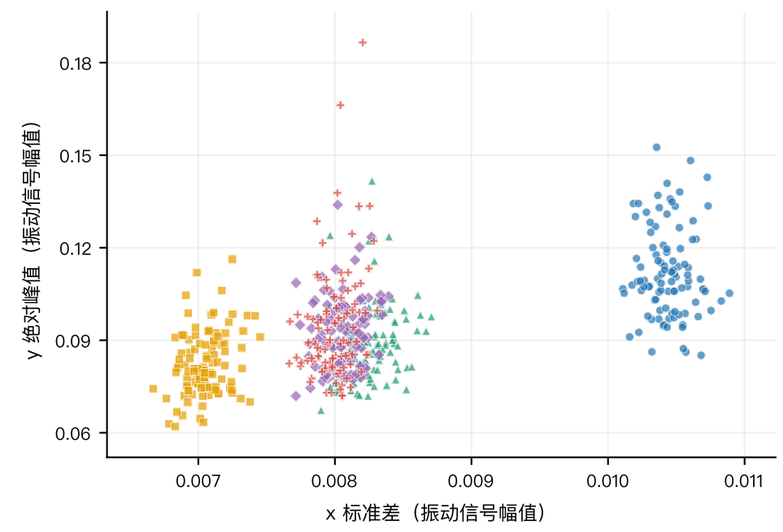
031 x 标准差 × y 标准差



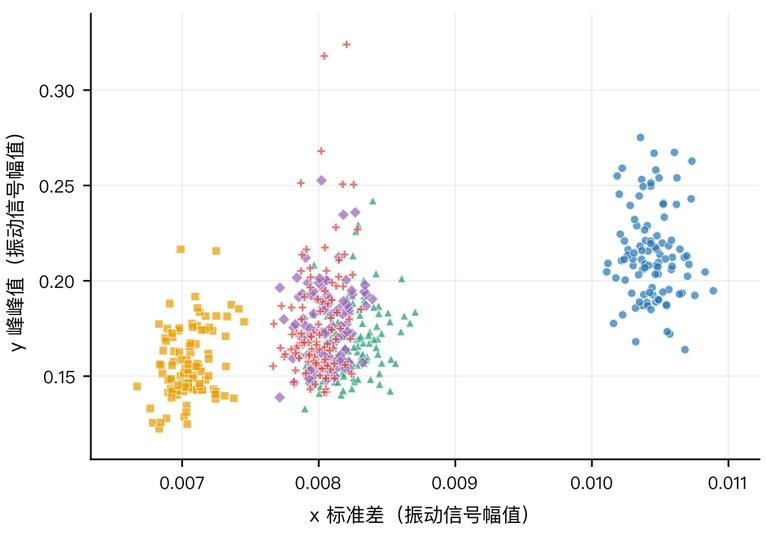
032 x 标准差 × y RMS



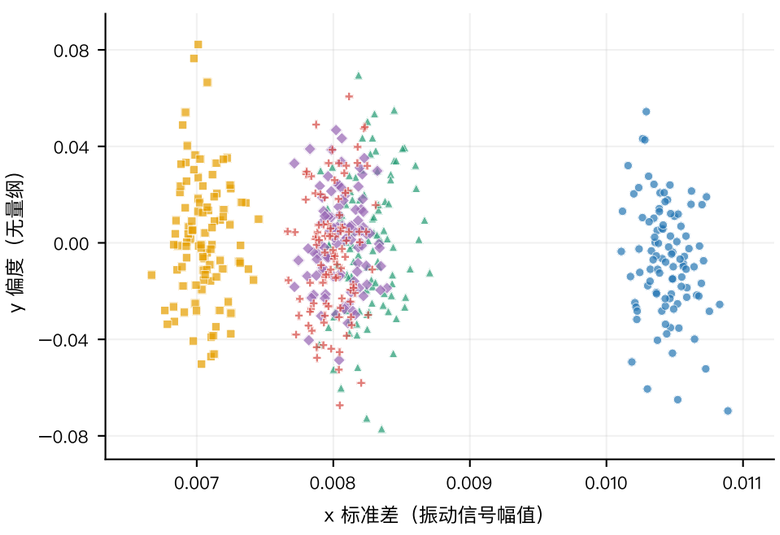
033 x 标准差 × y 绝对峰值



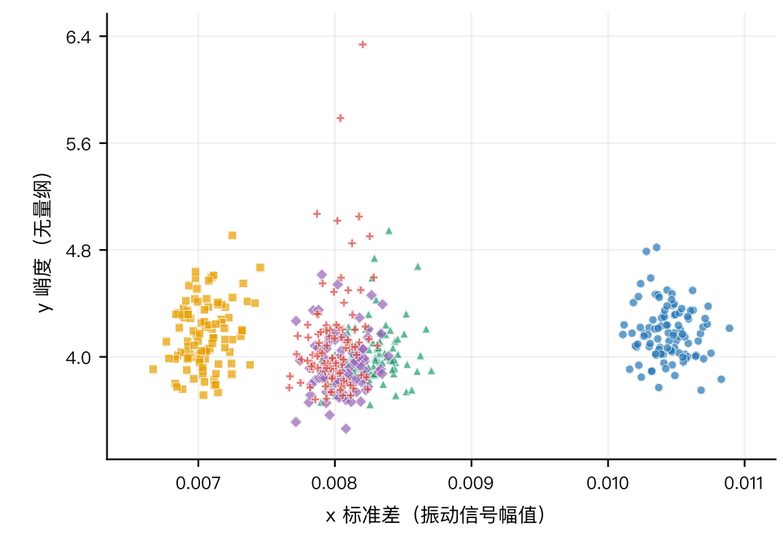
034 x 标准差 × y 峰峰值



035 x 标准差 × y 偏度



036 x 标准差 × y 峭度



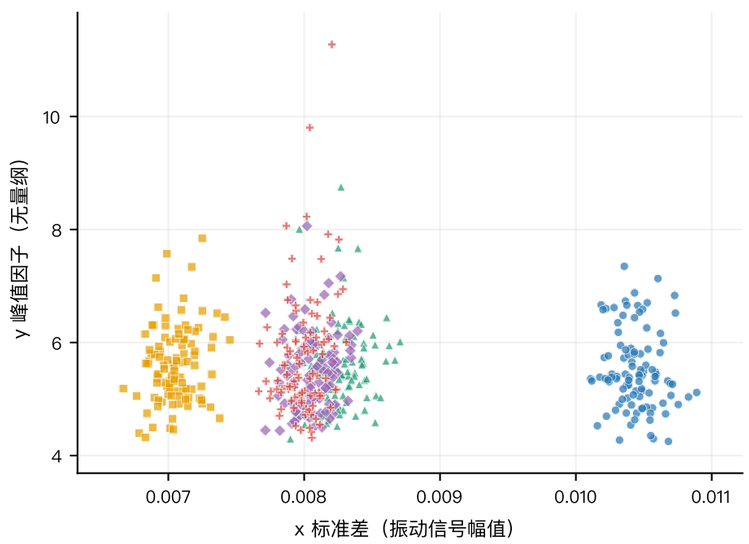
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

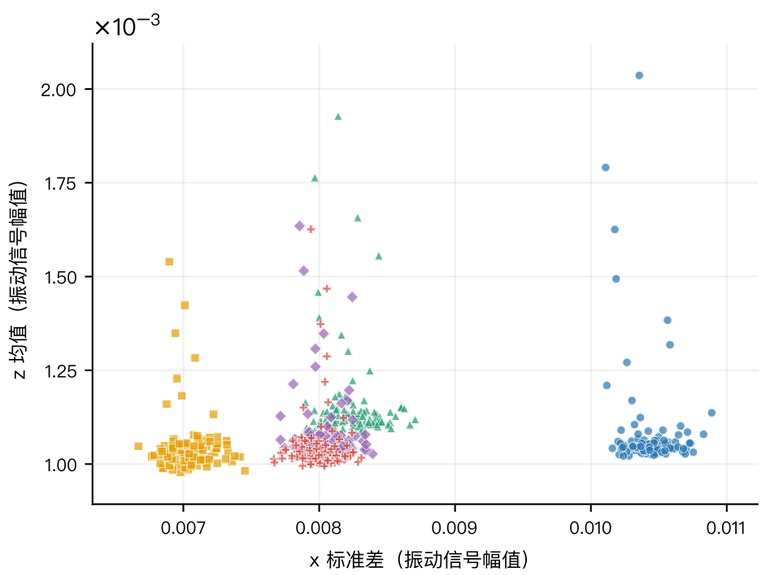
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

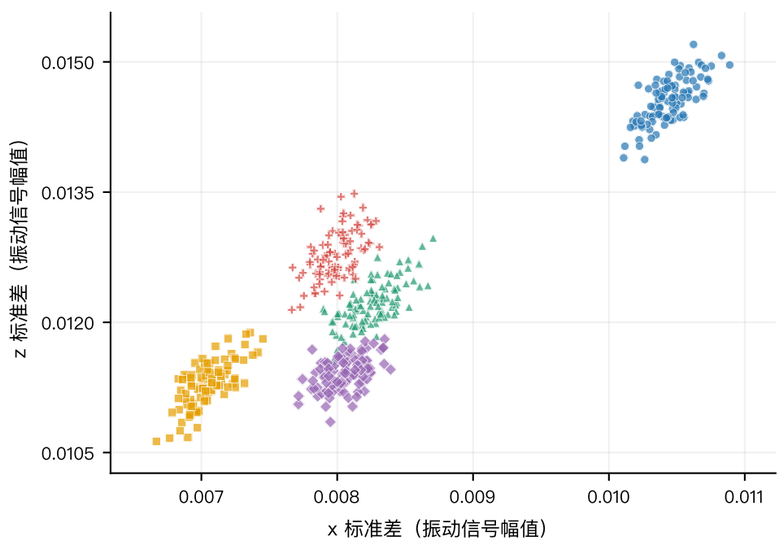
037 x 标准差 × y 峰值因子



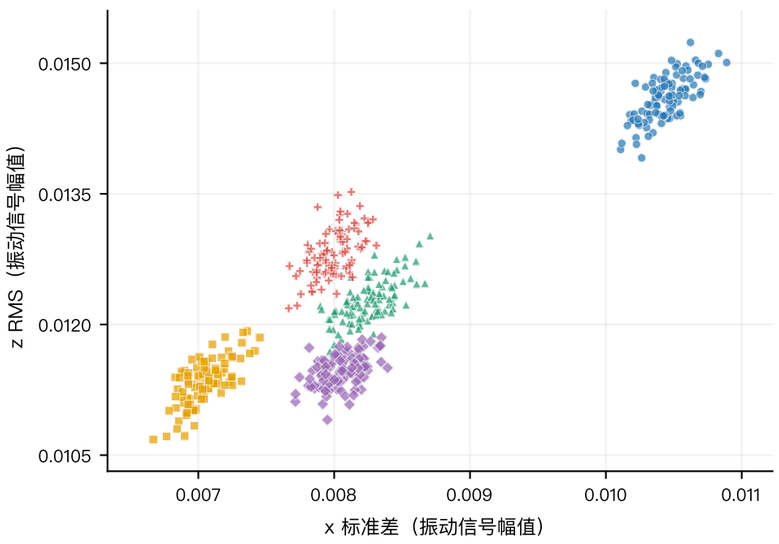
038 x 标准差 × z 均值



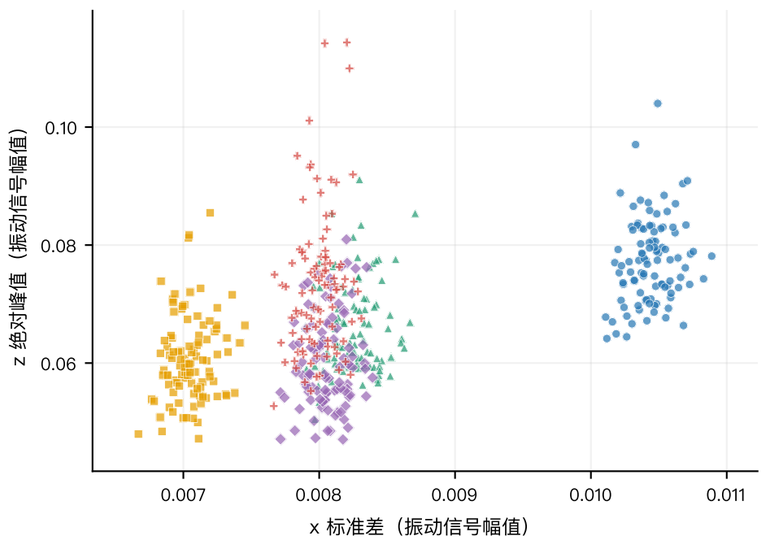
039 x 标准差 × z 标准差



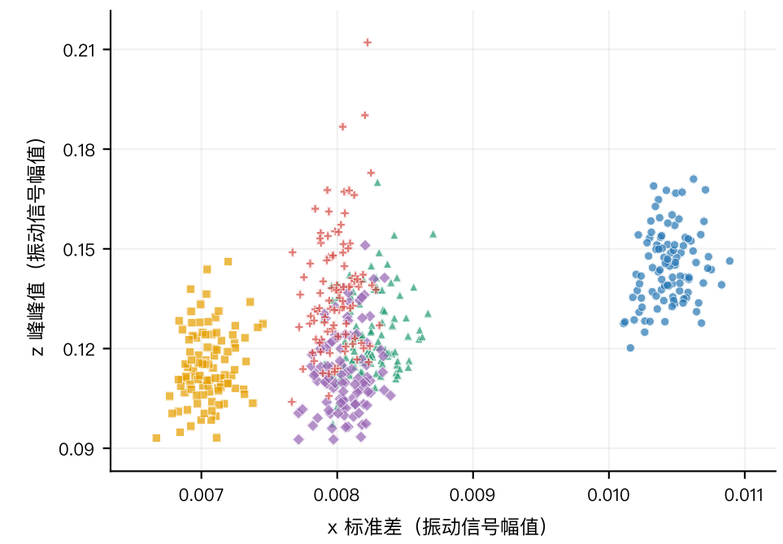
040 x 标准差 × z RMS



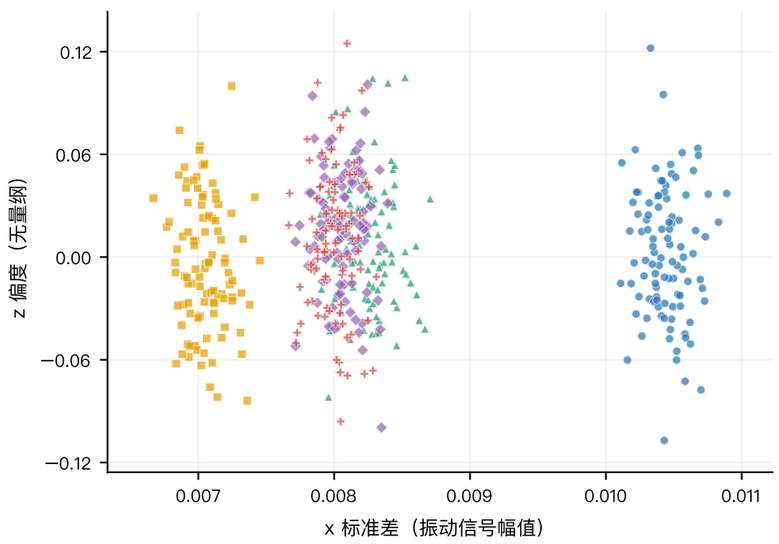
041 x 标准差 × z 绝对峰值



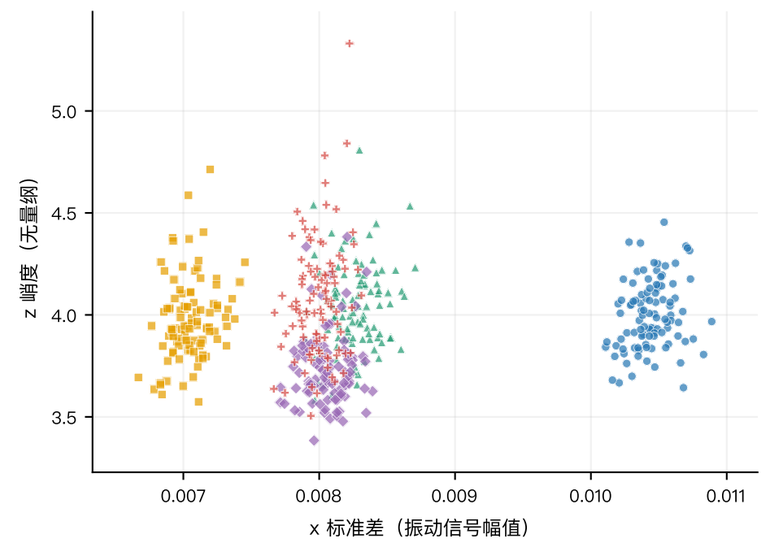
042 x 标准差 × z 峰峰值



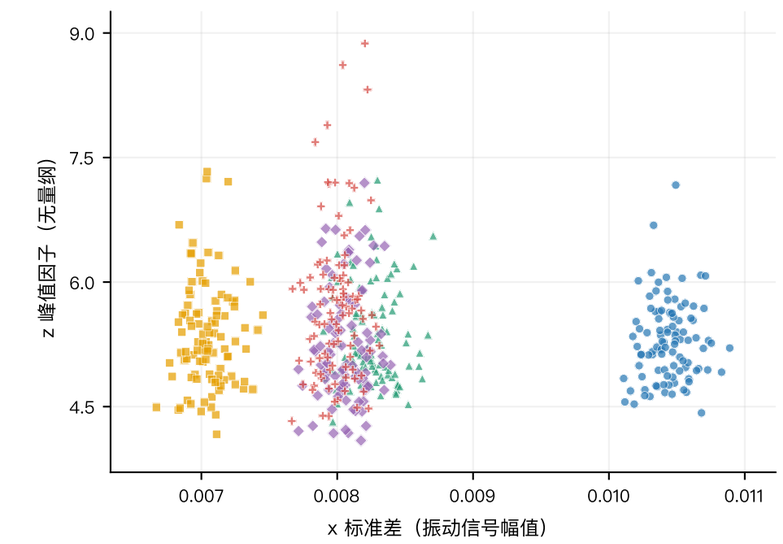
043 x 标准差 × z 偏度



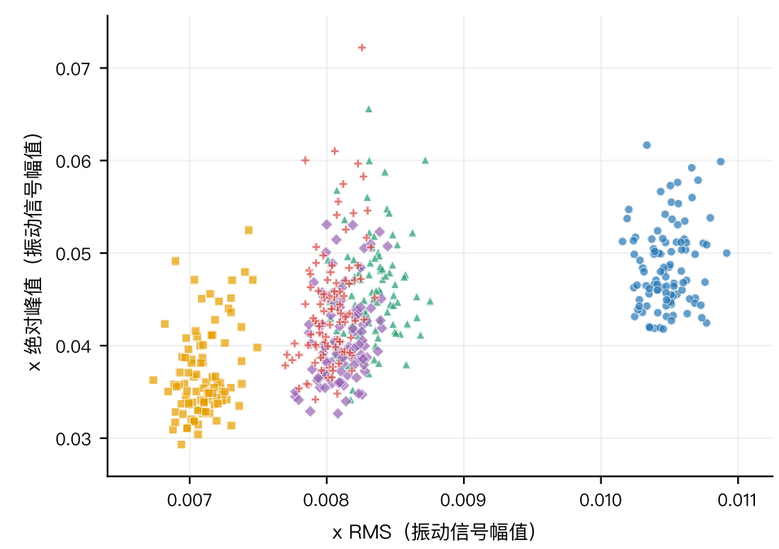
044 x 标准差 × z 峭度



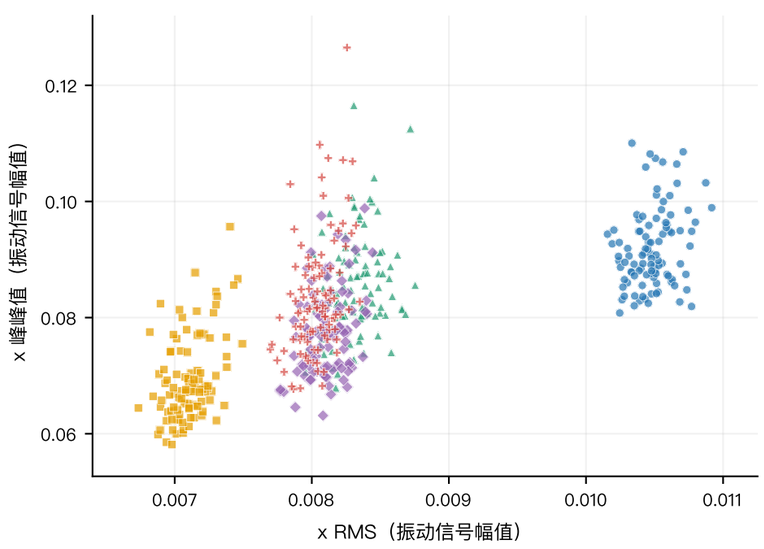
045 x 标准差 × z 峰值因子



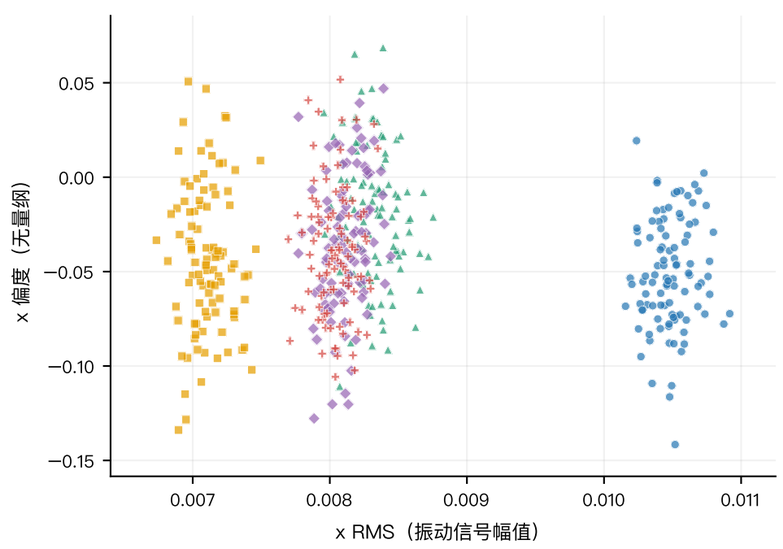
046 x RMS × x 绝对峰值



047 x RMS × x 峰峰值



048 x RMS × x 偏度



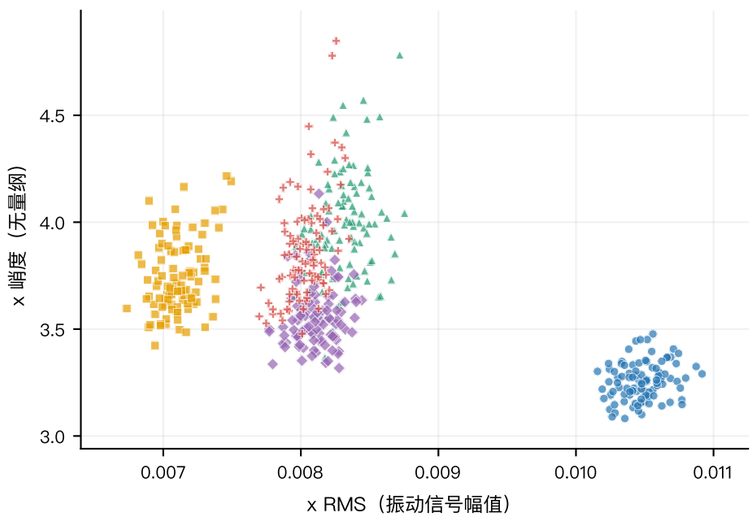
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

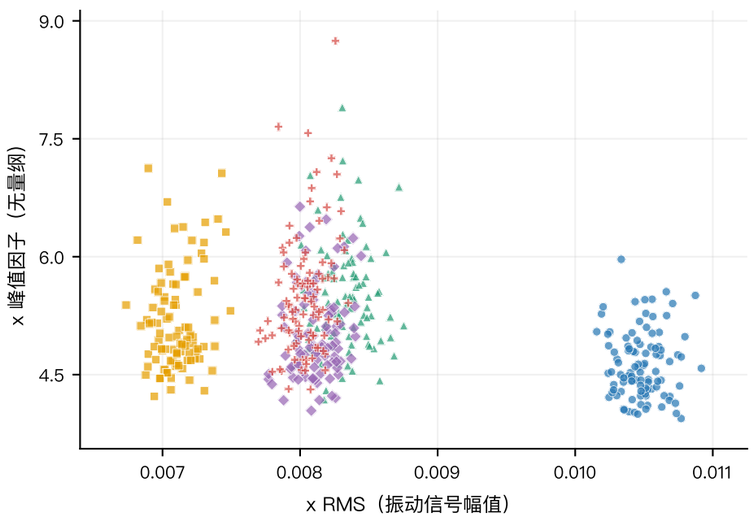
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

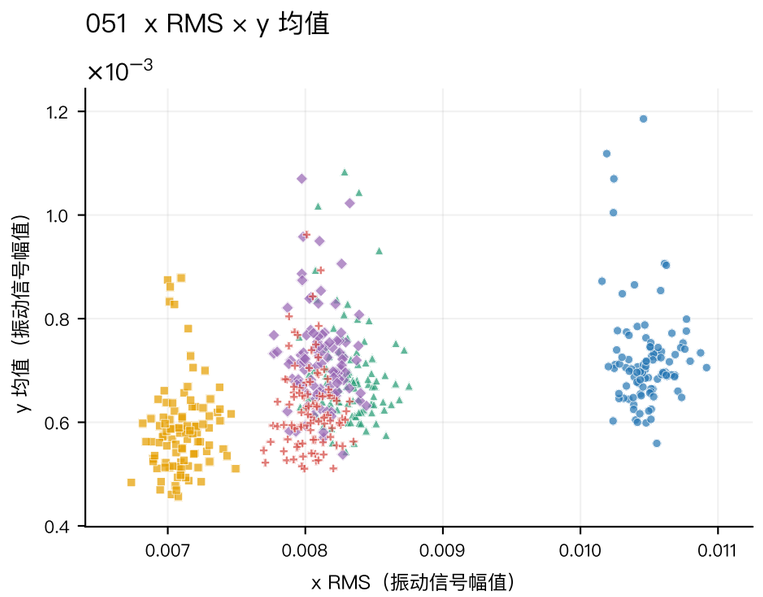
049 x RMS × x 峭度



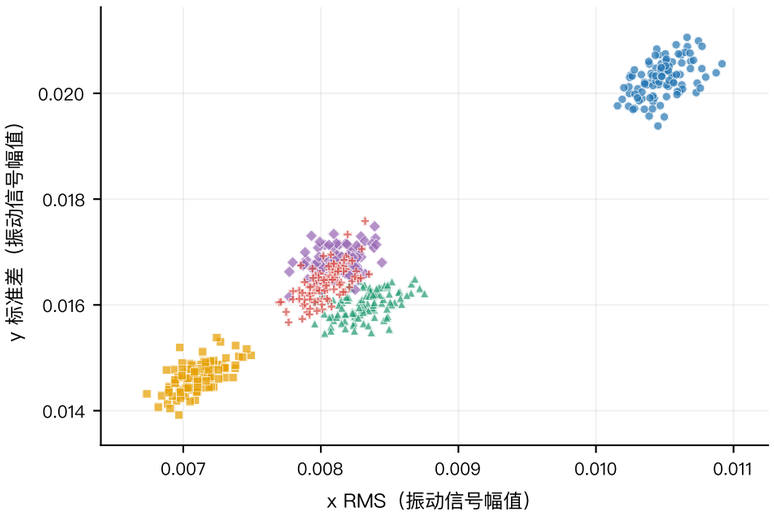
050 x RMS × x 峰值因子



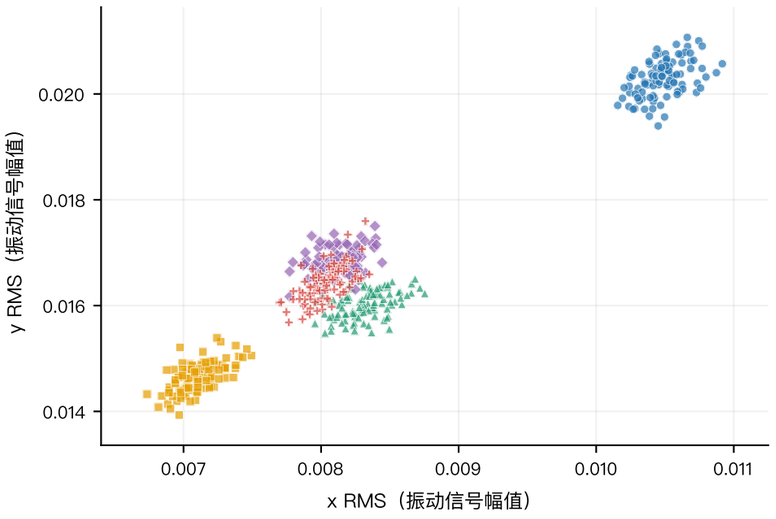
051 x RMS × y 均值



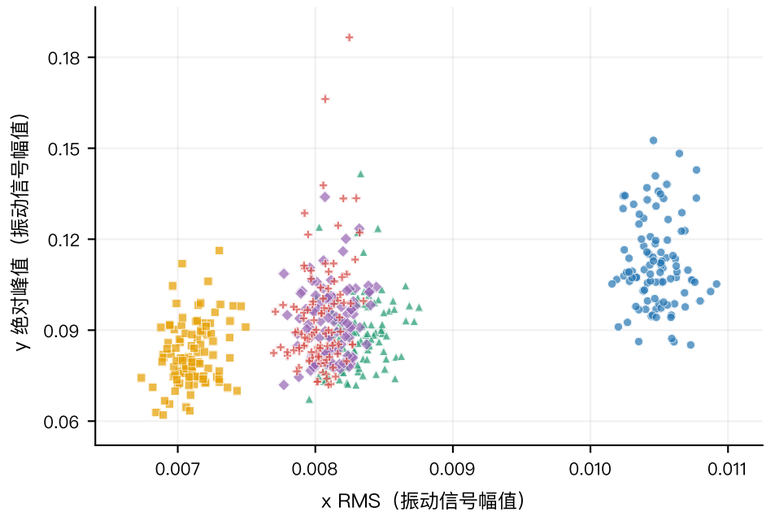
052 x RMS × y 标准差



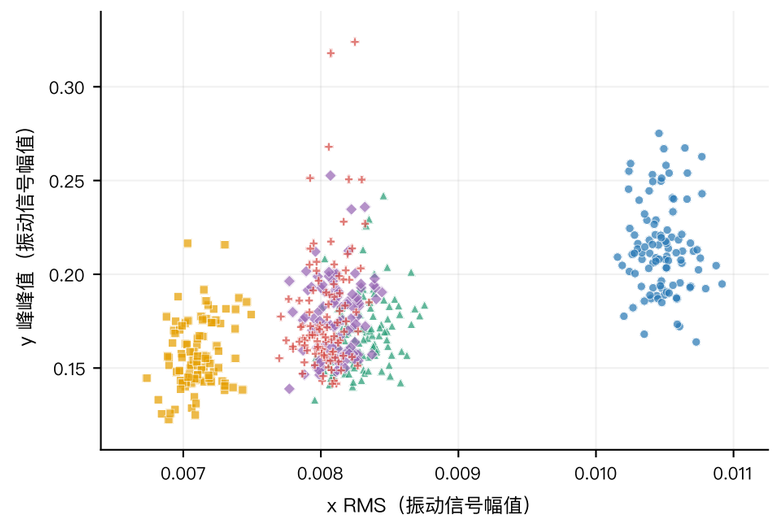
053 x RMS × y RMS



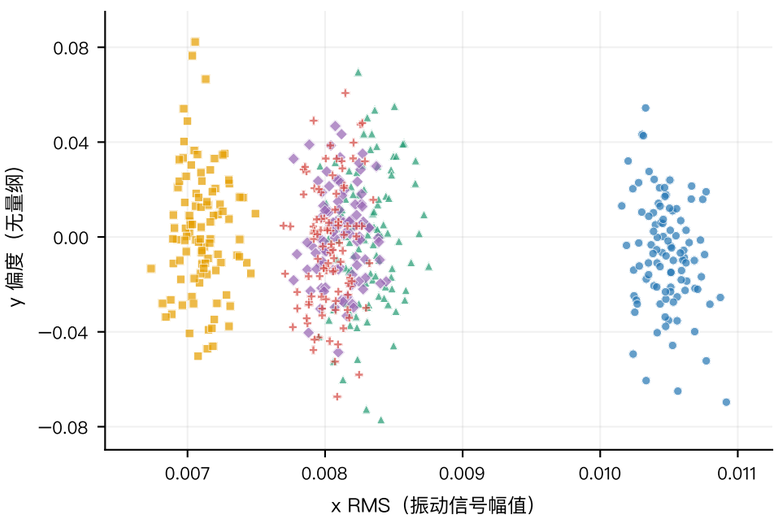
054 x RMS × y 绝对峰值



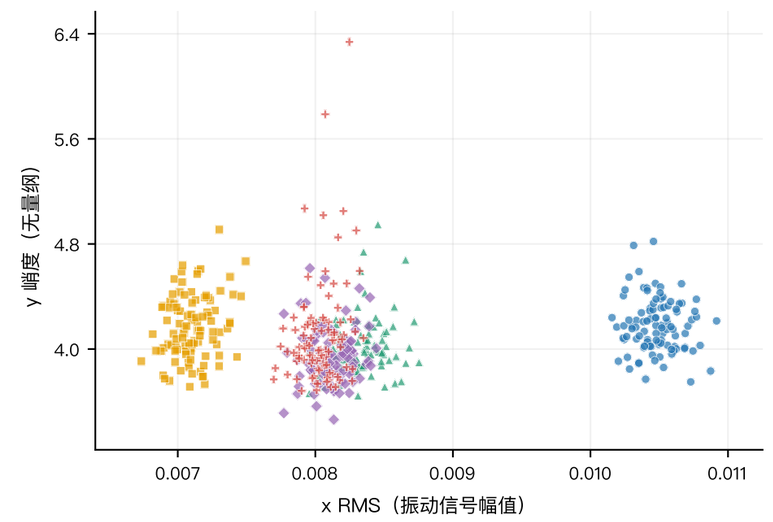
055 x RMS × y 峰峰值



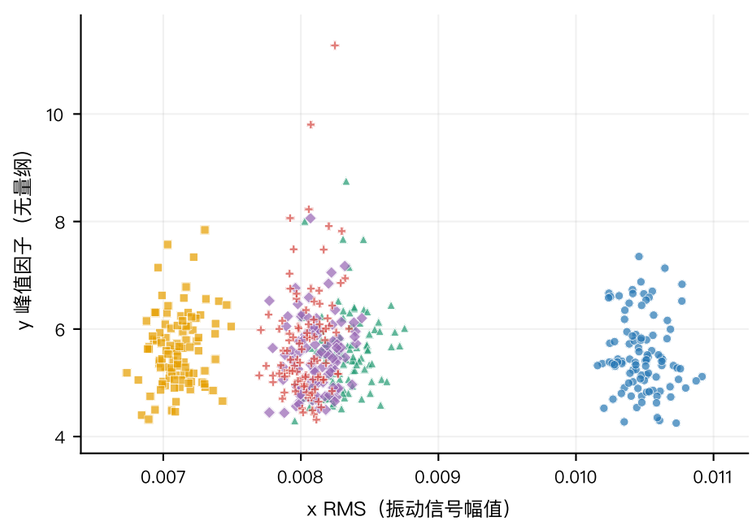
056 x RMS × y 偏度



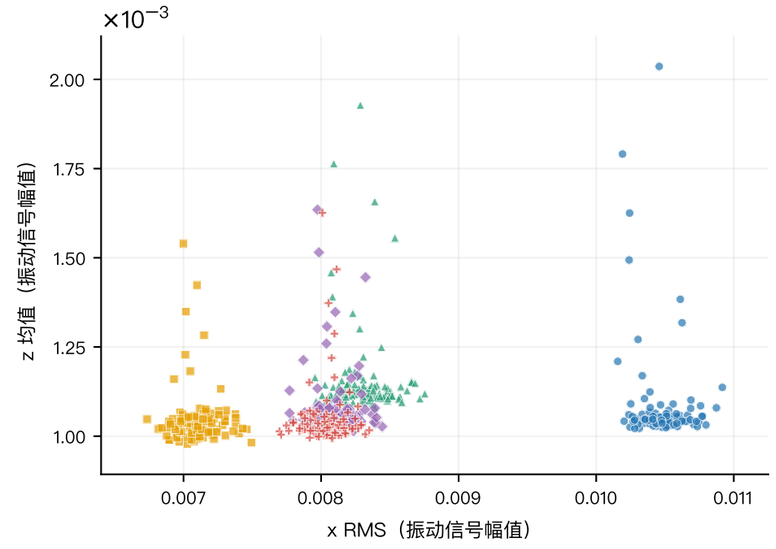
057 x RMS × y 峭度



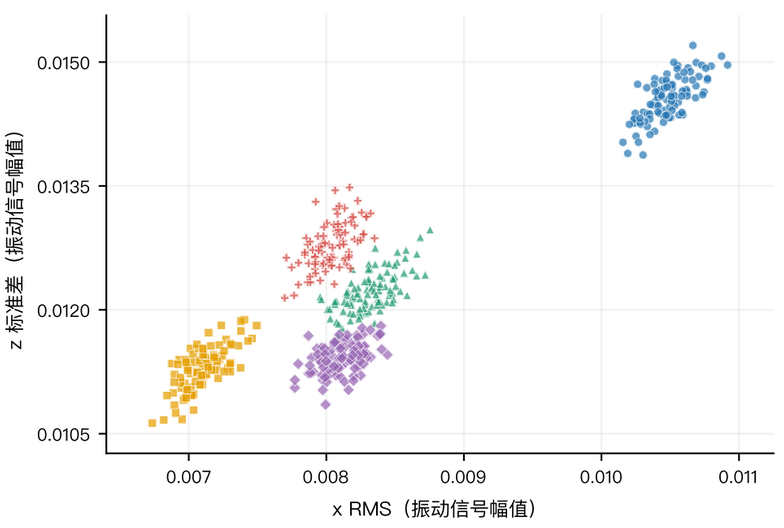
058 x RMS × y 峰值因子



059 x RMS × z 均值



060 x RMS × z 标准差



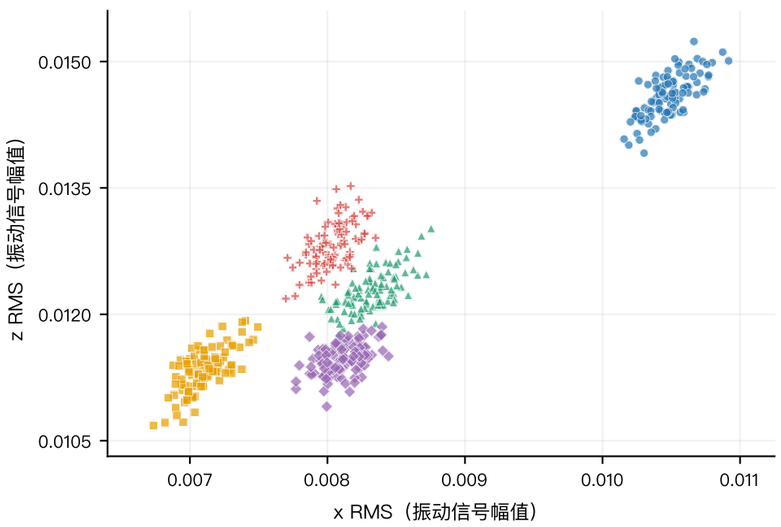
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

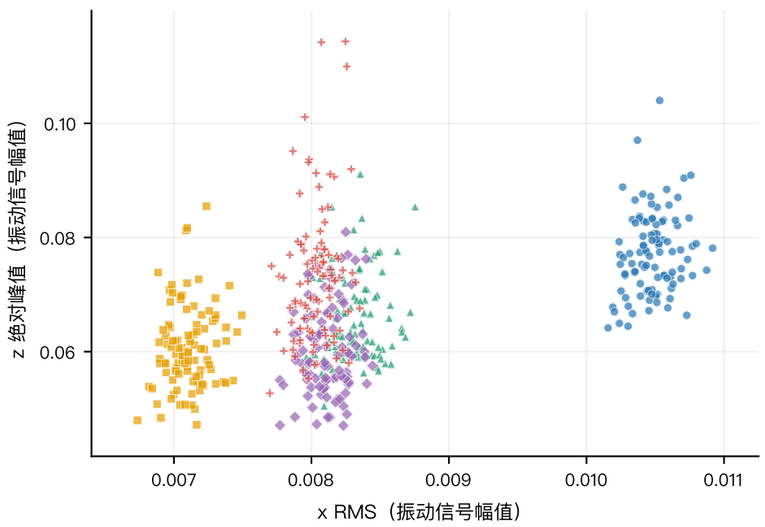
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

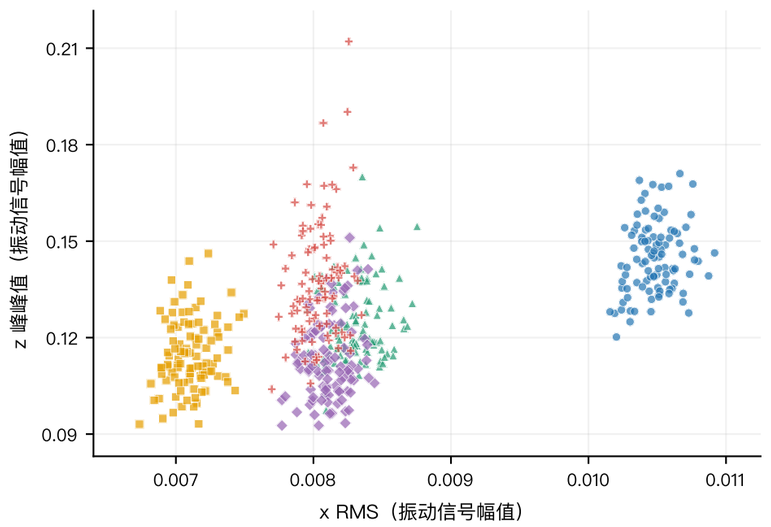
061 x RMS × z RMS



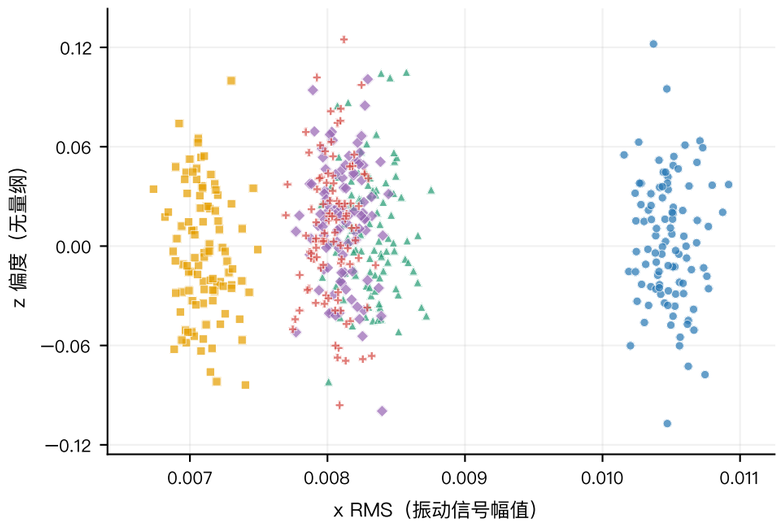
062 x RMS × z 绝对峰值



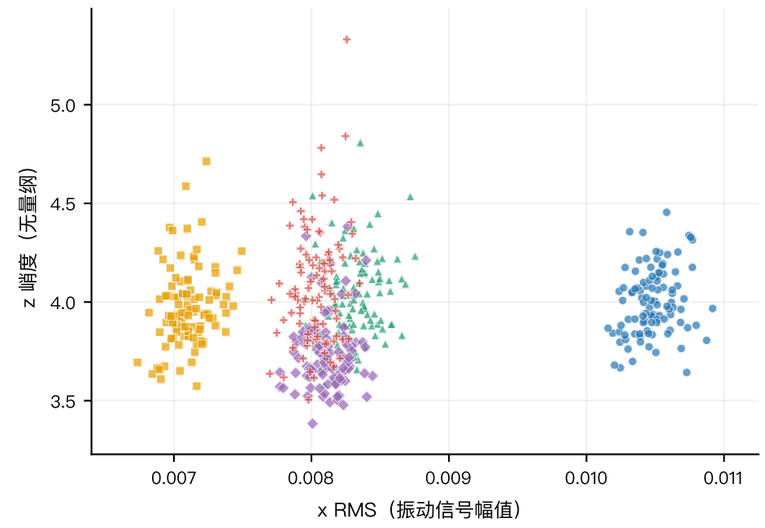
063 x RMS × z 峰峰值



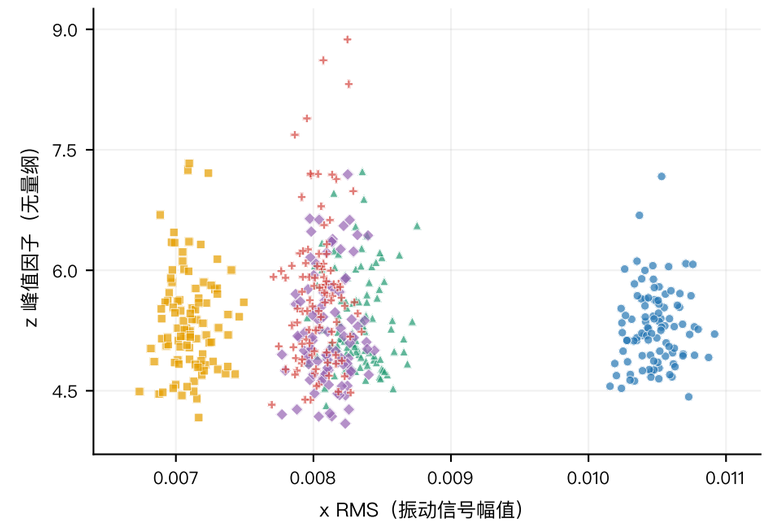
064 x RMS × z 偏度



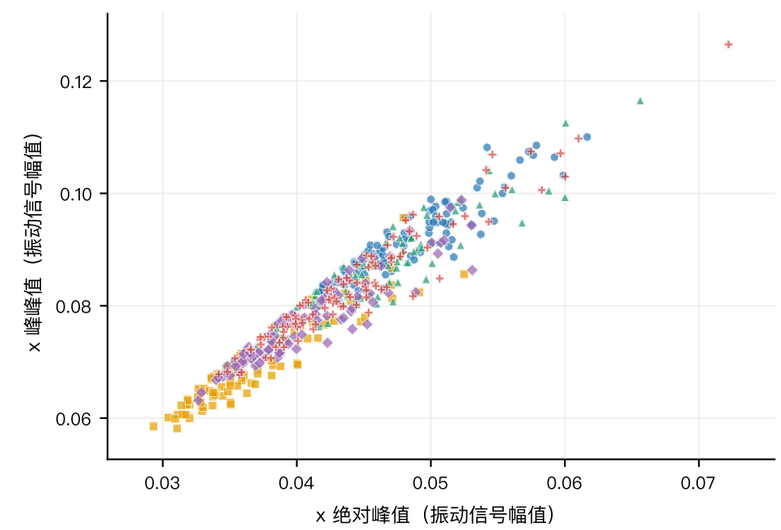
065 x RMS × z 峭度



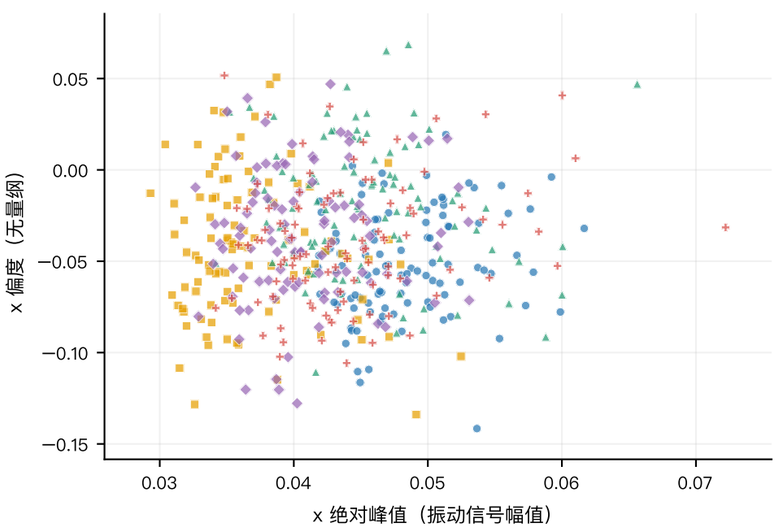
066 x RMS × z 峰值因子



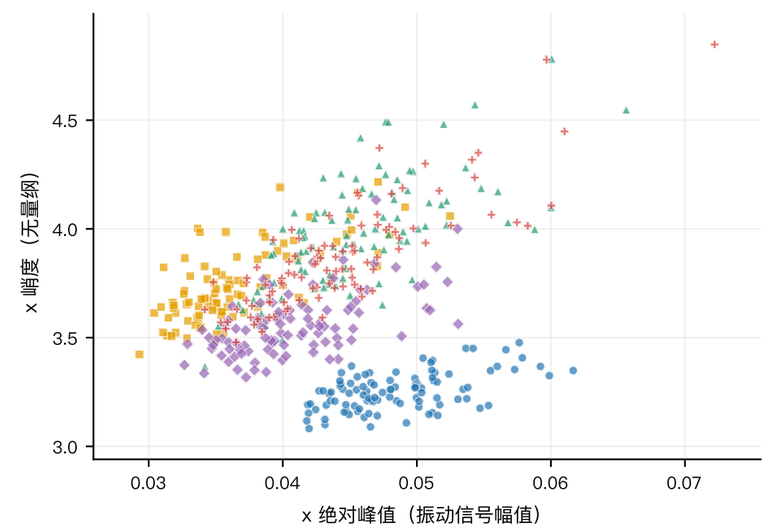
067 x 绝对峰值 × x 峰峰值



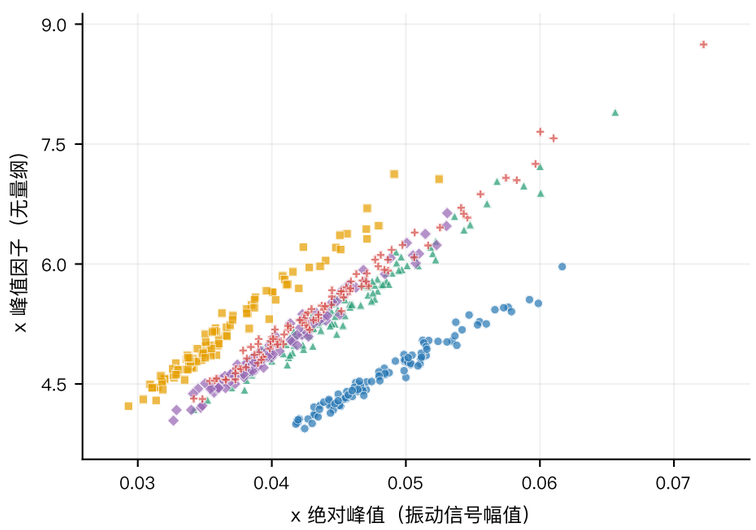
068 x 绝对峰值 × x 偏度



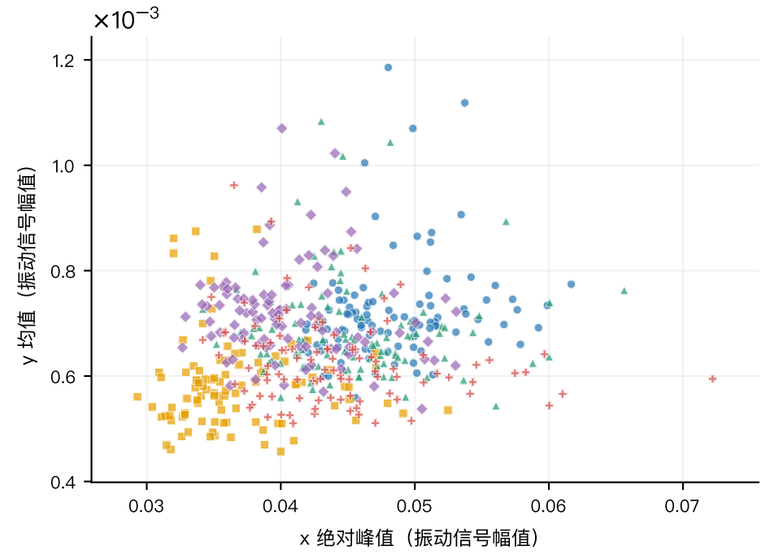
069 x 绝对峰值 × x 峭度



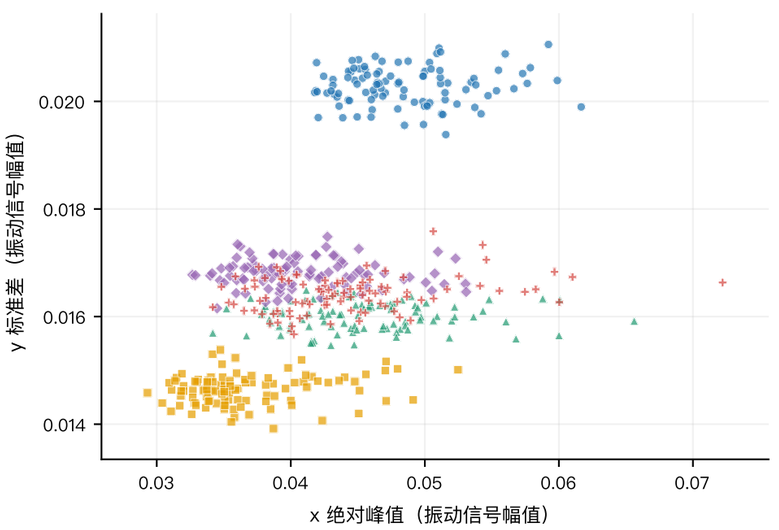
070 x 绝对峰值 × x 峰值因子



071 x 绝对峰值 × y 均值



072 x 绝对峰值 × y 标准差



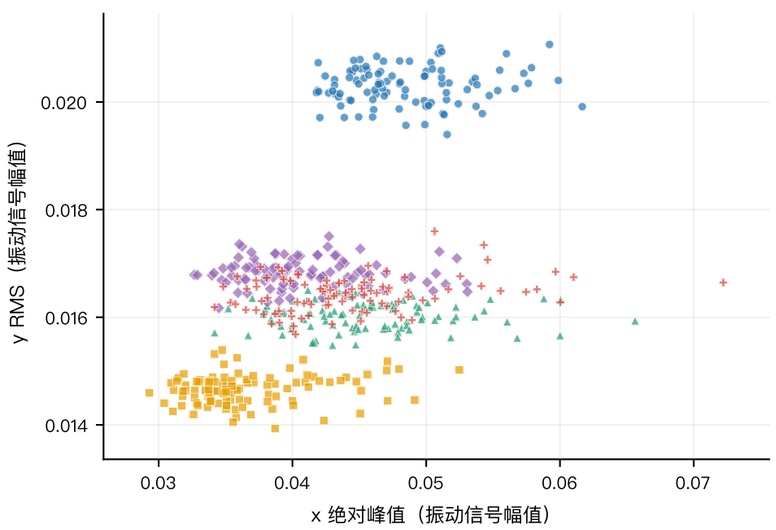
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

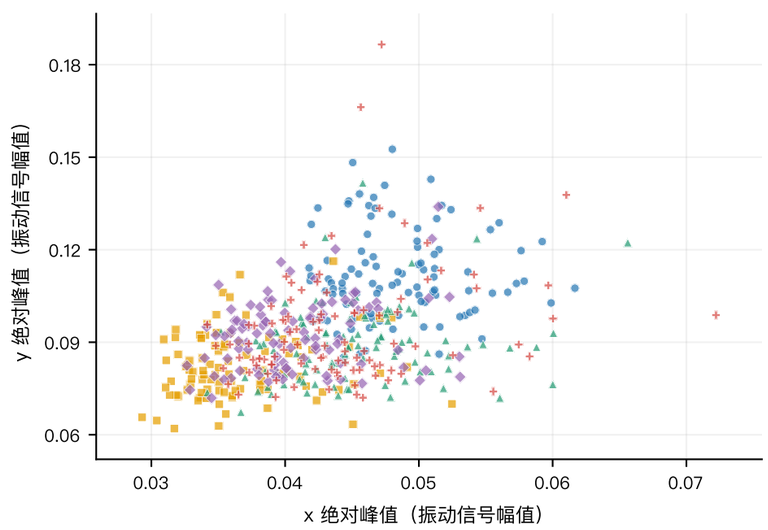
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

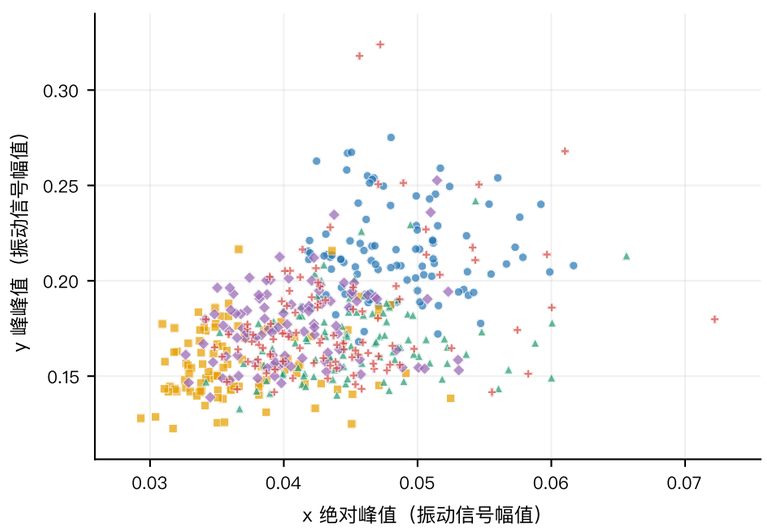
073 x 绝对峰值 × y RMS



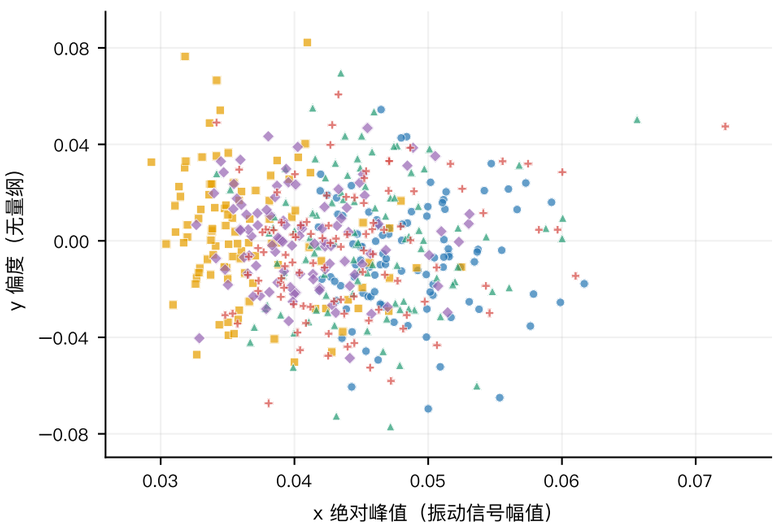
074 x 绝对峰值 × y 绝对峰值



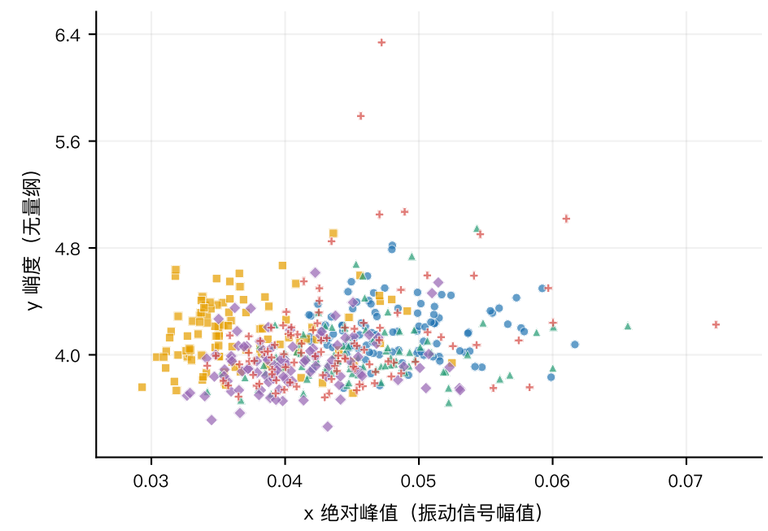
075 x 绝对峰值 × y 峰峰值



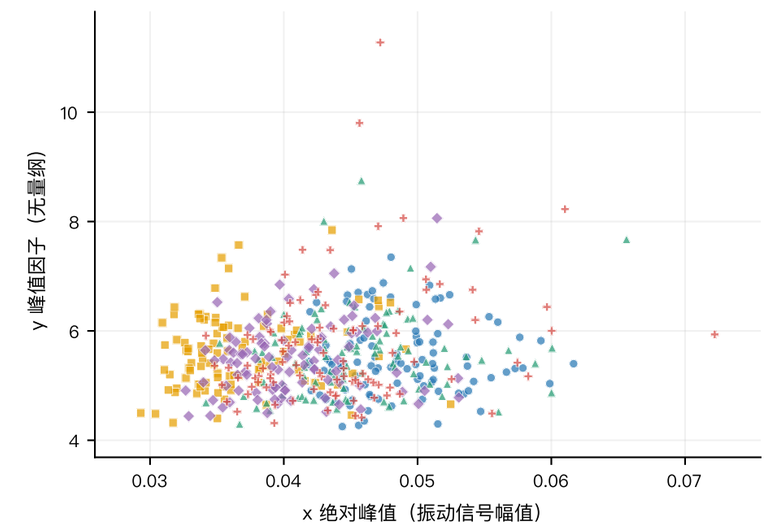
076 x 绝对峰值 × y 偏度



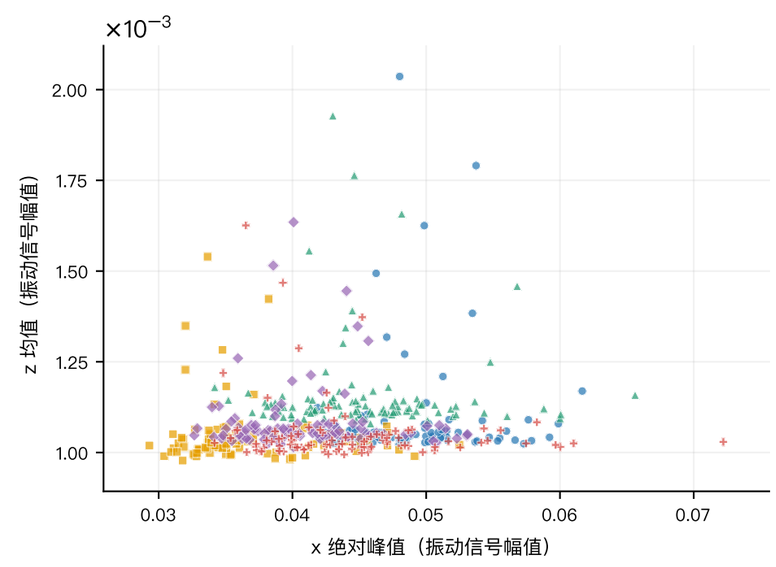
077 x 绝对峰值 × y 峭度



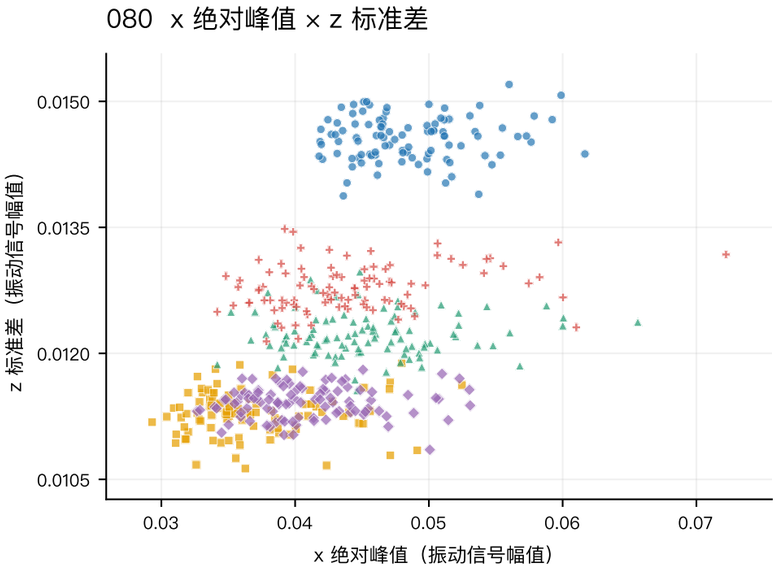
078 x 绝对峰值 × y 峰值因子



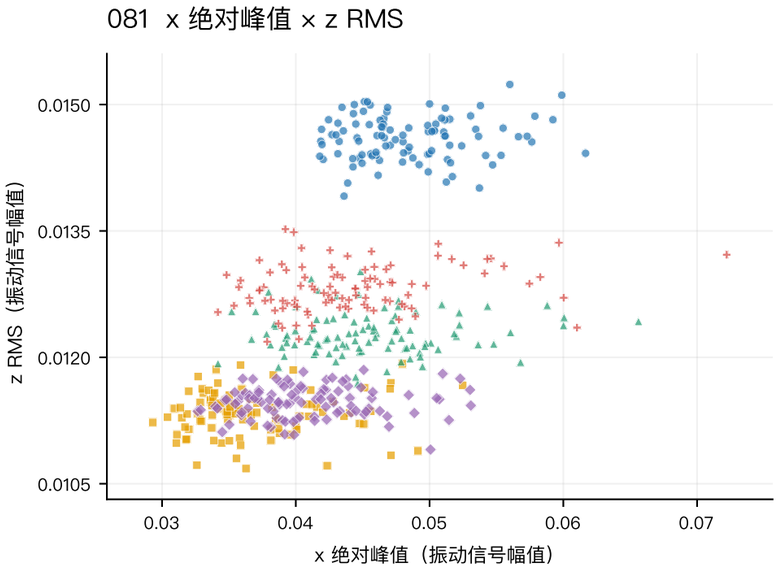
079 x 绝对峰值 × z 均值



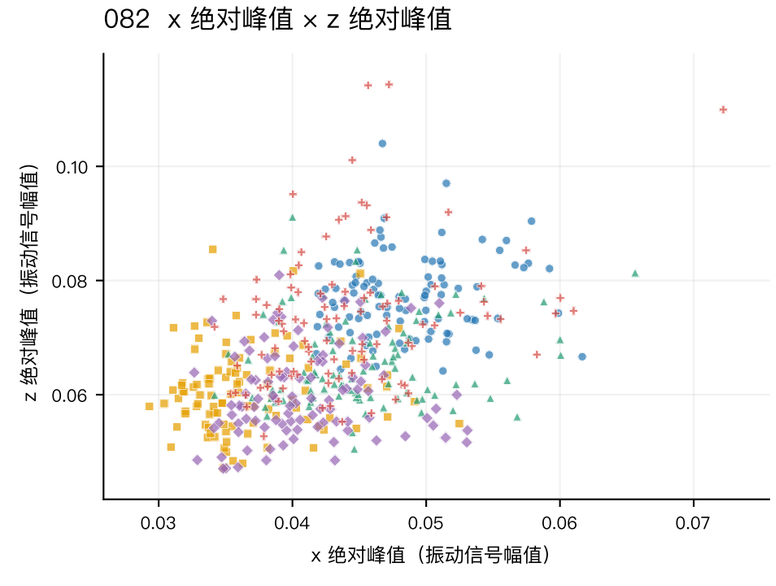
080 x 绝对峰值 × z 标准差



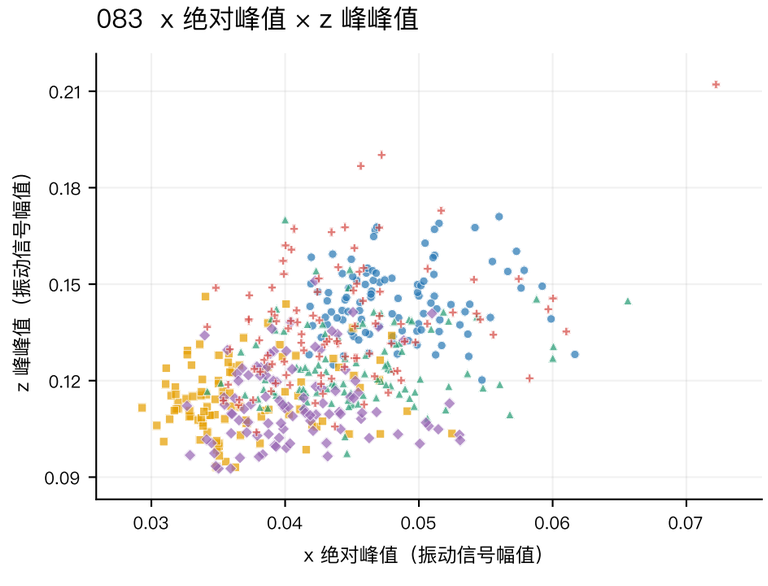
081 x 绝对峰值 × z RMS



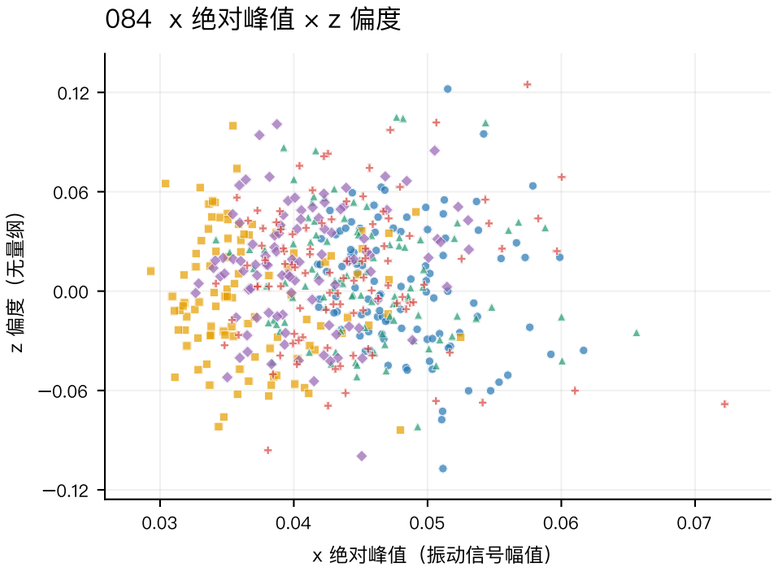
082 x 绝对峰值 × z 绝对峰值



083 x 绝对峰值 × z 峰峰值



084 x 绝对峰值 × z 偏度



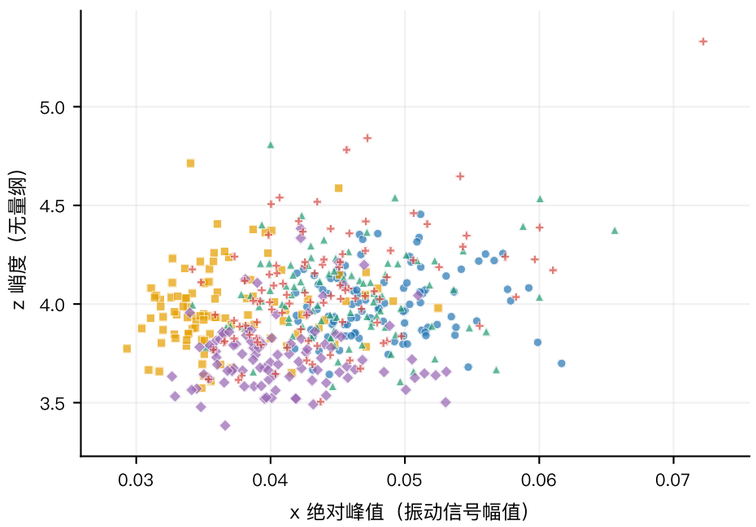
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

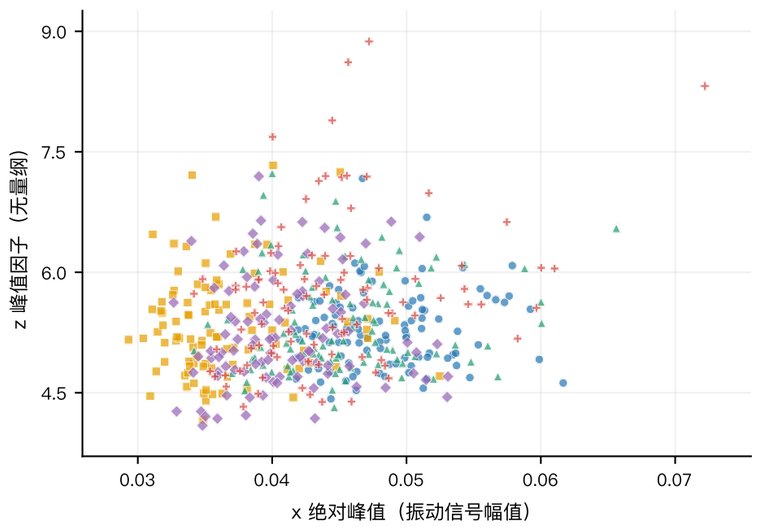
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

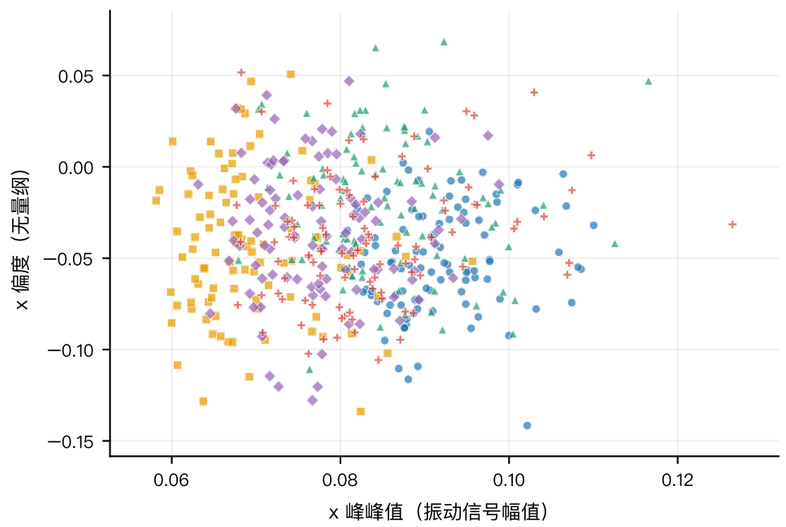
085 x 绝对峰值 × z 峭度



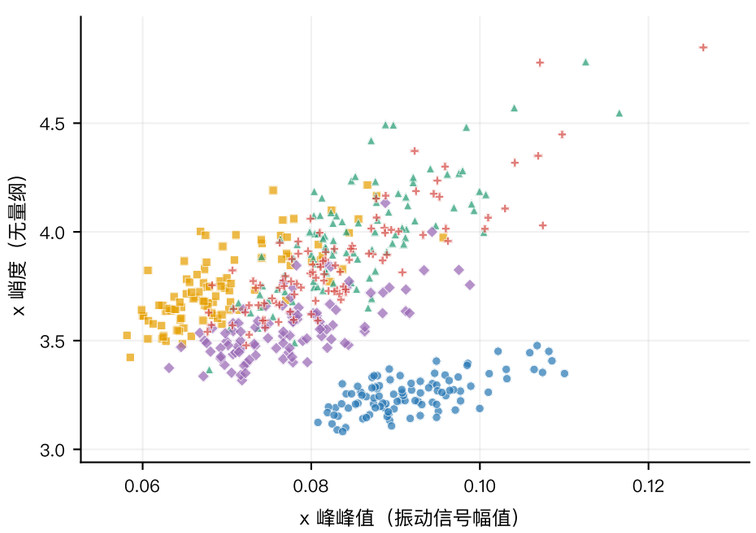
086 x 绝对峰值 × z 峰值因子



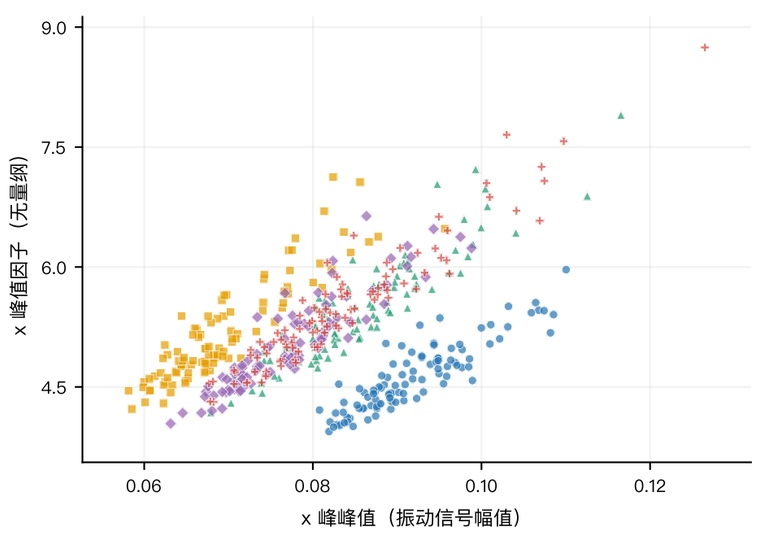
087 x 峰峰值 × x 偏度



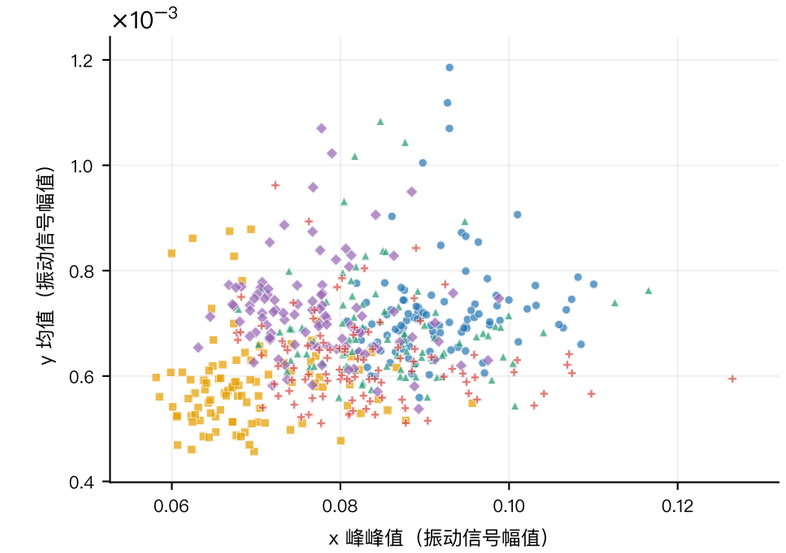
088 x 峰峰值 × x 峭度



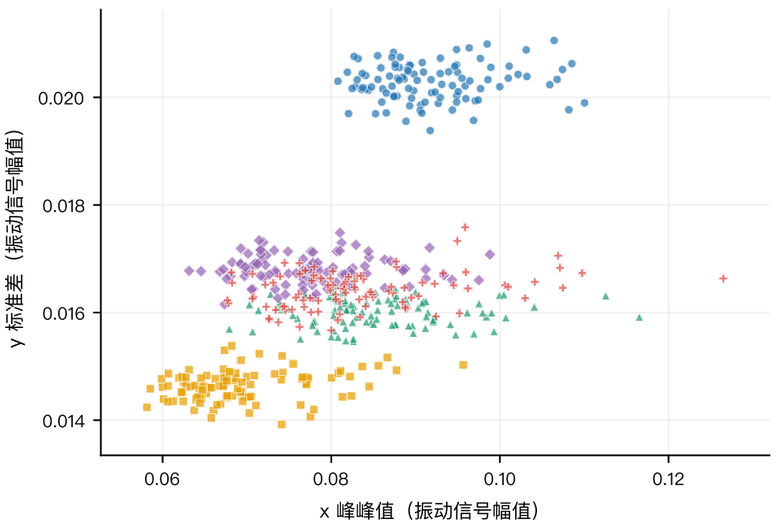
089 x 峰峰值 × x 峰值因子



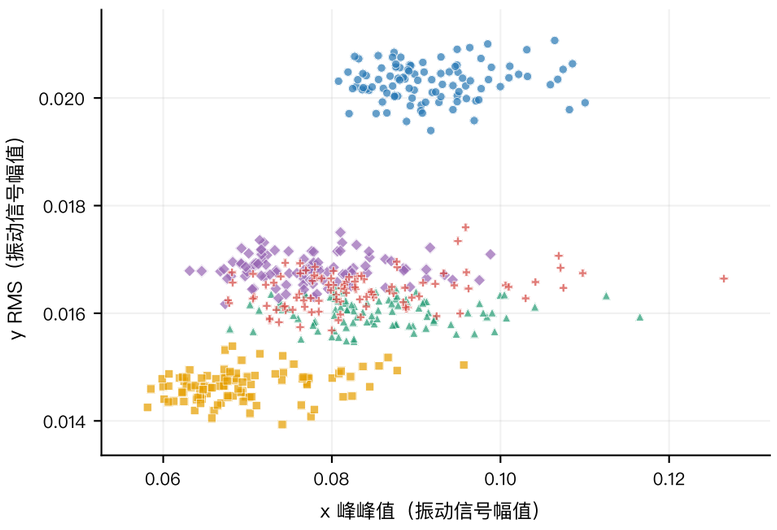
090 x 峰峰值 × y 均值



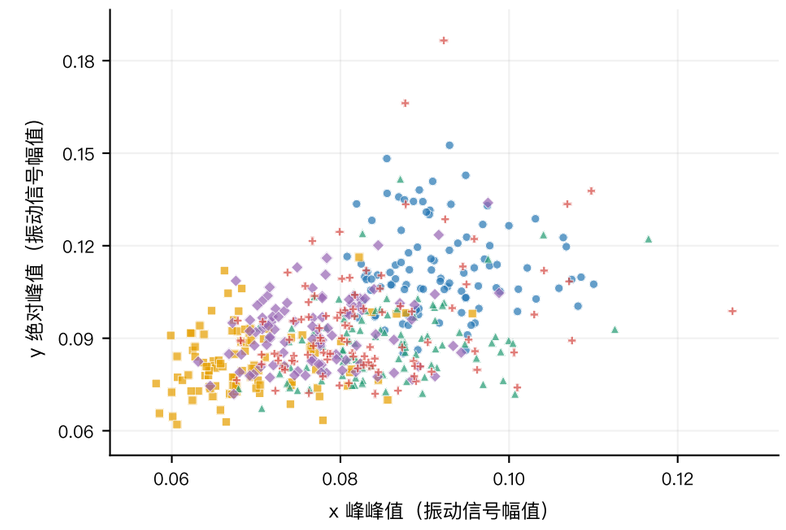
091 x 峰峰值 × y 标准差



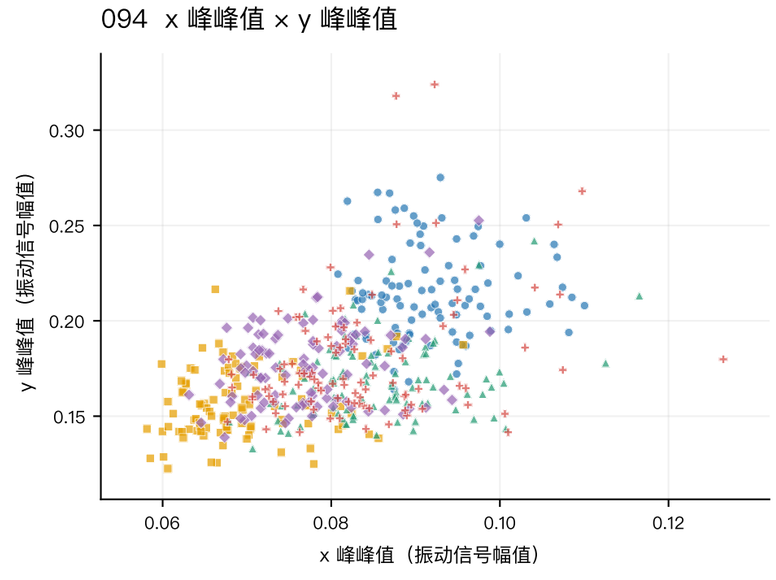
092 x 峰峰值 × y RMS



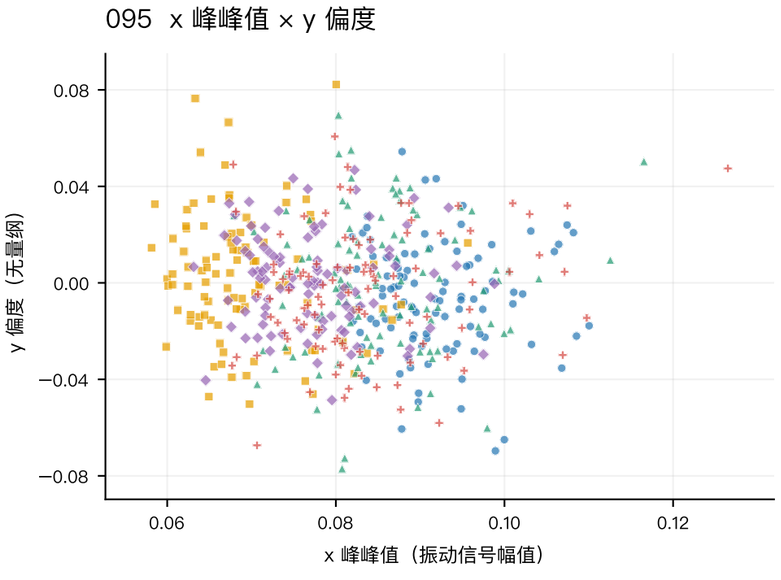
093 x 峰峰值 × y 绝对峰值



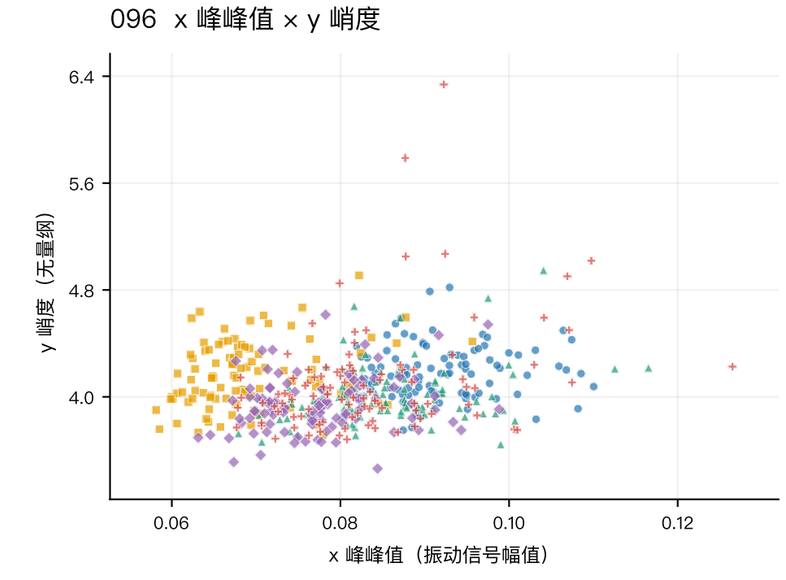
094 x 峰峰值 × y 峰峰值



095 x 峰峰值 × y 偏度



096 x 峰峰值 × y 峭度



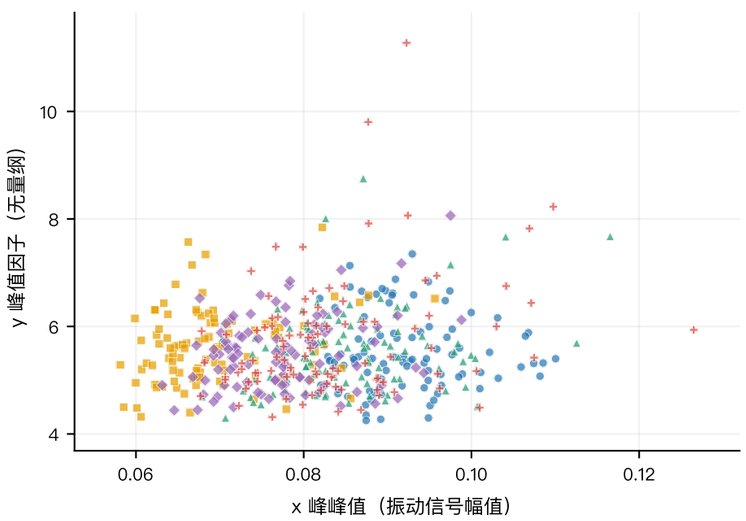
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

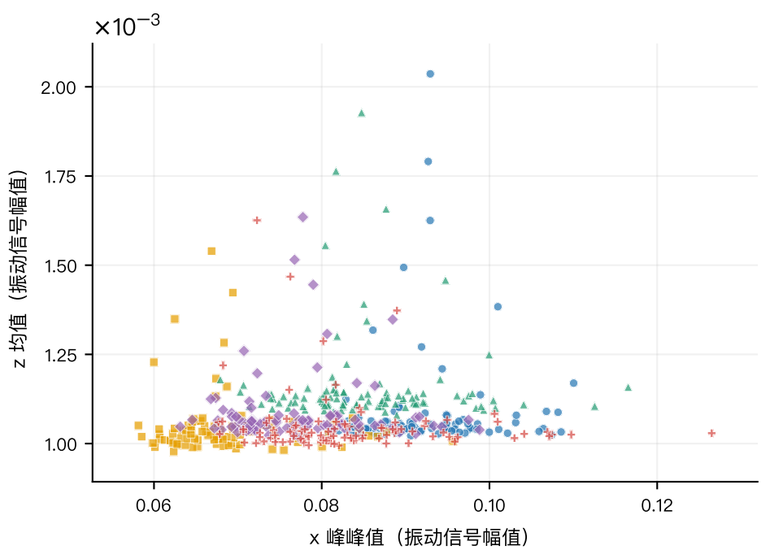
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

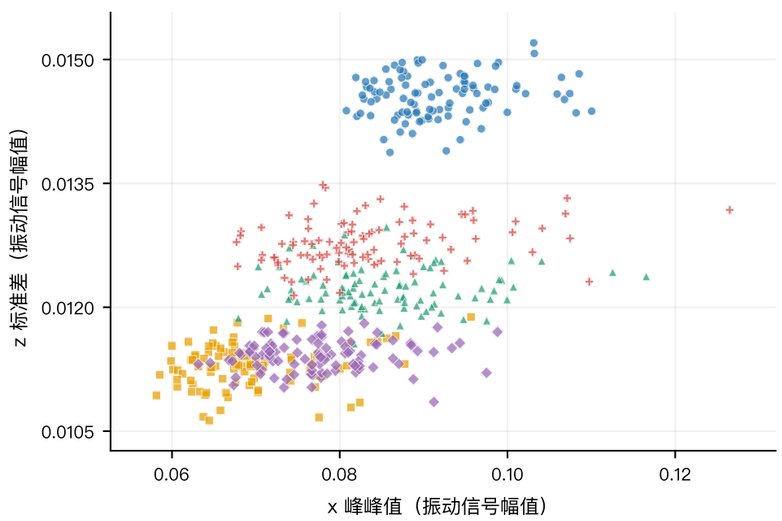
097 x 峰峰值 × y 峰值因子



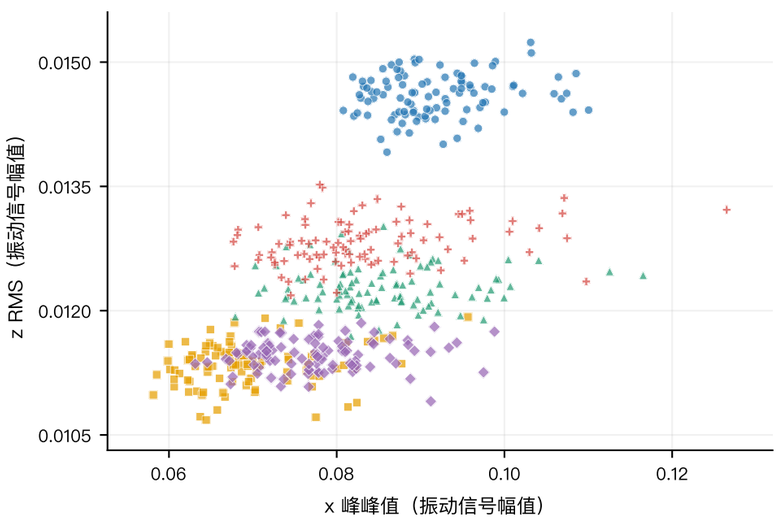
098 x 峰峰值 × z 均值



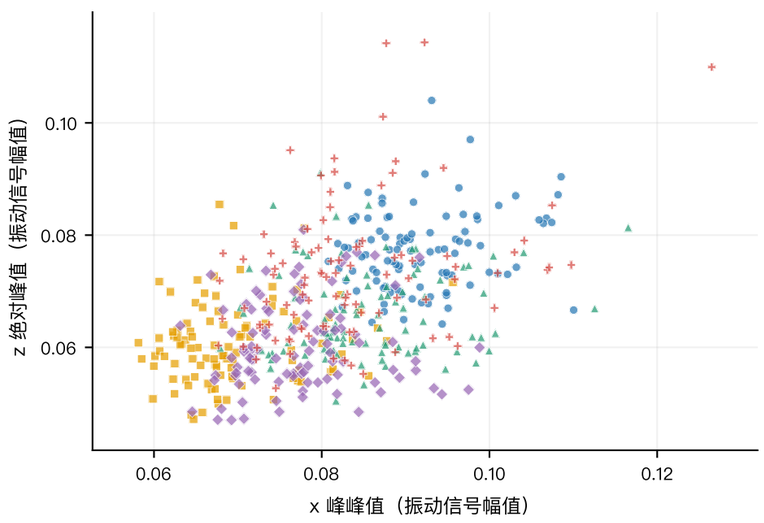
099 x 峰峰值 × z 标准差



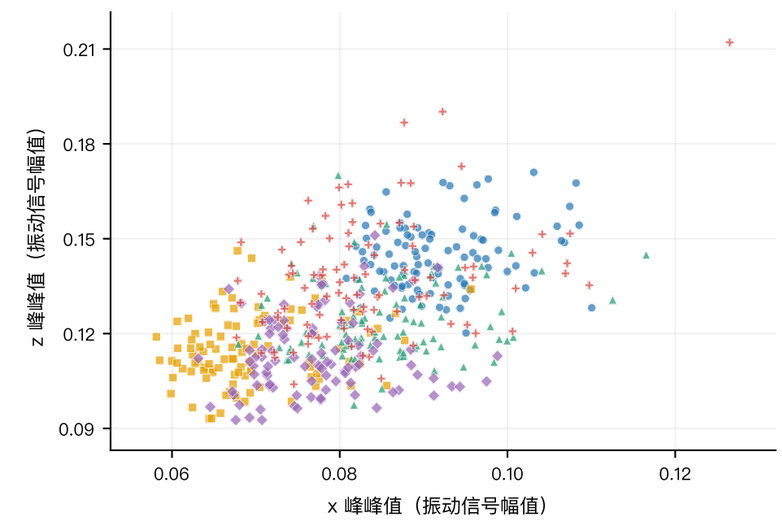
100 x 峰峰值 × z RMS



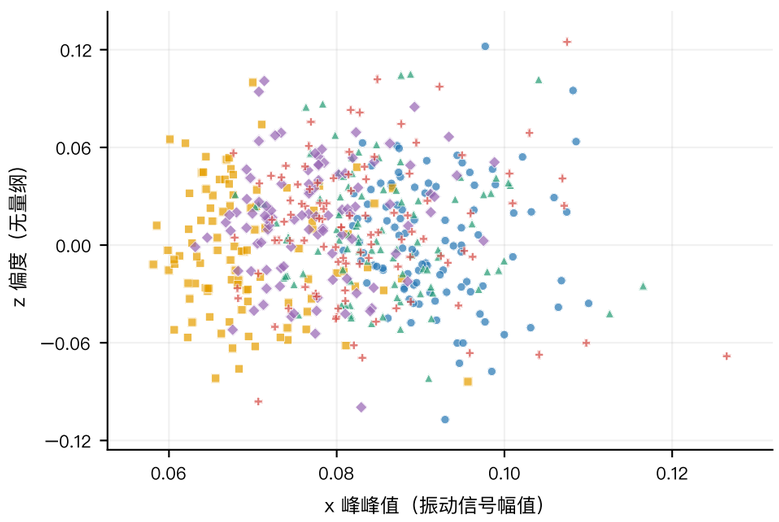
101 x 峰峰值 × z 绝对峰值



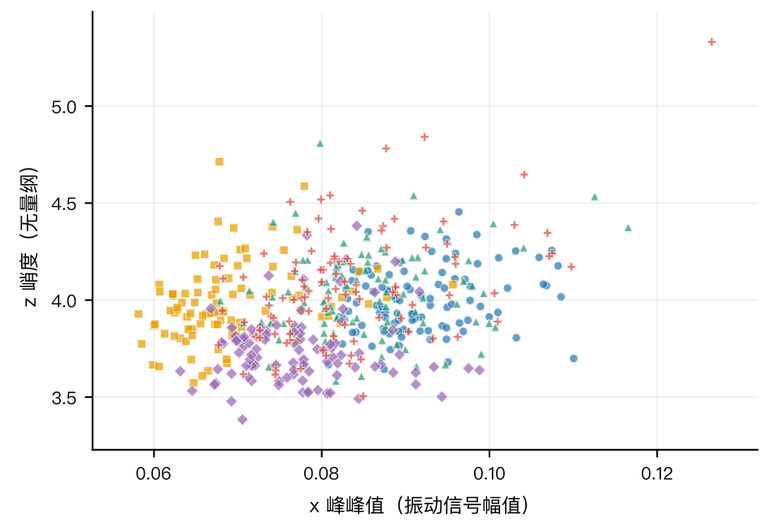
102 x 峰峰值 × z 峰峰值



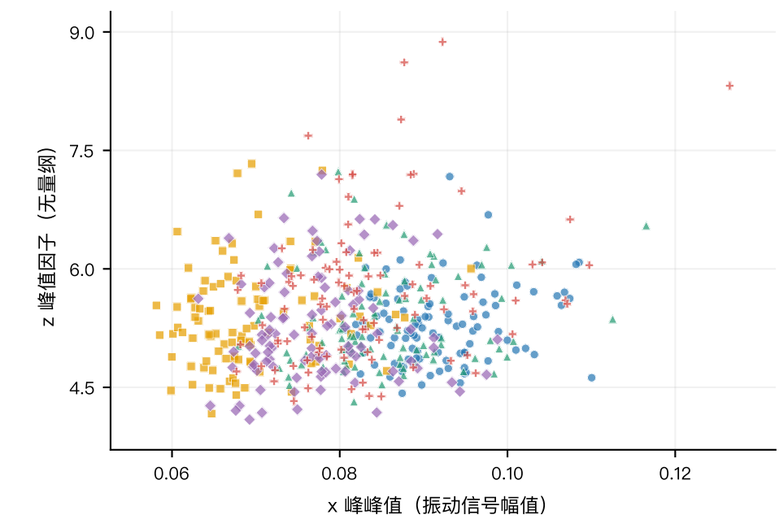
103 x 峰峰值 × z 偏度



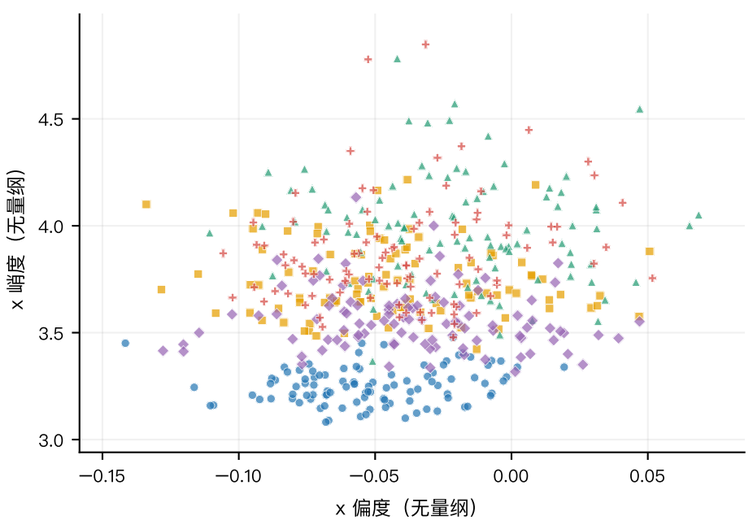
104 x 峰峰值 × z 峭度



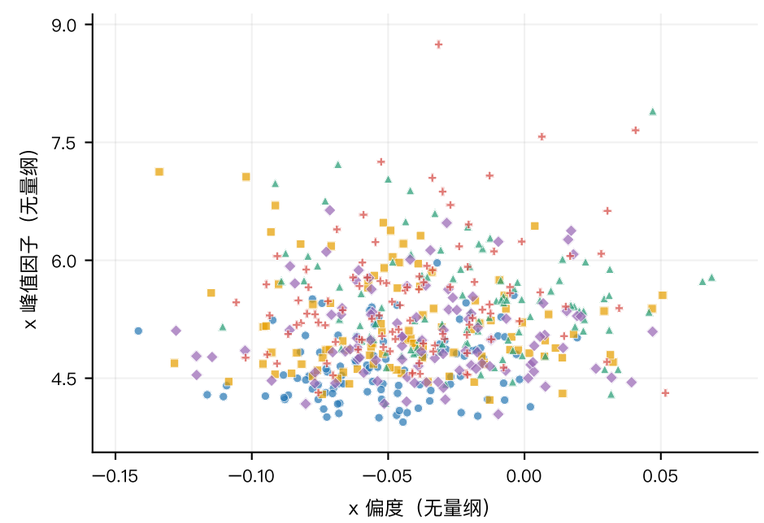
105 x 峰峰值 × z 峰值因子



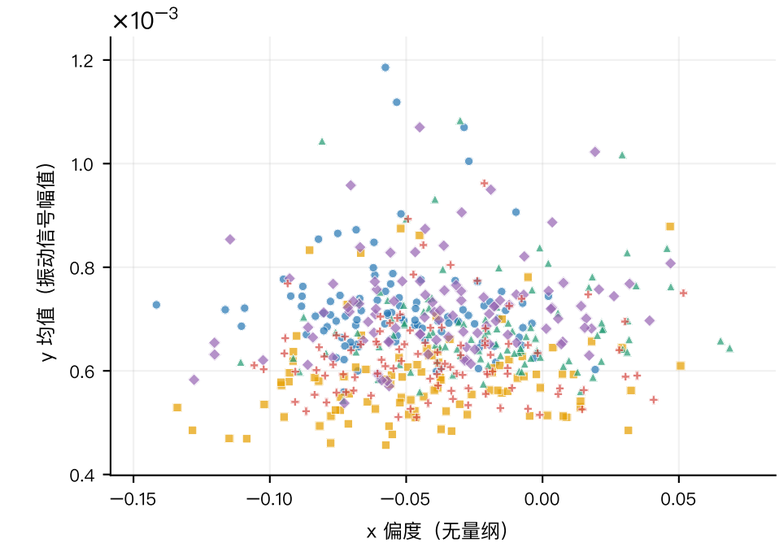
106 x 偏度 × x 峭度



107 x 偏度 × x 峰值因子



108 x 偏度 × y 均值



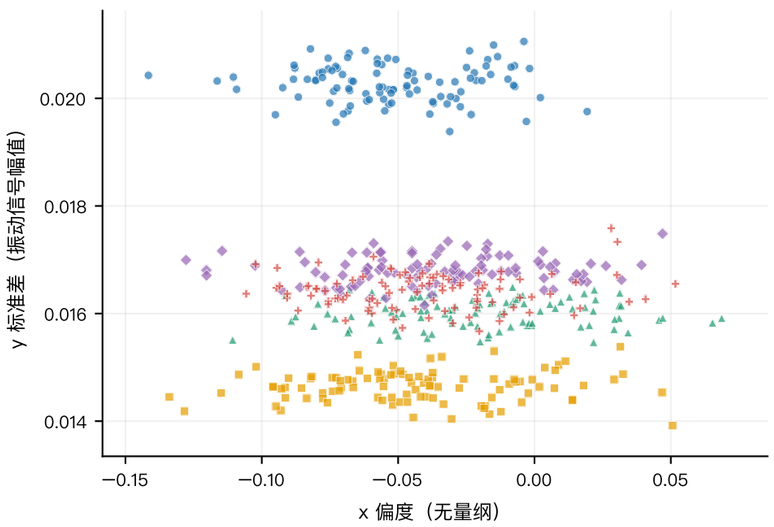
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

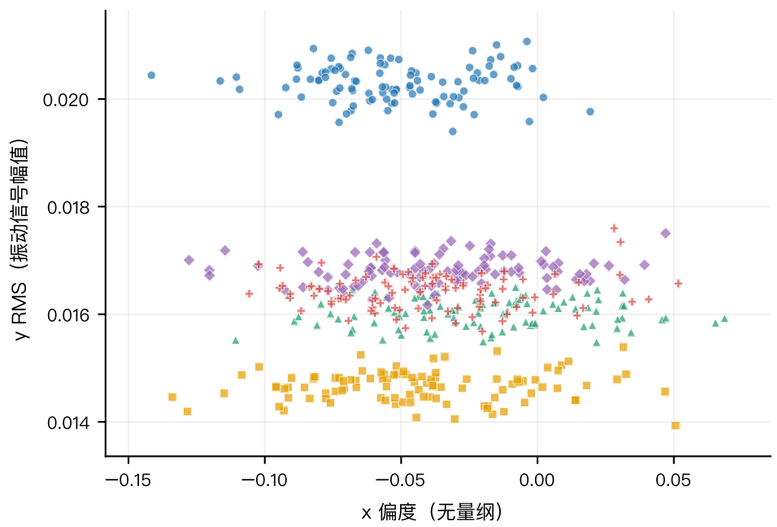
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

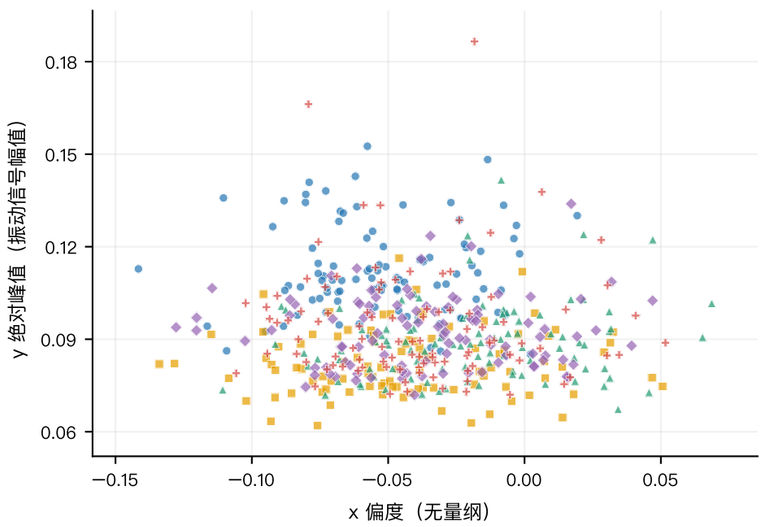
109 x 偏度 × y 标准差



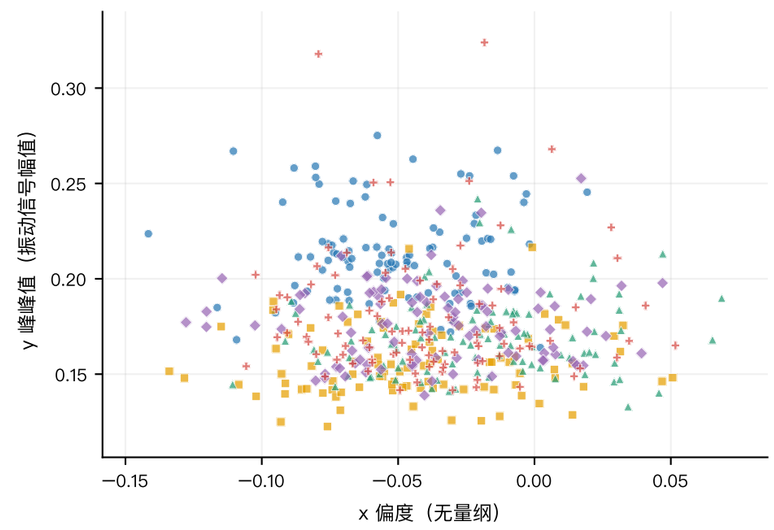
110 x 偏度 × y RMS



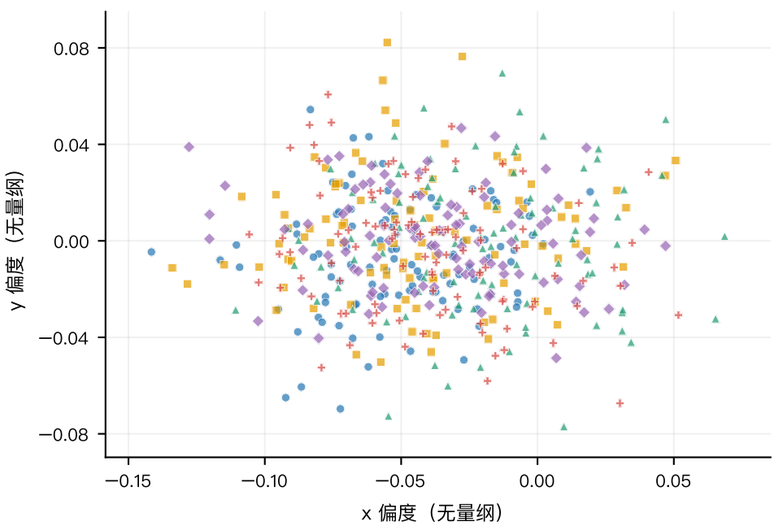
111 x 偏度 × y 绝对峰值



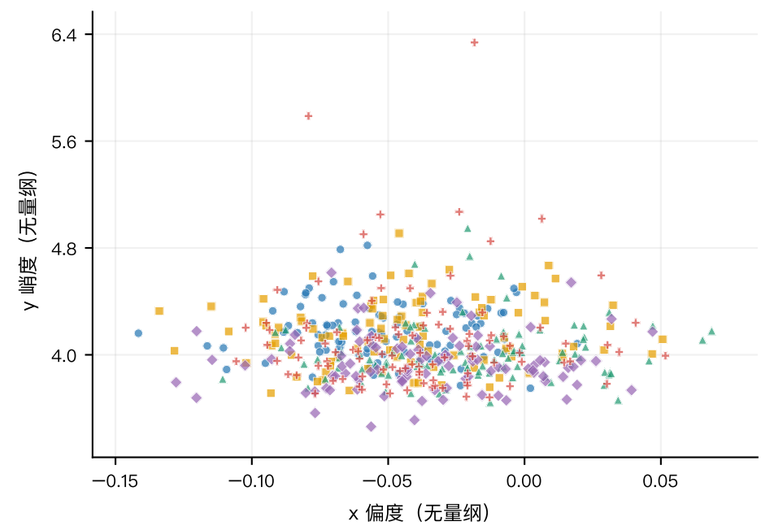
112 x 偏度 × y 峰峰值



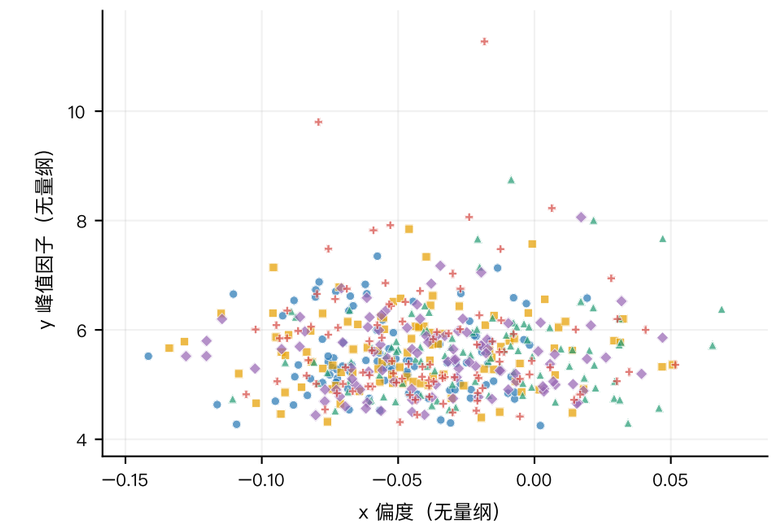
113 x 偏度 × y 偏度



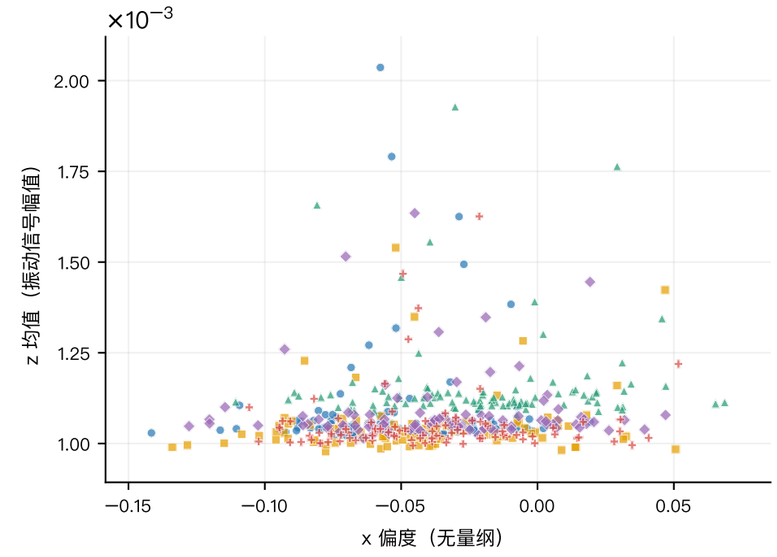
114 x 偏度 × y 峭度



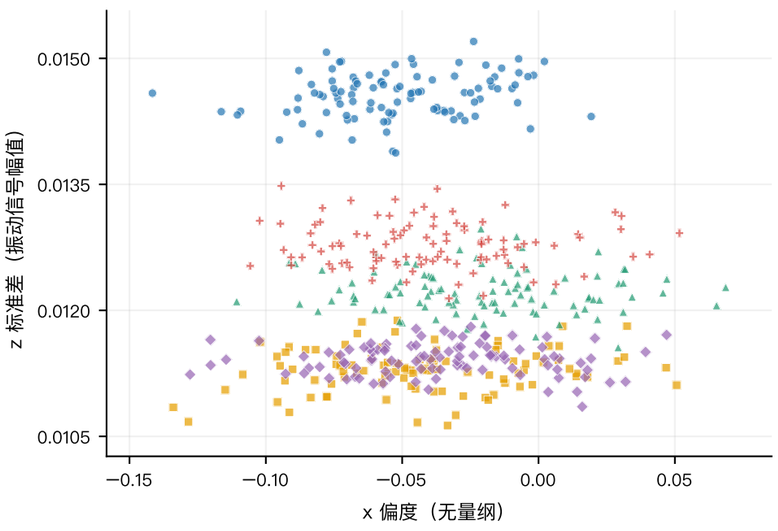
115 x 偏度 × y 峰值因子



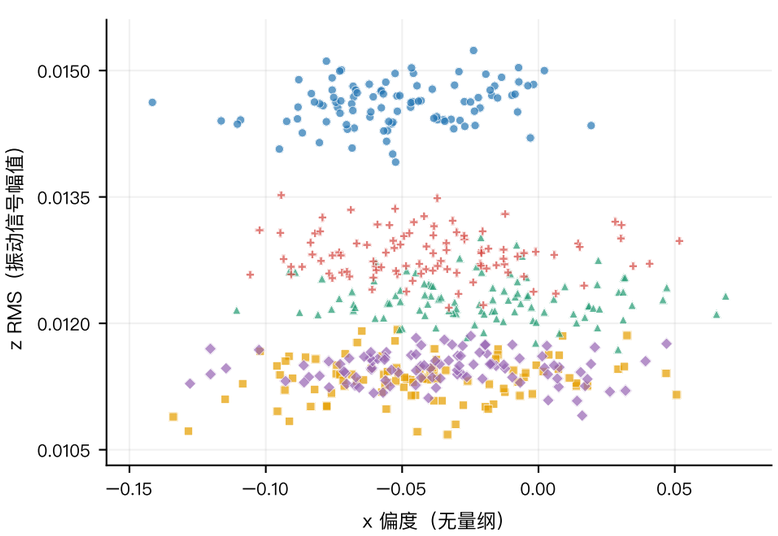
116 x 偏度 × z 均值



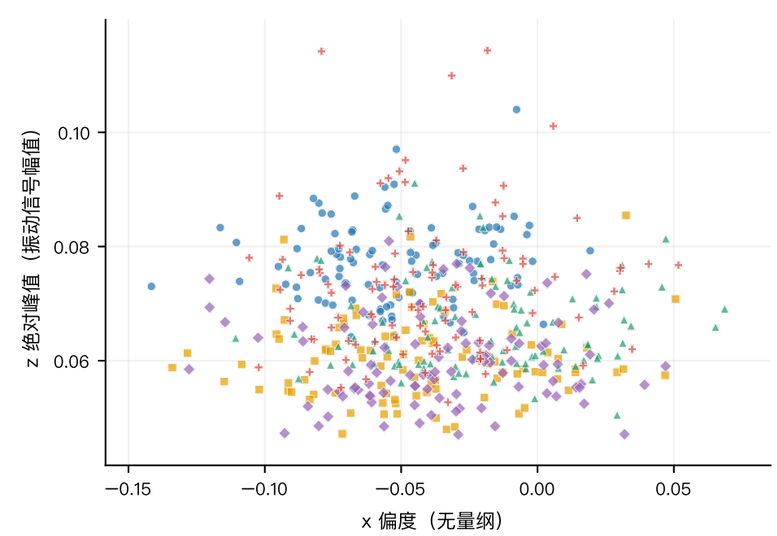
117 x 偏度 × z 标准差



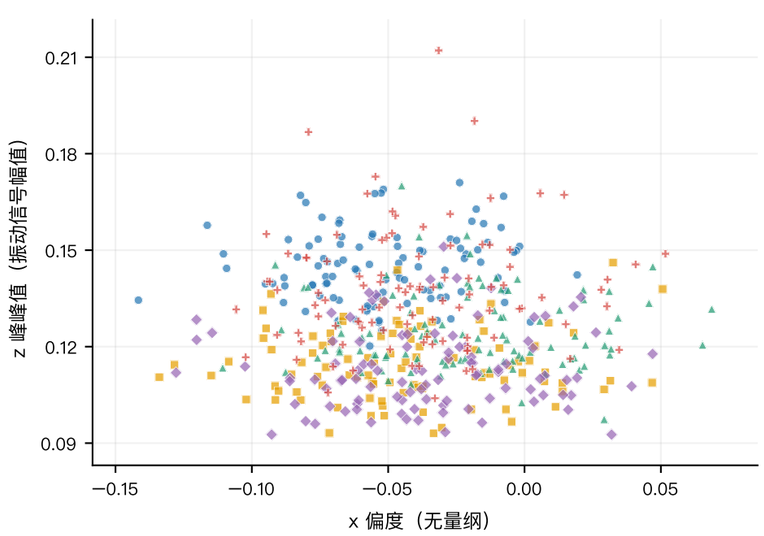
118 x 偏度 × z RMS



119 x 偏度 × z 绝对峰值



120 x 偏度 × z 峰峰值



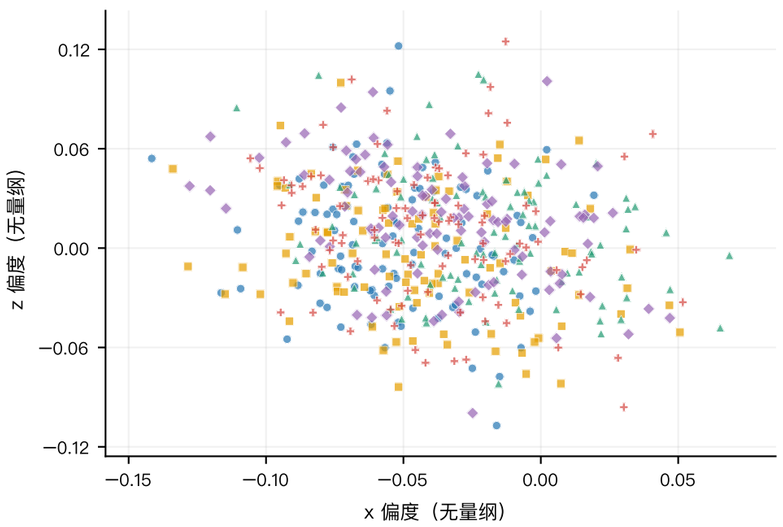
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

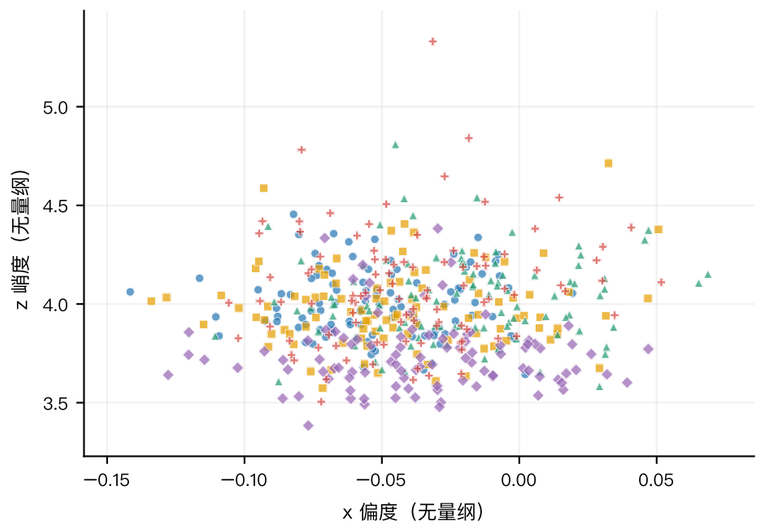
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

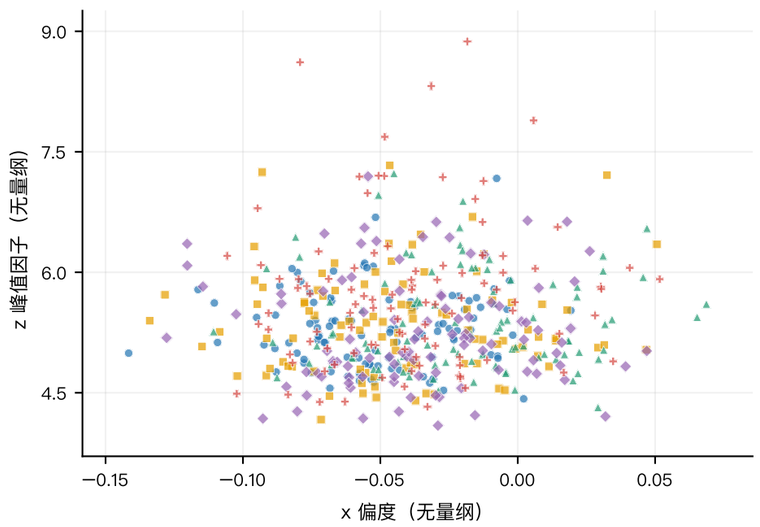
121 x 偏度 × z 偏度



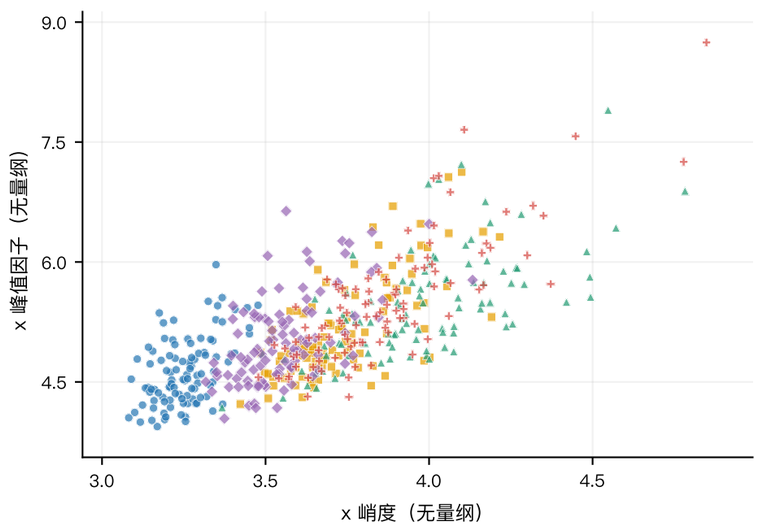
122 x 偏度 × z 峭度



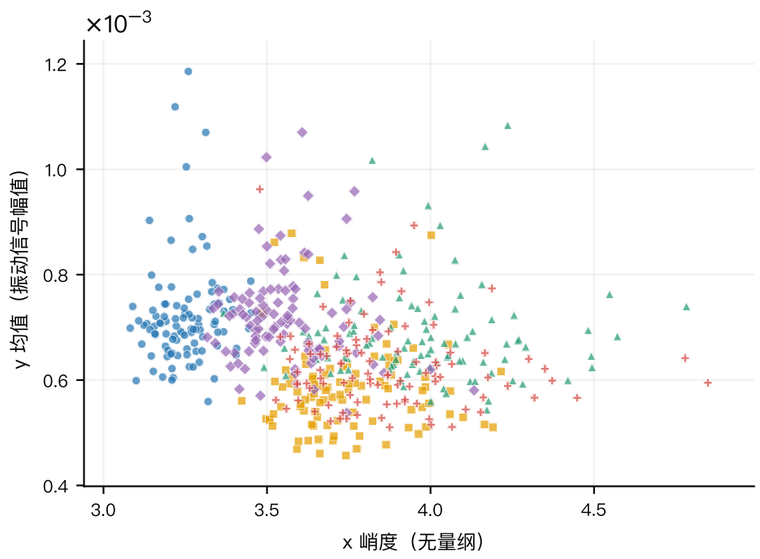
123 x 偏度 × z 峰值因子



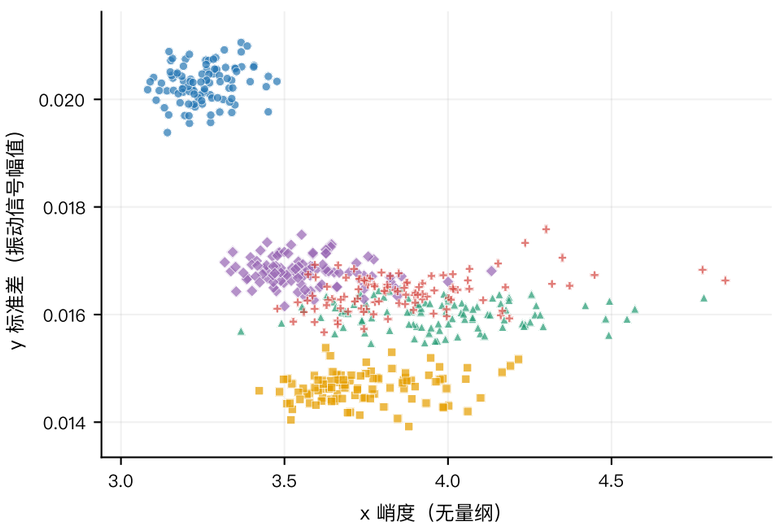
124 x 峭度 × x 峰值因子



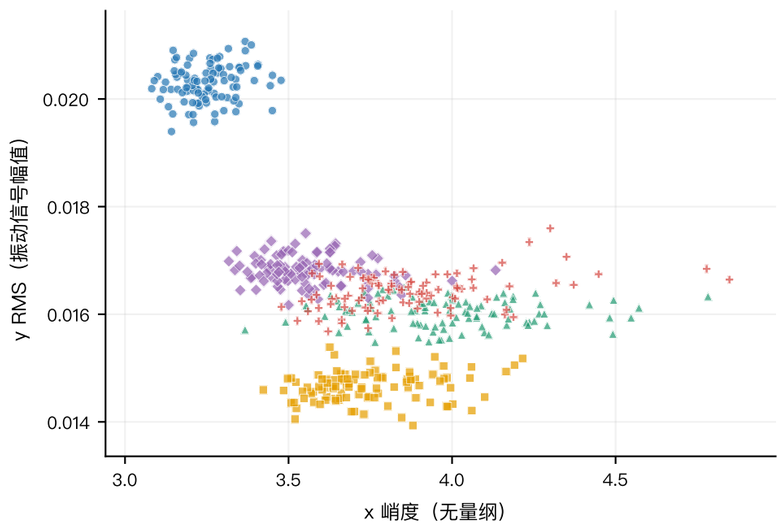
125 x 峭度 × y 均值



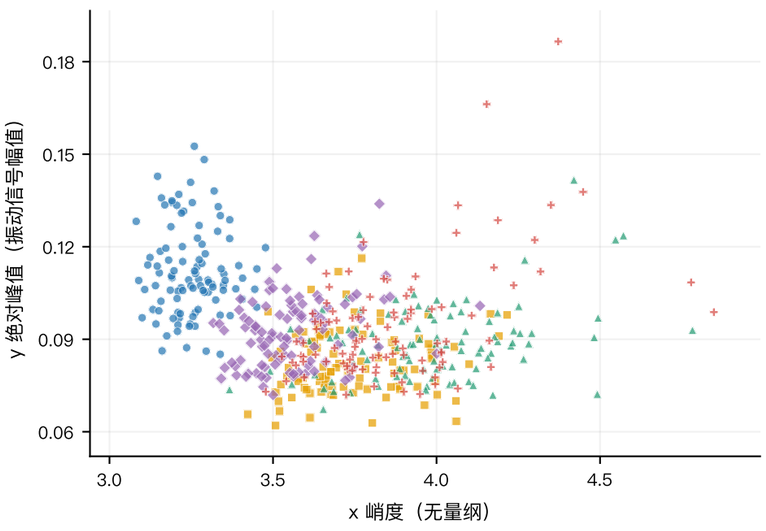
126 x 峭度 × y 标准差



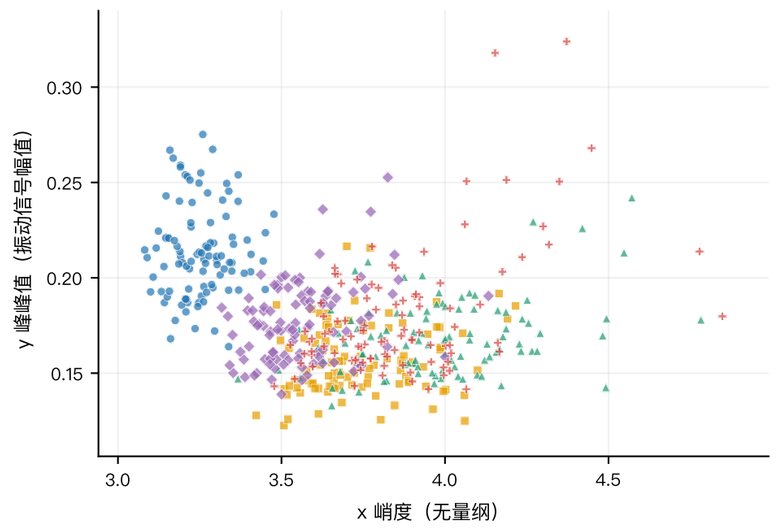
127 x 峭度 × y RMS



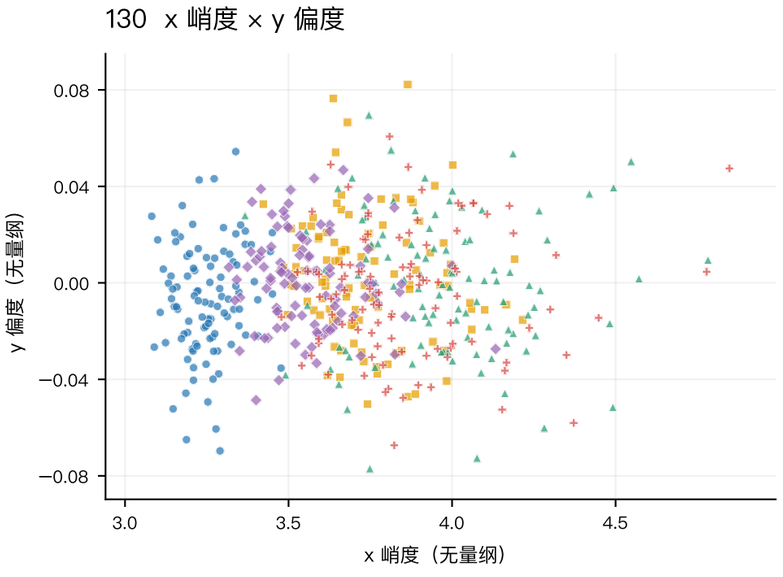
128 x 峭度 × y 绝对峰值



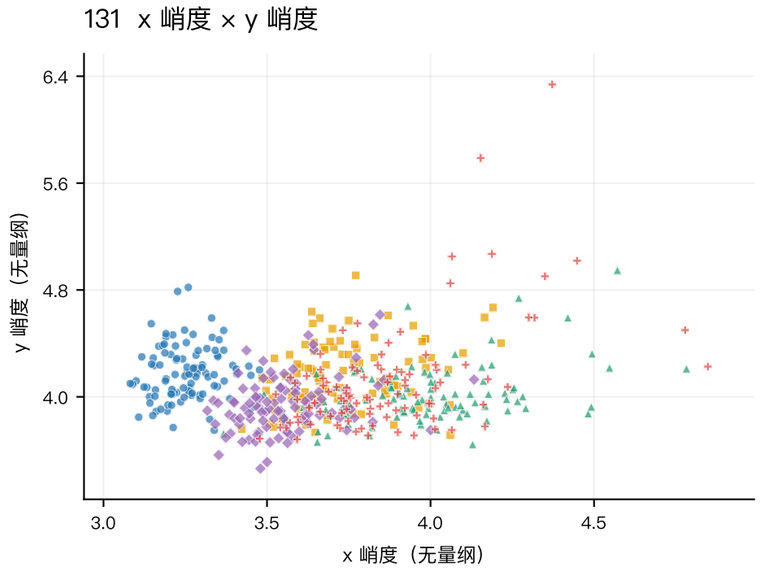
129 x 峭度 × y 峰峰值



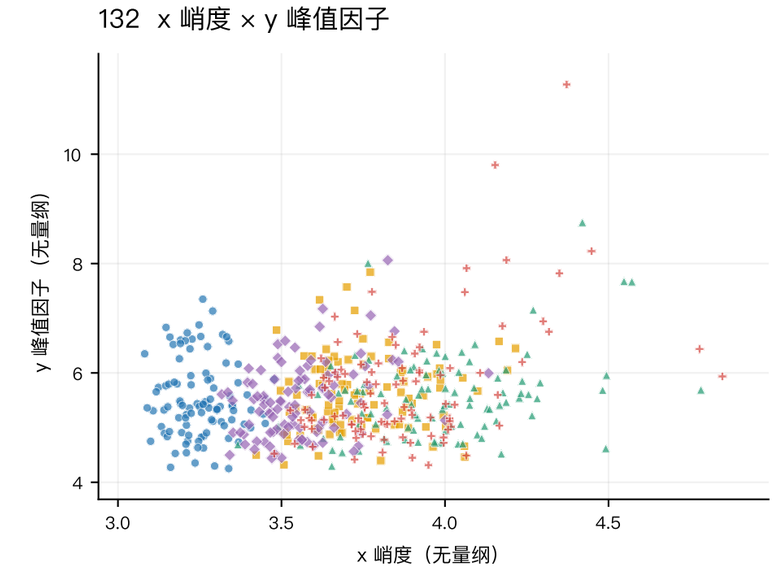
130 x 峭度 × y 偏度



131 x 峭度 × y 峭度



132 x 峭度 × y 峰值因子



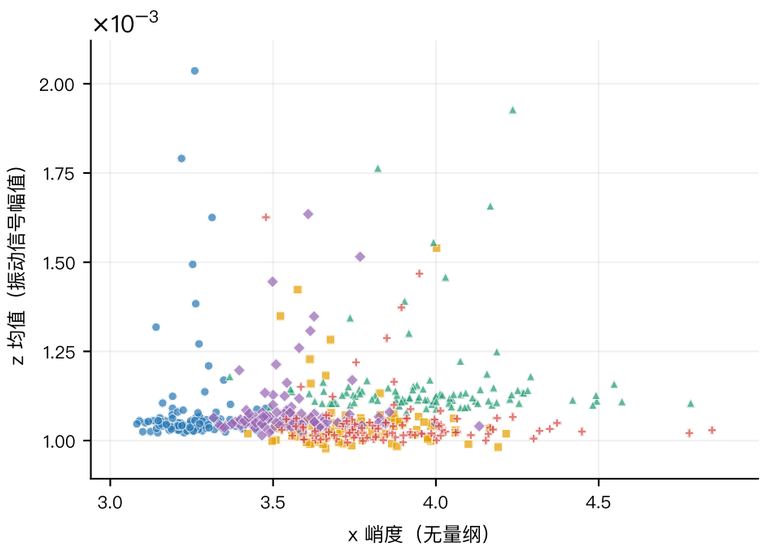
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

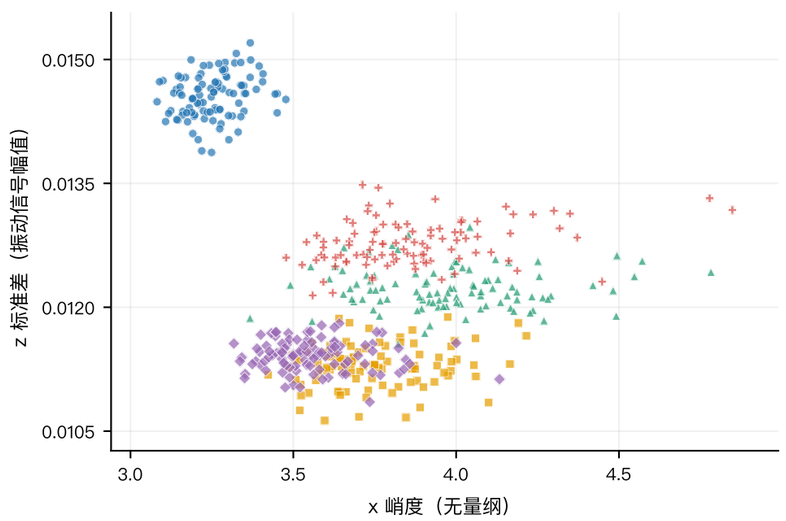
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

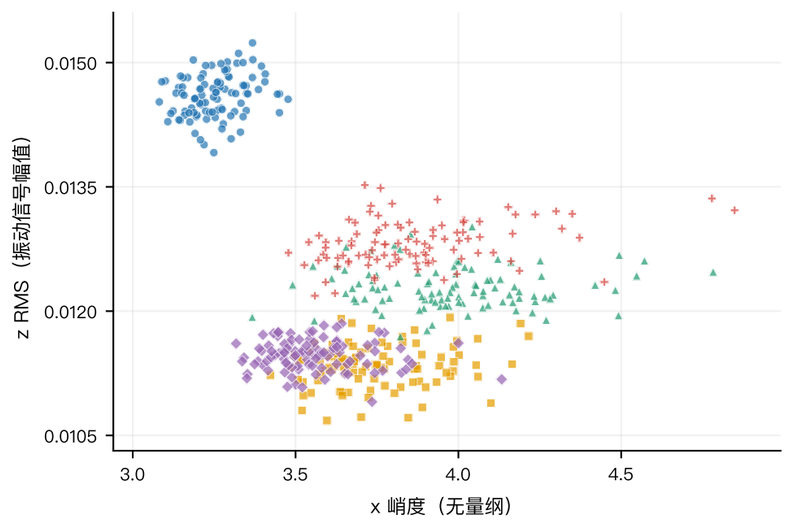
133 x 峭度 × z 均值



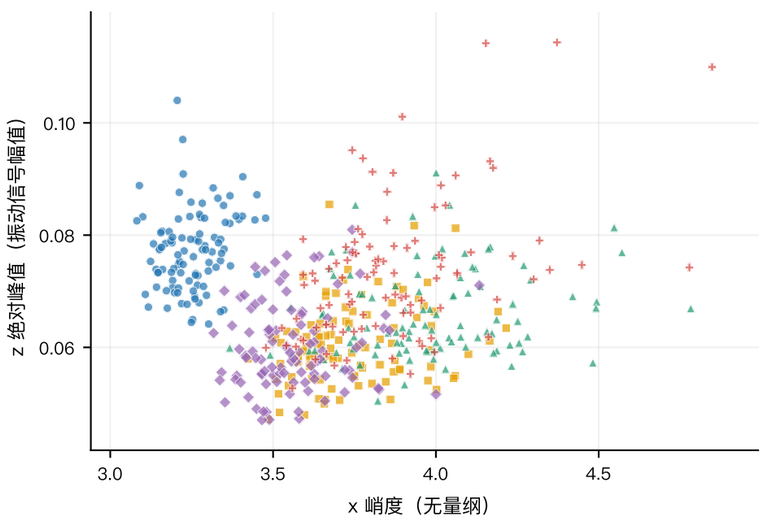
134 x 峭度 × z 标准差



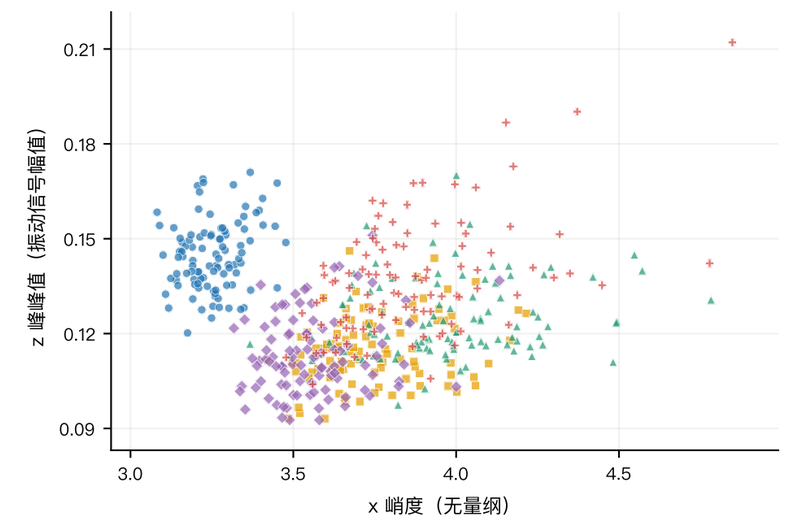
135 x 峭度 × z RMS



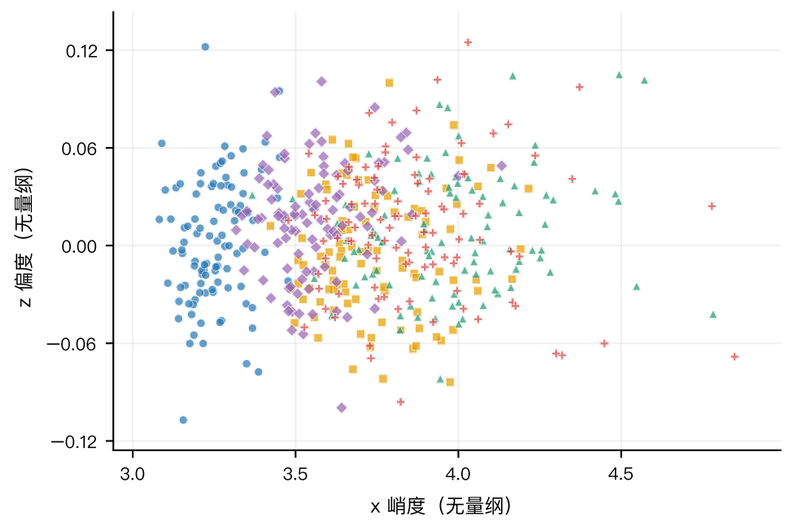
136 x 峭度 × z 绝对峰值



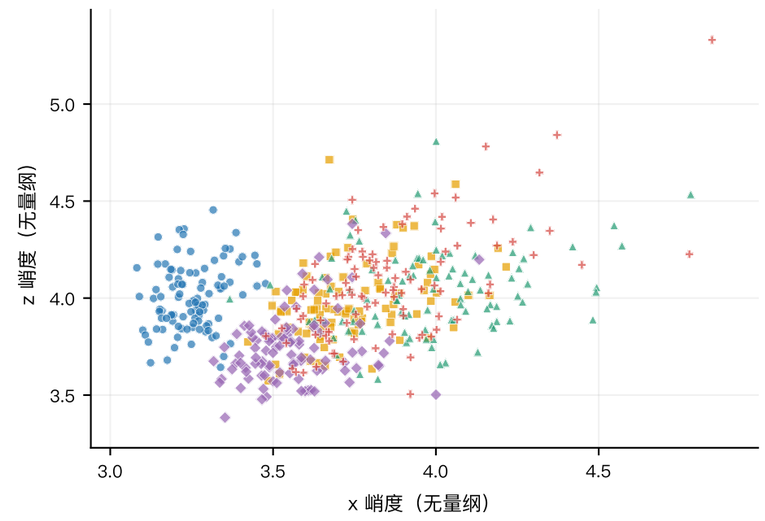
137 x 峭度 × z 峰峰值



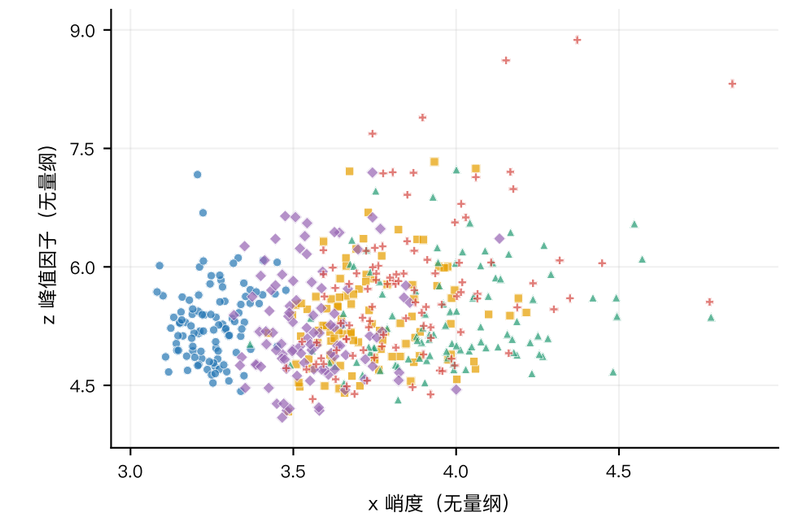
138 x 峭度 × z 偏度



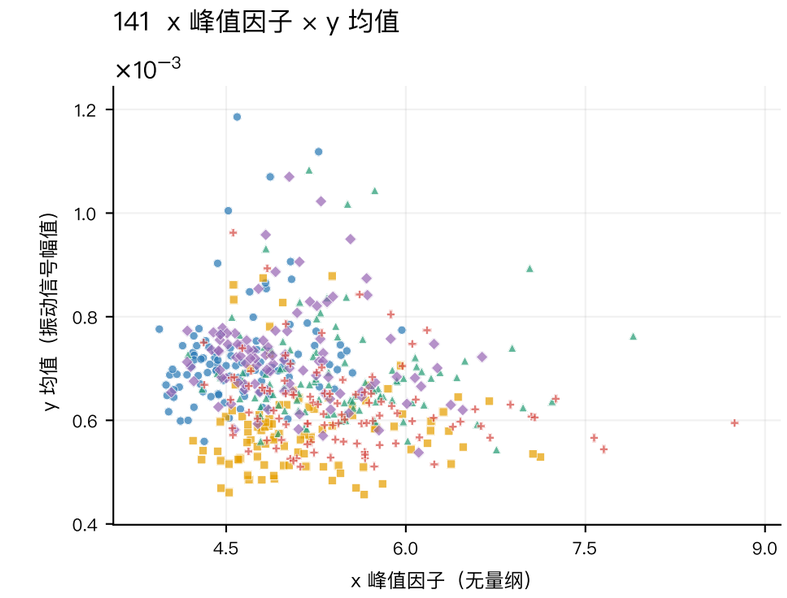
139 x 峭度 × z 峭度



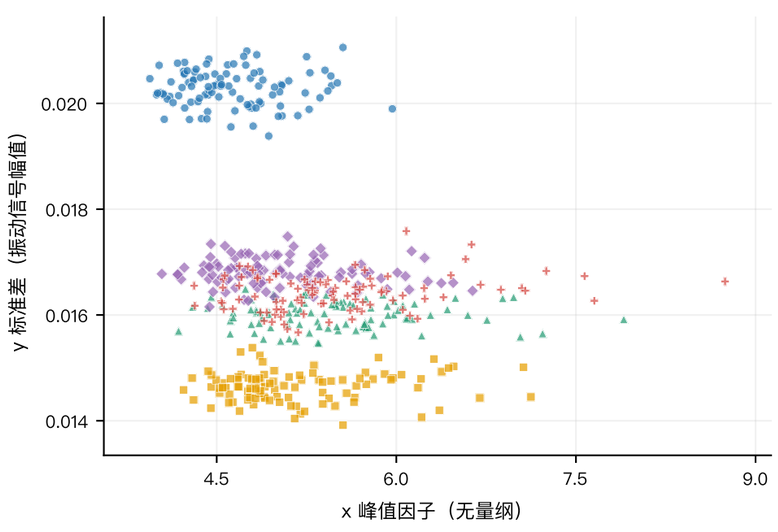
140 x 峭度 × z 峰值因子



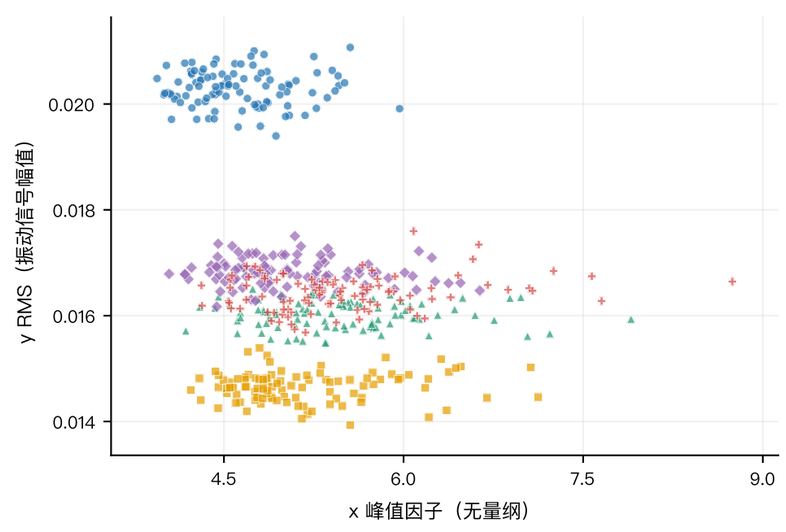
141 x 峰值因子 × y 均值



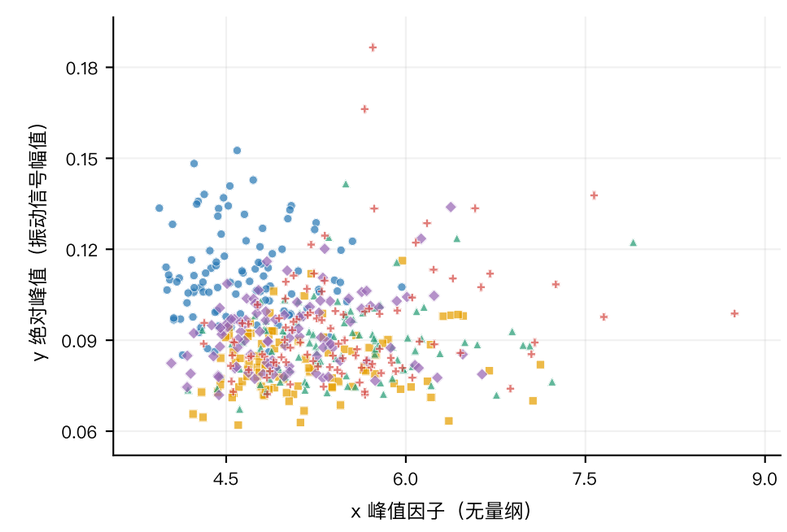
142 x 峰值因子 × y 标准差



143 x 峰值因子 × y RMS



144 x 峰值因子 × y 绝对峰值



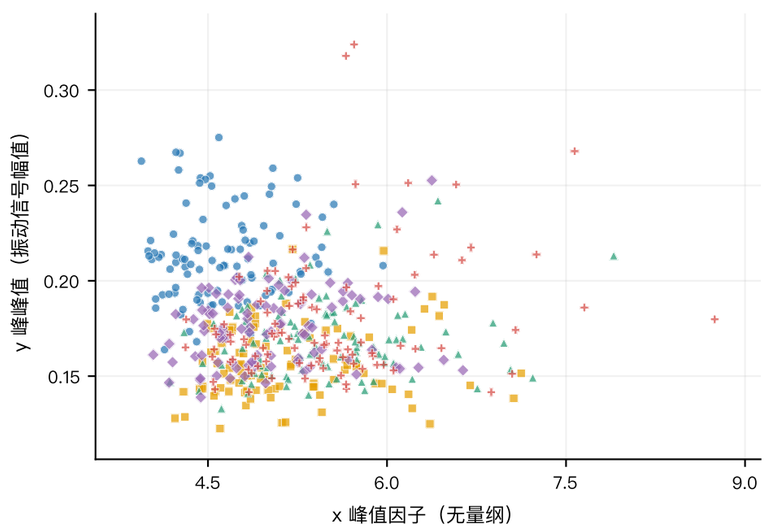
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

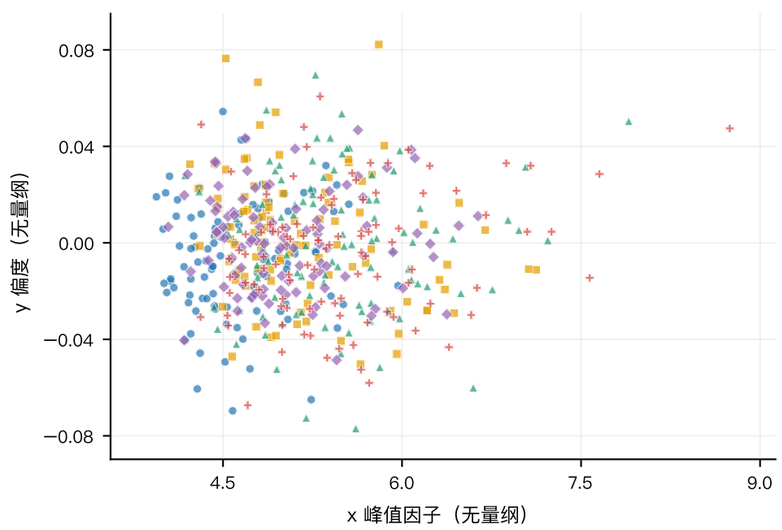
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

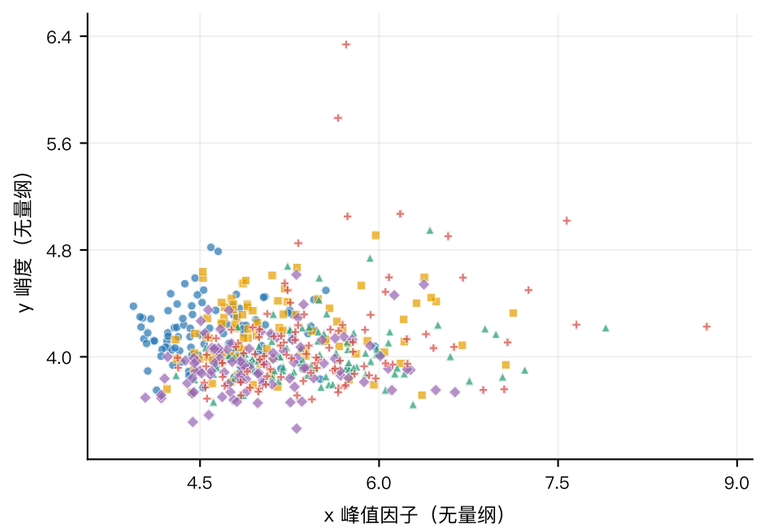
145 x 峰值因子 × y 峰峰值



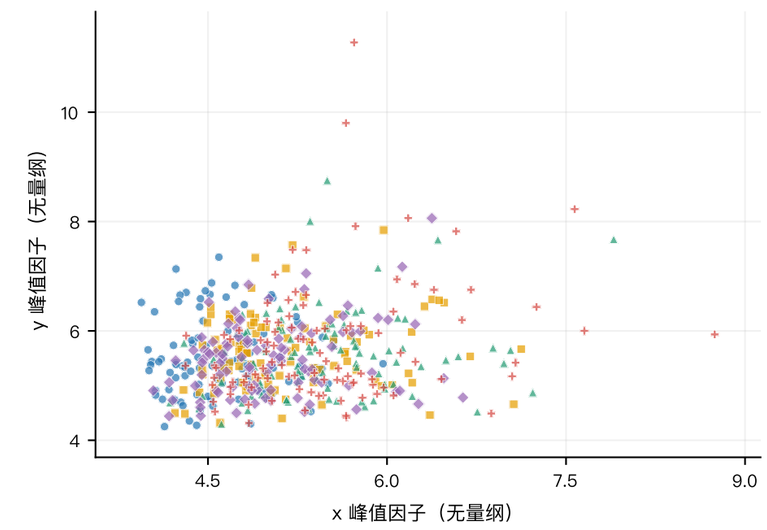
146 x 峰值因子 × y 偏度



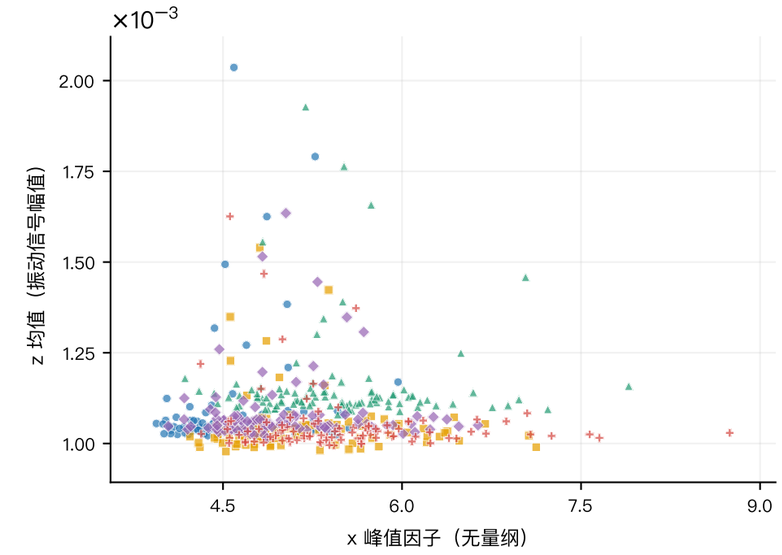
147 x 峰值因子 × y 峭度



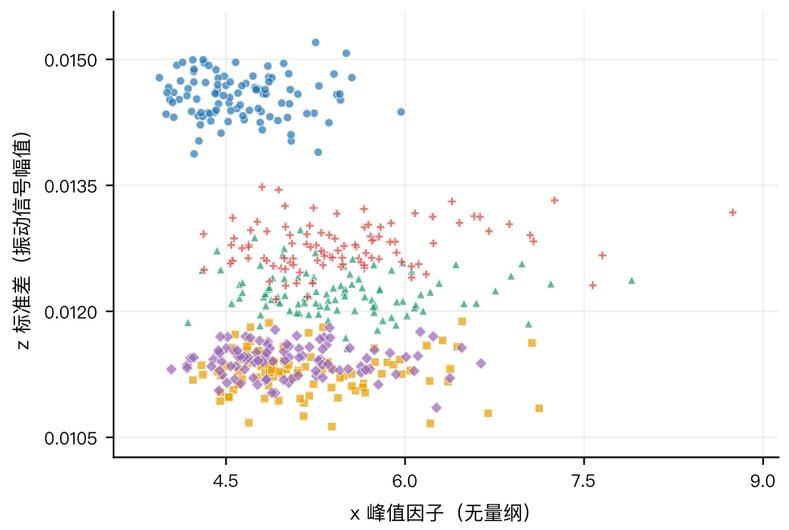
148 x 峰值因子 × y 峰值因子



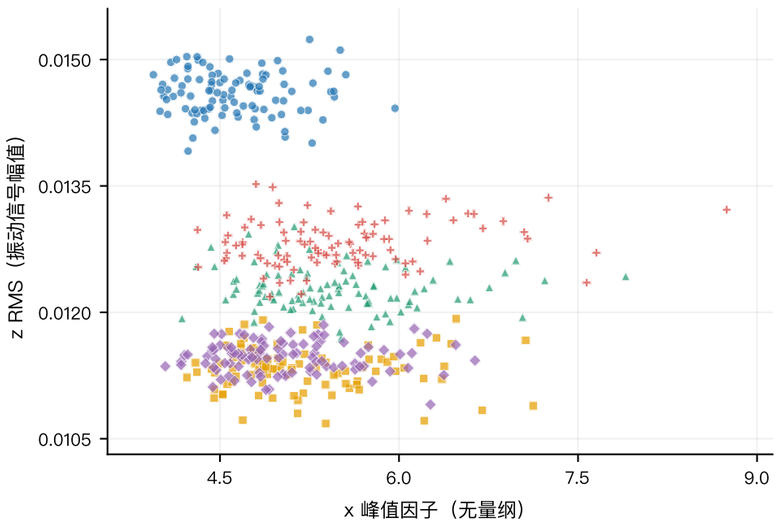
149 x 峰值因子 × z 均值



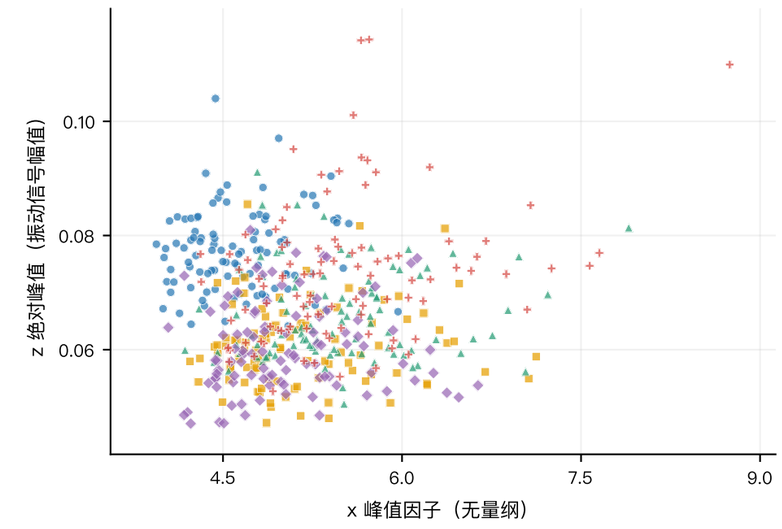
150 x 峰值因子 × z 标准差



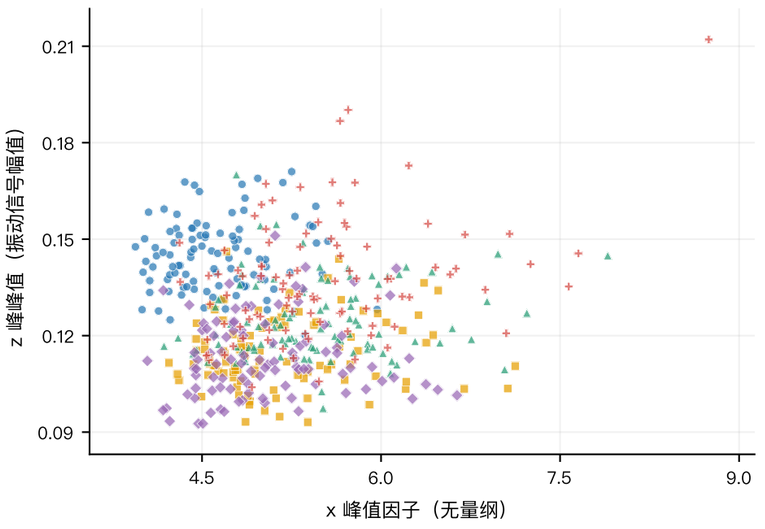
151 x 峰值因子 × z RMS



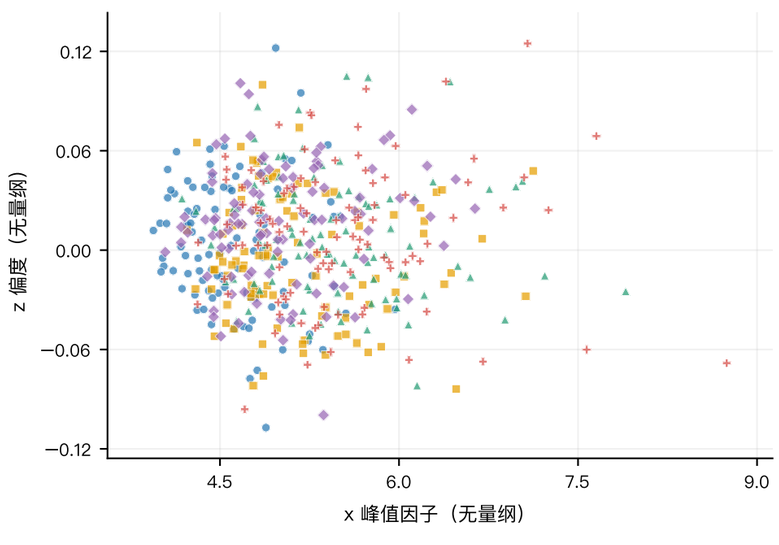
152 x 峰值因子 × z 绝对峰值



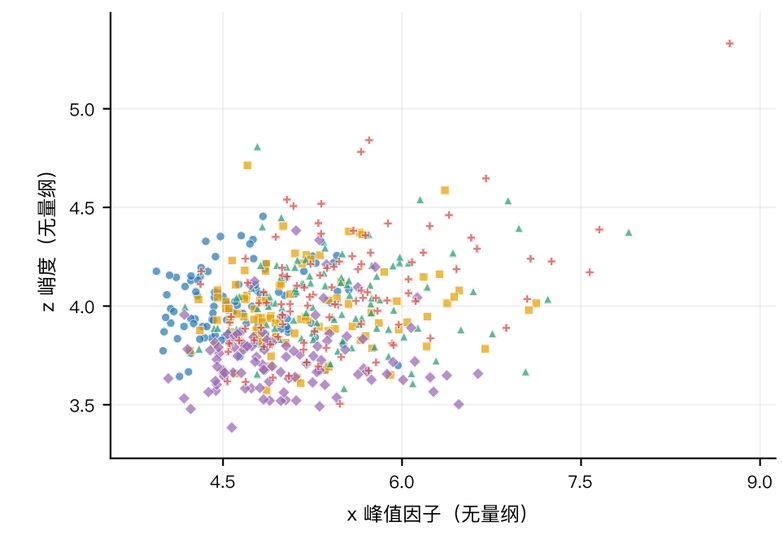
153 x 峰值因子 × z 峰峰值



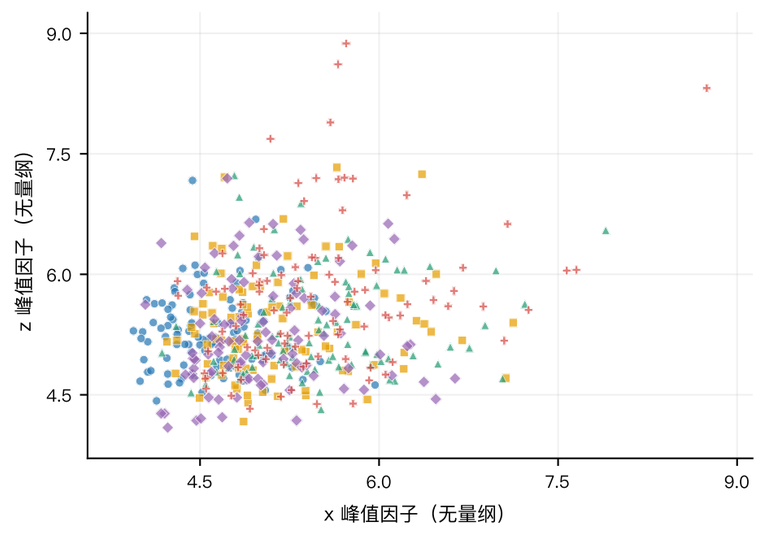
154 x 峰值因子 × z 偏度



155 x 峰值因子 × z 峭度



156 x 峰值因子 × z 峰值因子



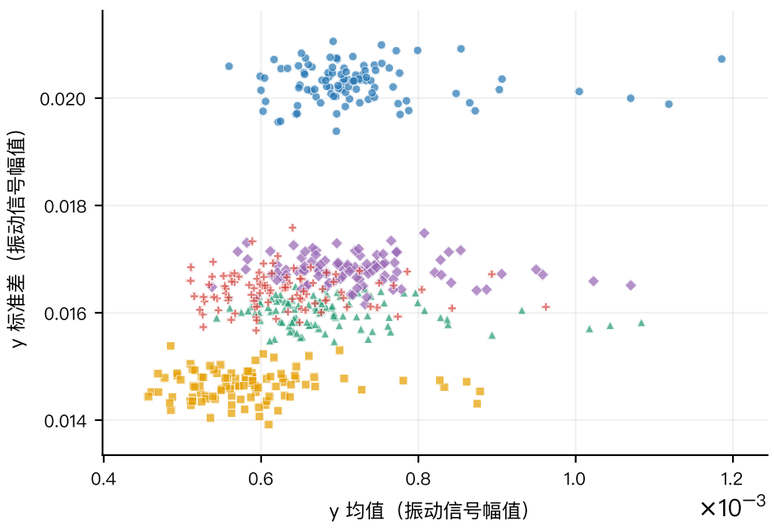
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

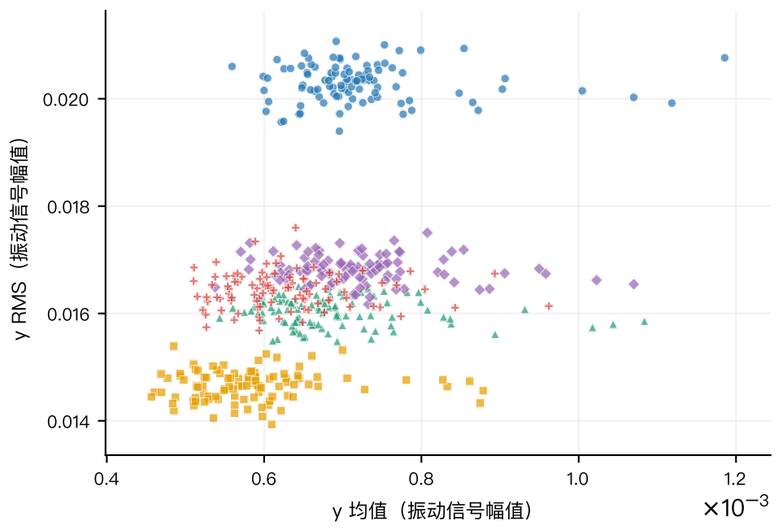
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

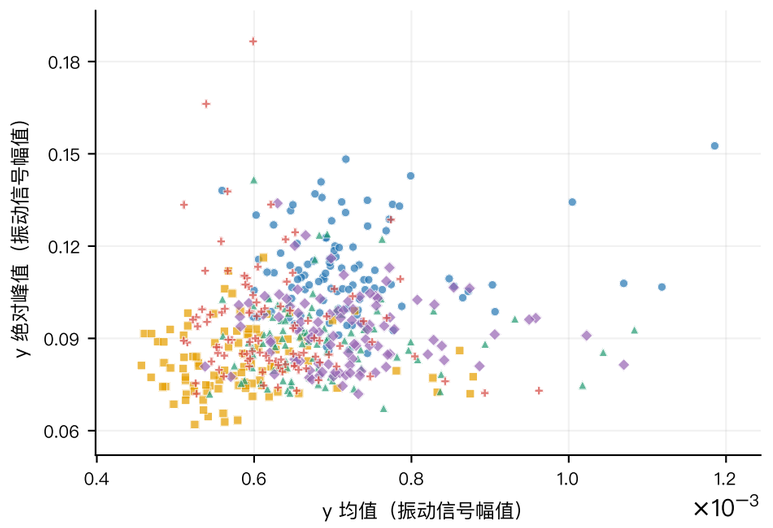
157 y 均值 × y 标准差



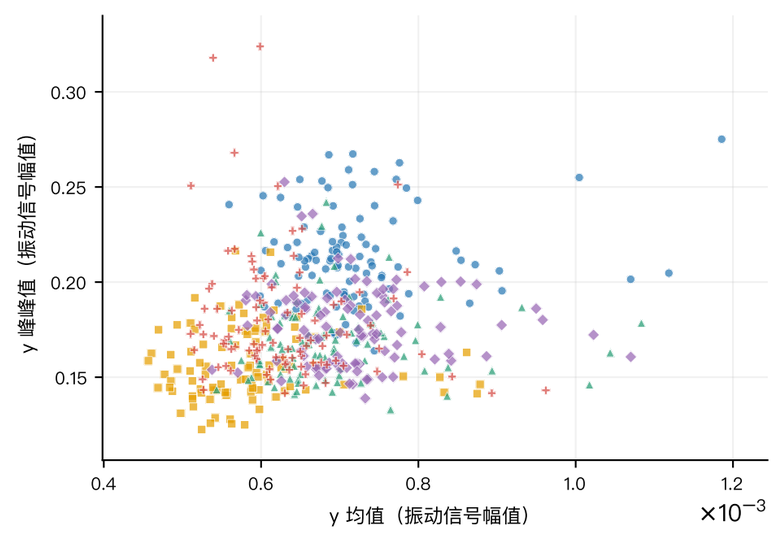
158 y 均值 × y RMS



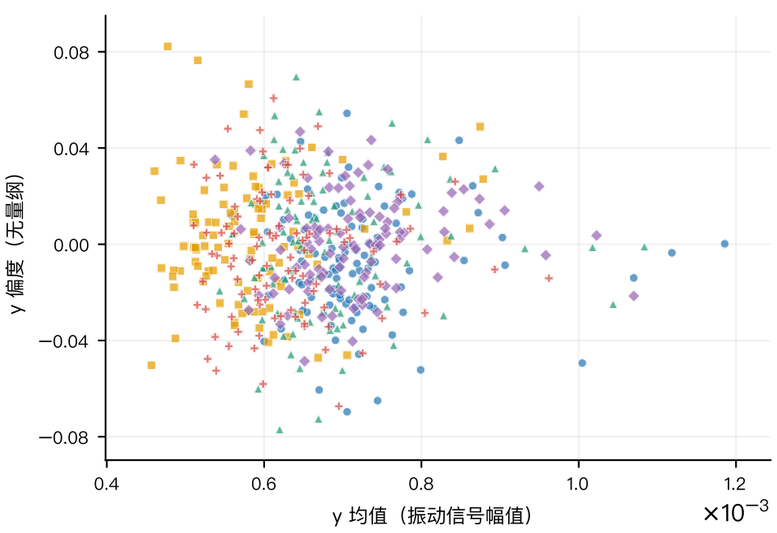
159 y 均值 × y 绝对峰值



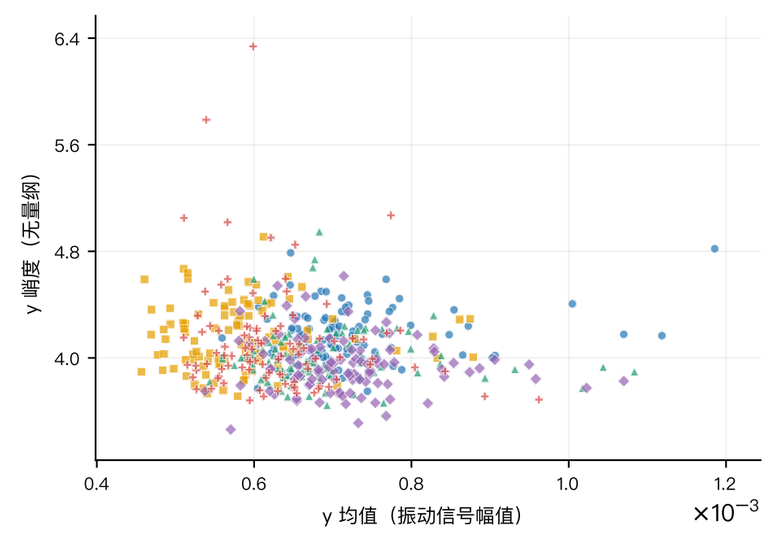
160 y 均值 × y 峰峰值



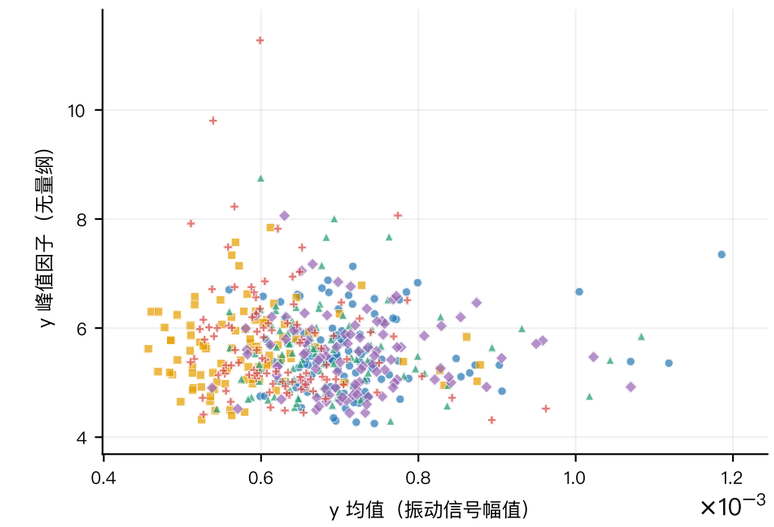
161 y 均值 × y 偏度



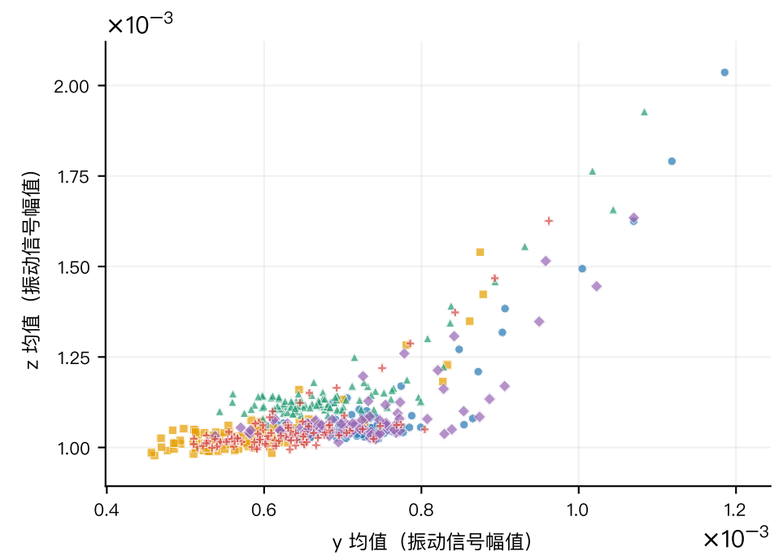
162 y 均值 × y 峭度



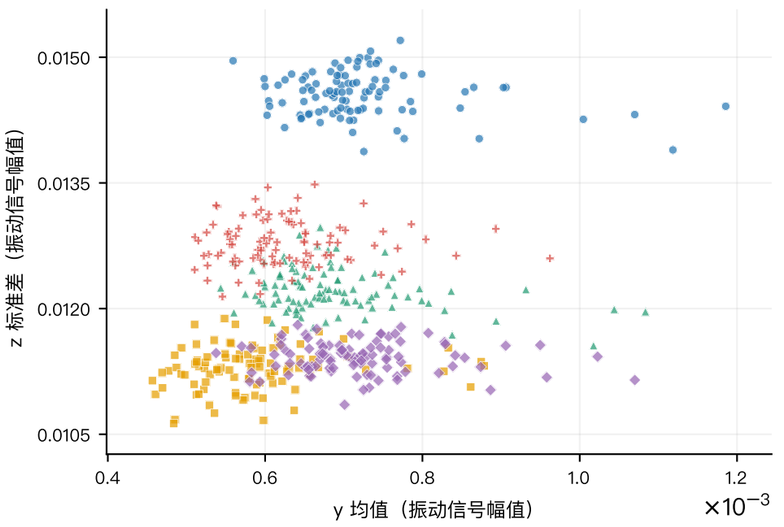
163 y 均值 × y 峰值因子



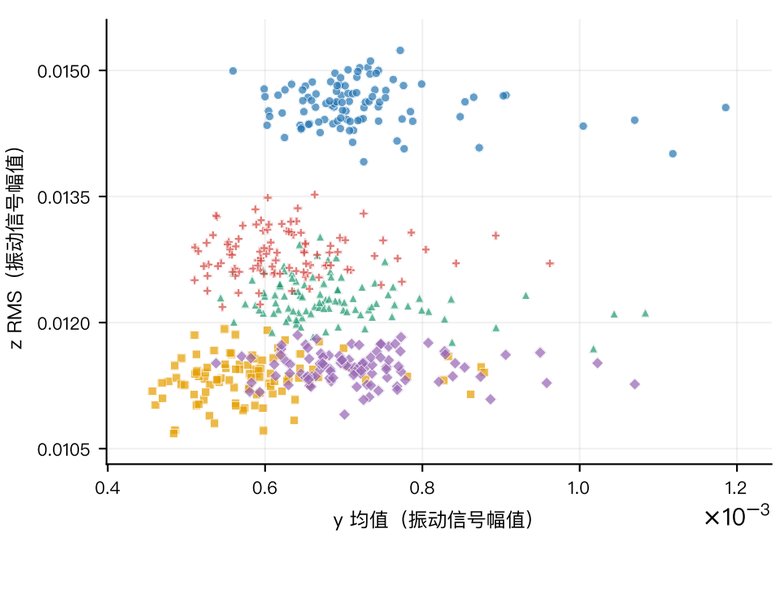
164 y 均值 × z 均值



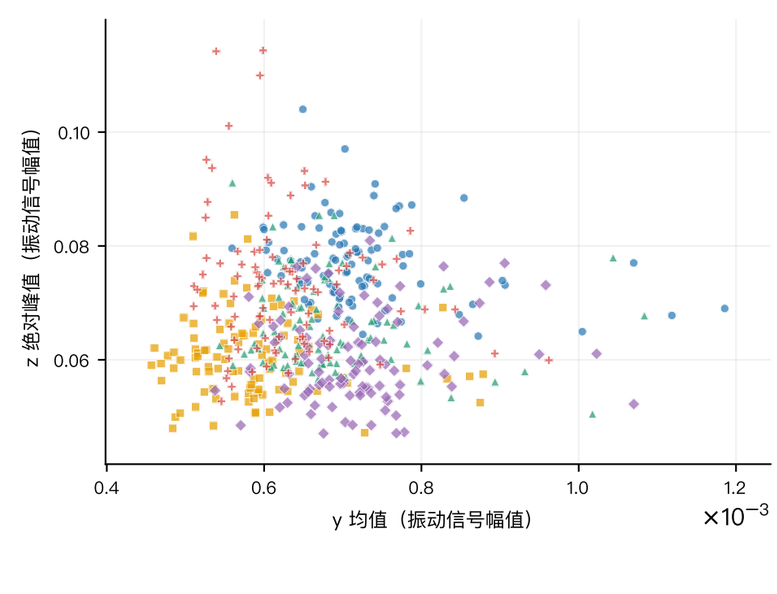
165 y 均值 × z 标准差



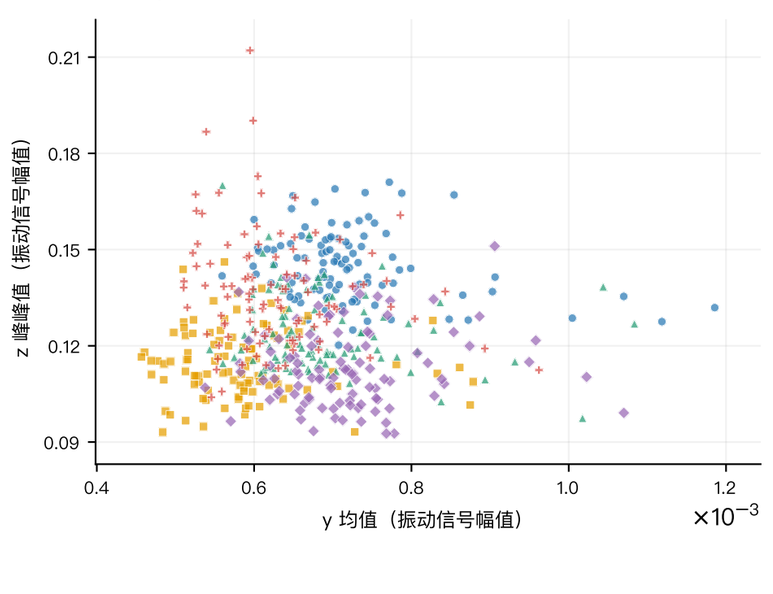
166 y 均值 × z RMS



167 y 均值 × z 绝对峰值



168 y 均值 × z 峰峰值



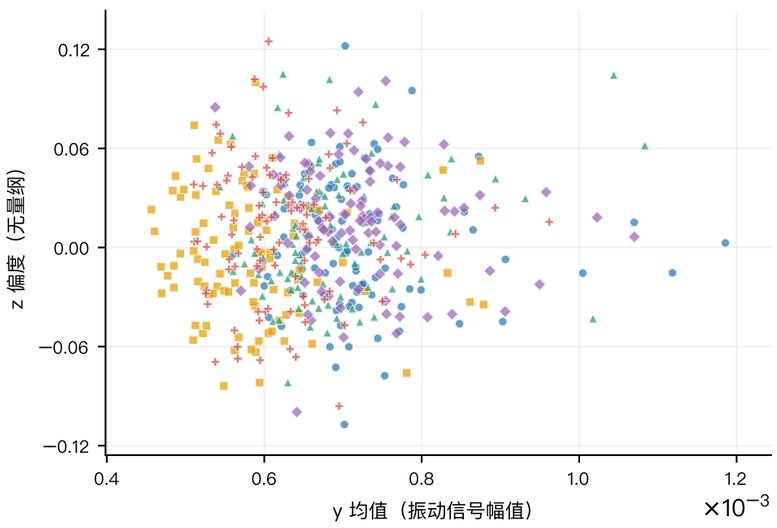
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

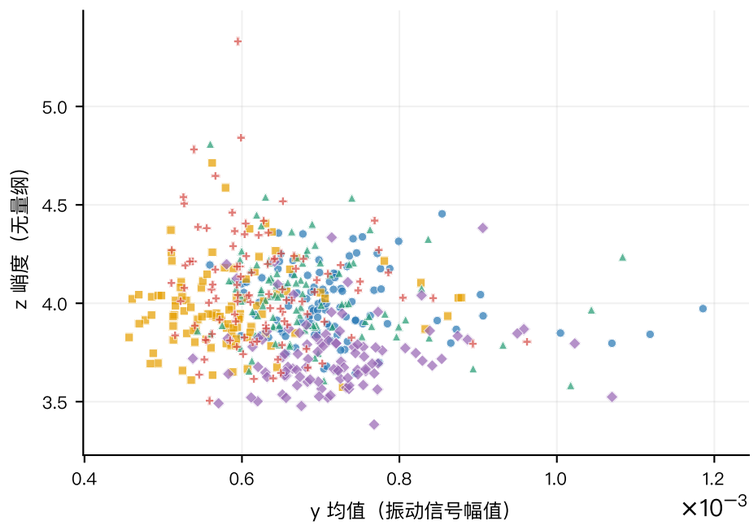
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

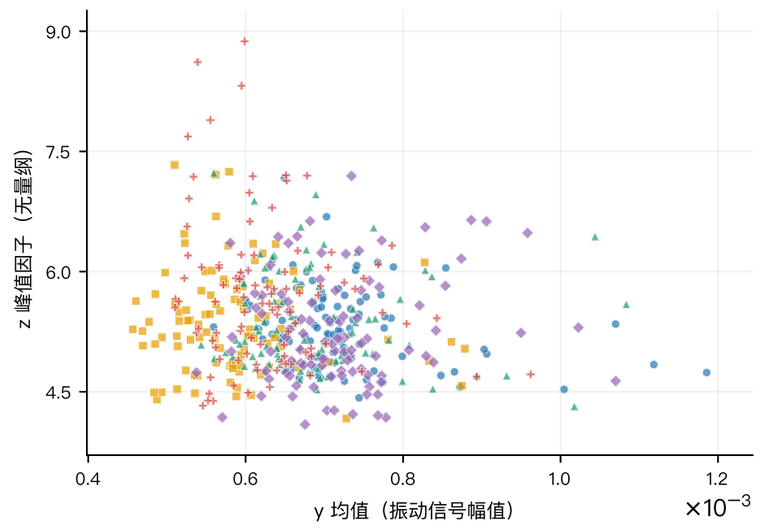
169 y 均值 × z 偏度



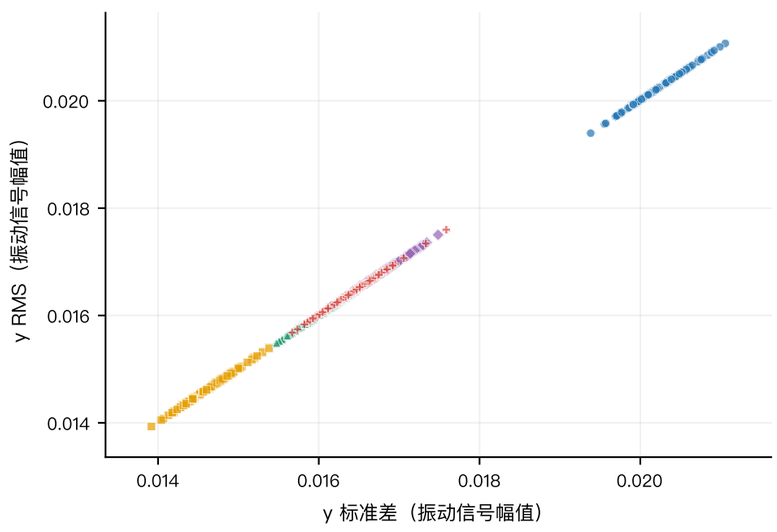
170 y 均值 × z 峭度



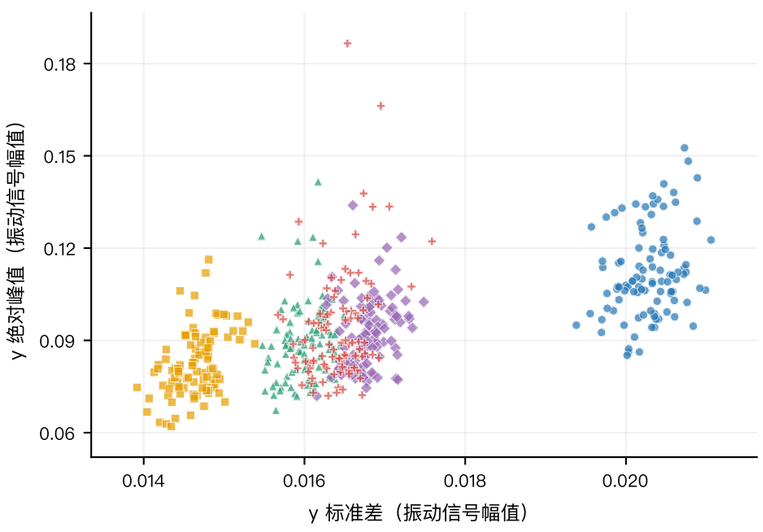
171 y 均值 × z 峰值因子



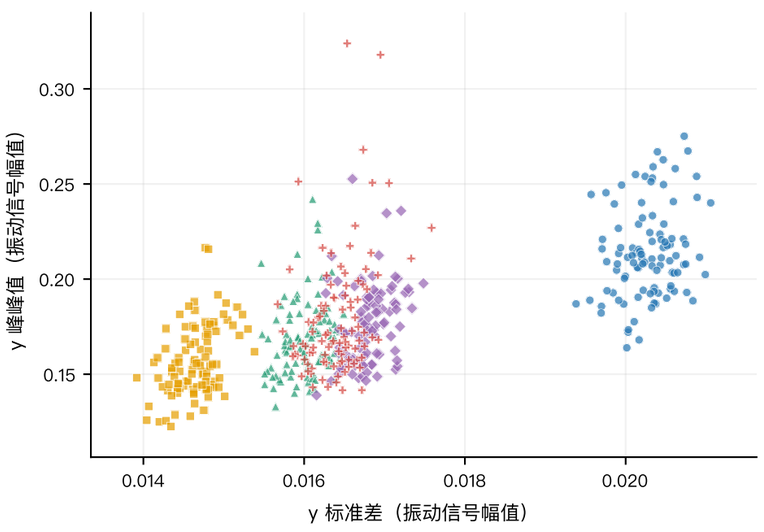
172 y 标准差 × y RMS



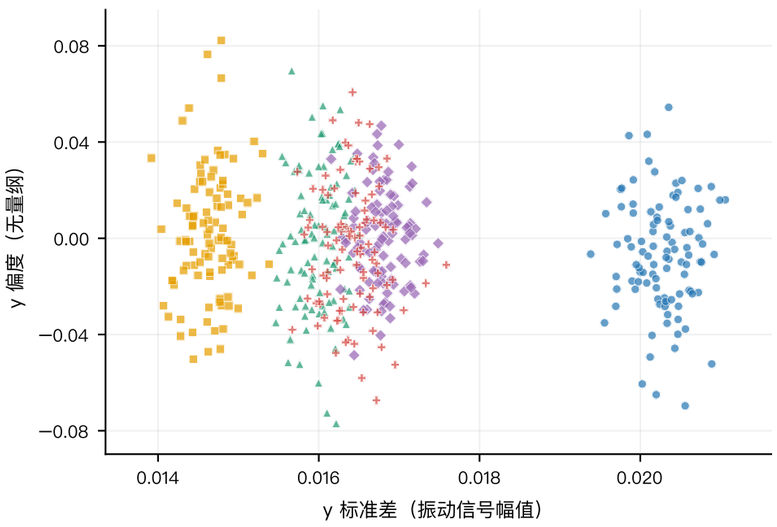
173 y 标准差 × y 绝对峰值



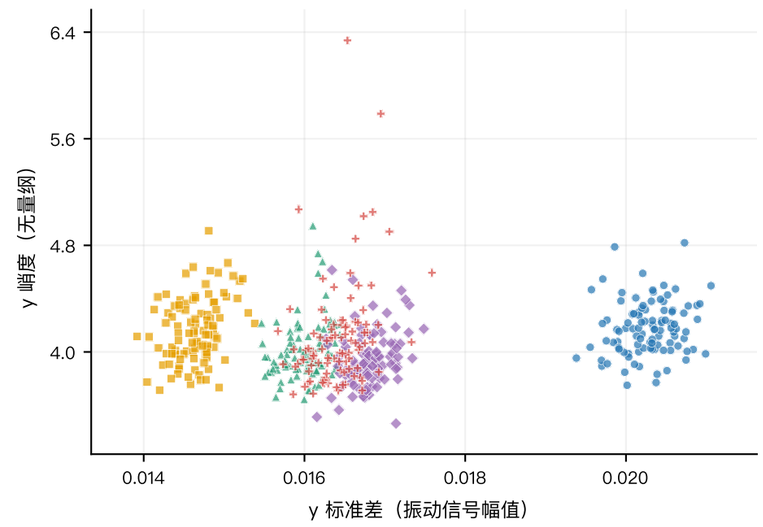
174 y 标准差 × y 峰峰值



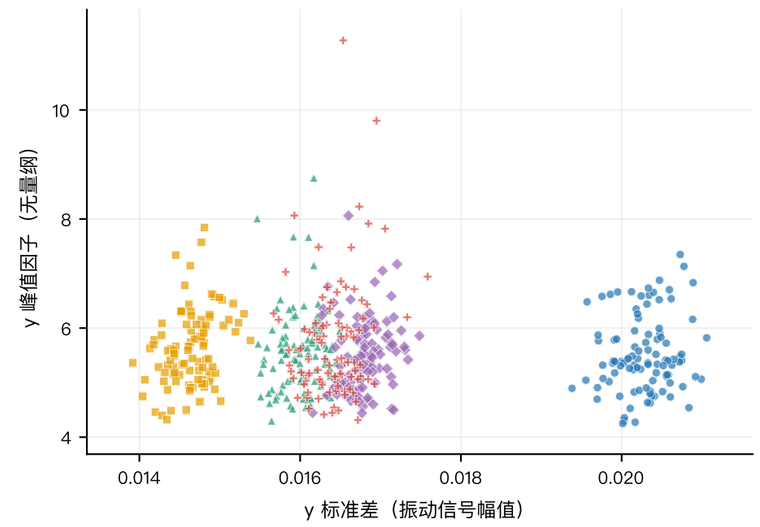
175 y 标准差 × y 偏度



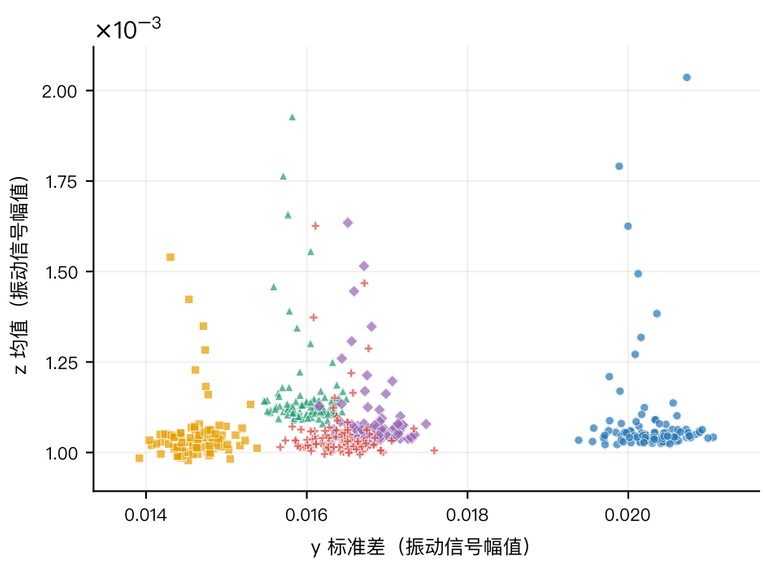
176 y 标准差 × y 峭度



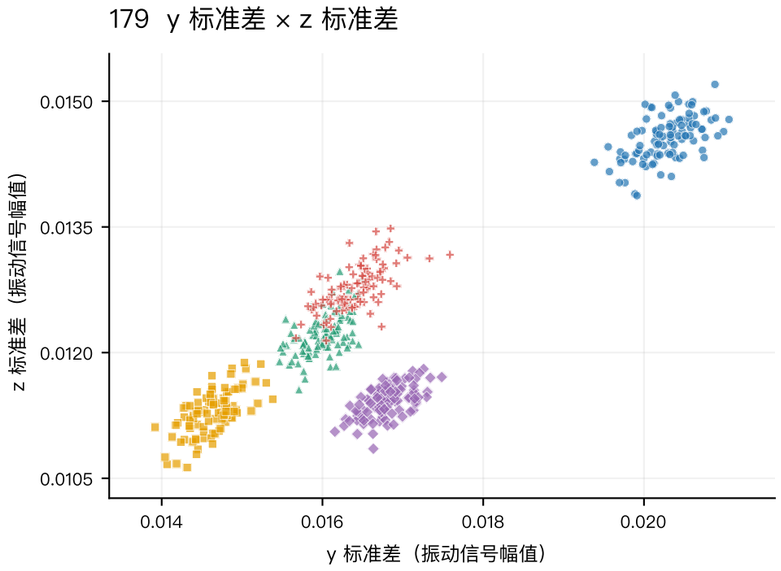
177 y 标准差 × y 峰值因子



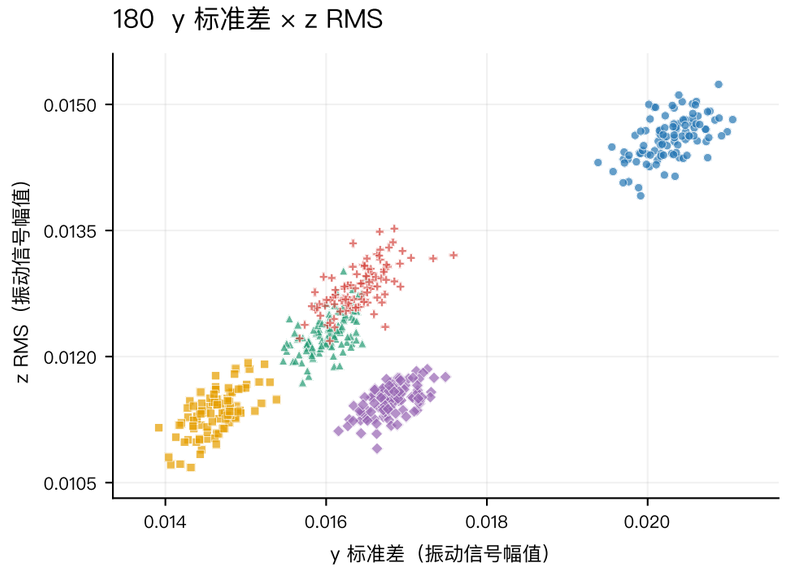
178 y 标准差 × z 均值



179 y 标准差 × z 标准差



180 y 标准差 × z RMS



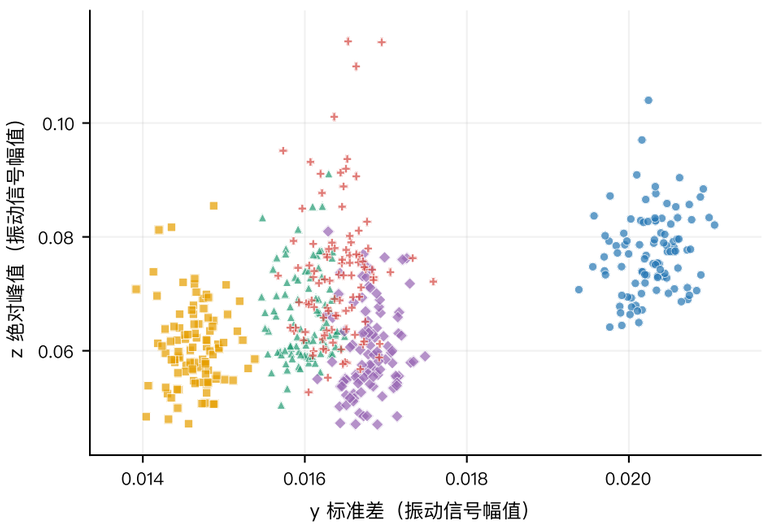
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

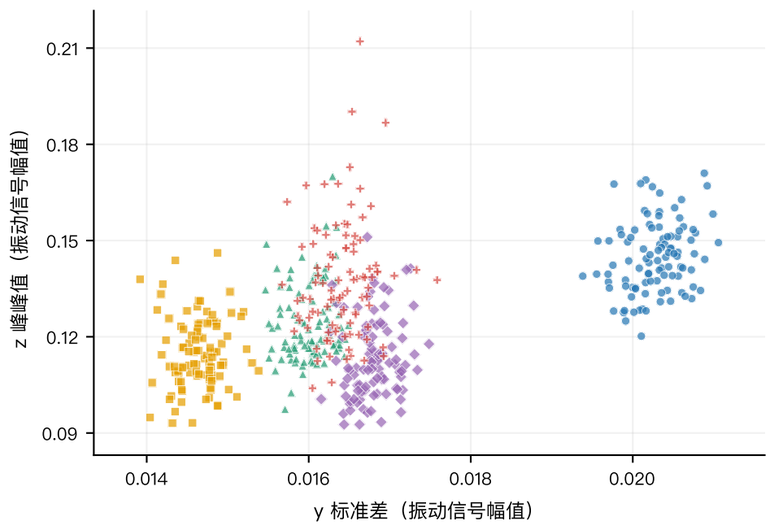
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

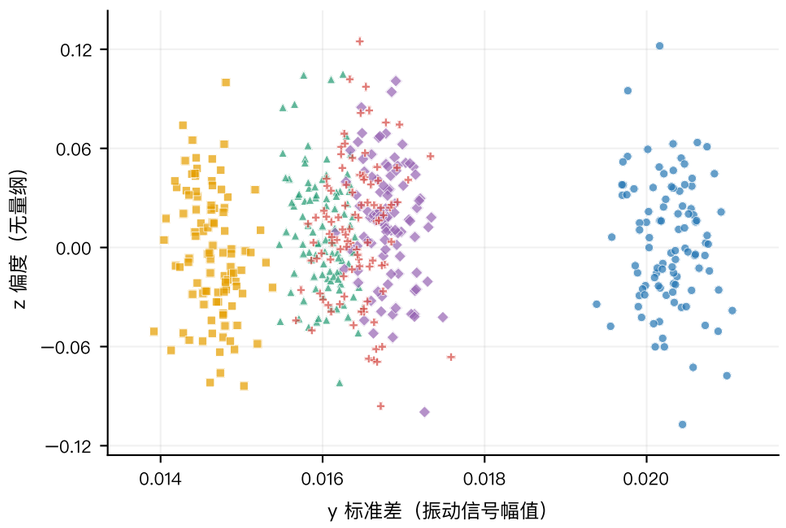
181 y 标准差 × z 绝对峰值



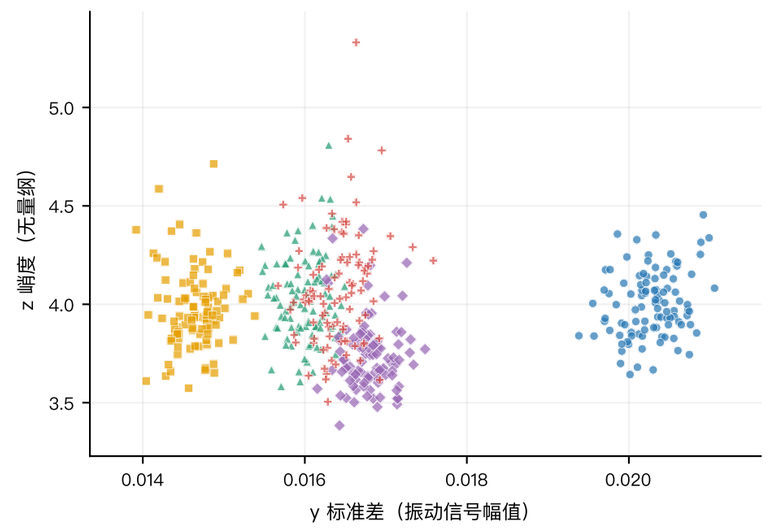
182 y 标准差 × z 峰峰值



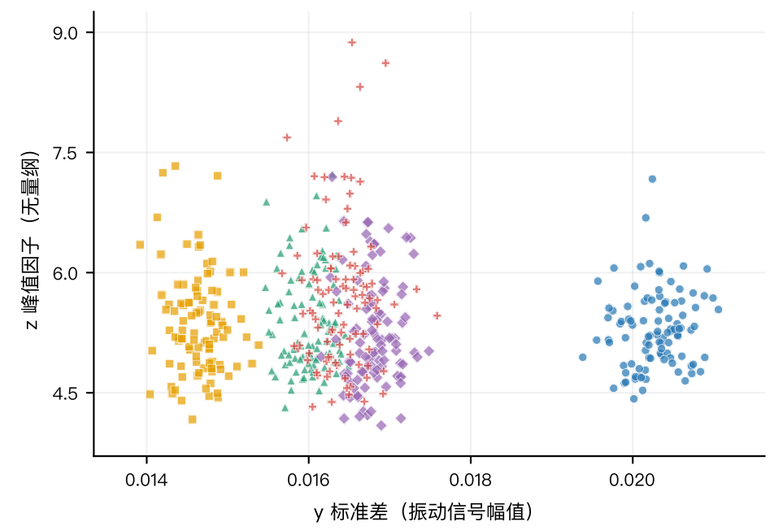
183 y 标准差 × z 偏度



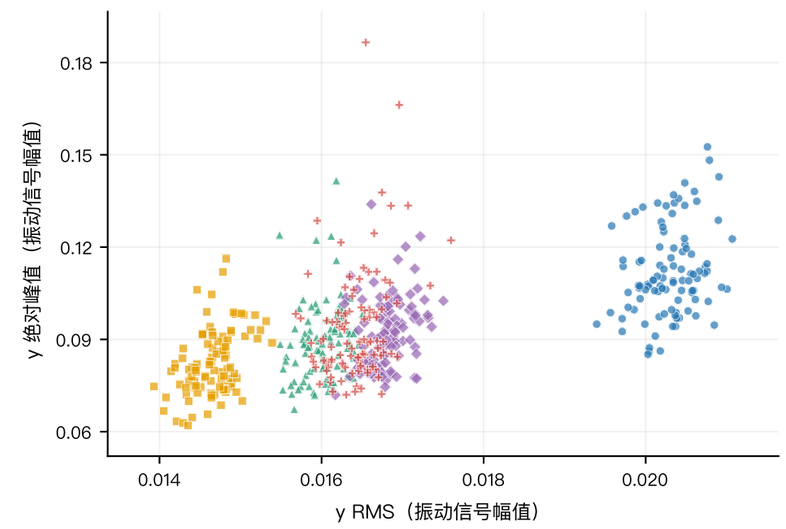
184 y 标准差 × z 峭度



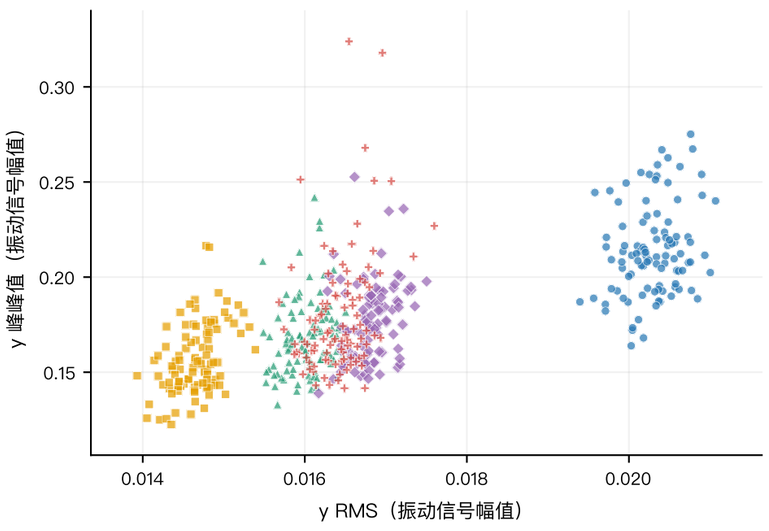
185 y 标准差 × z 峰值因子



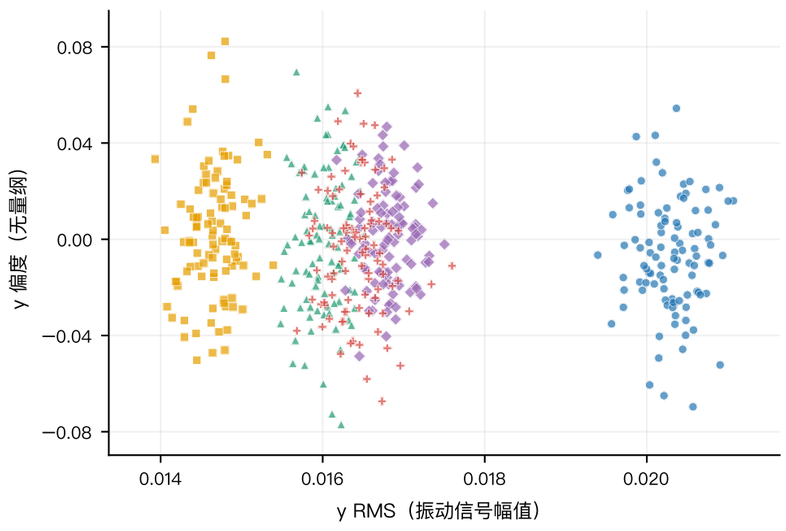
186 y RMS × y 绝对峰值



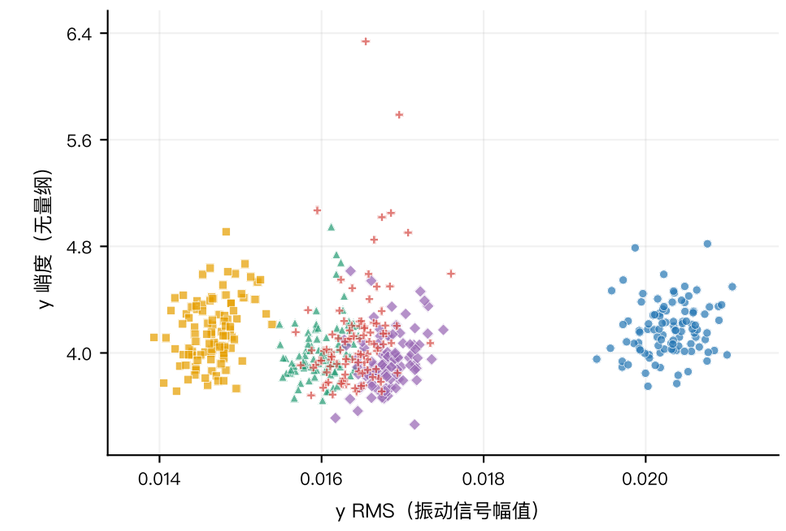
187 y RMS × y 峰峰值



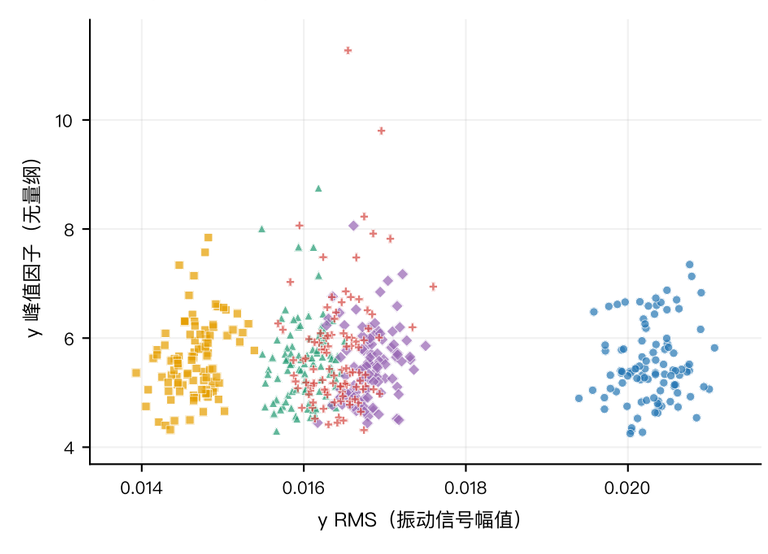
188 y RMS × y 偏度



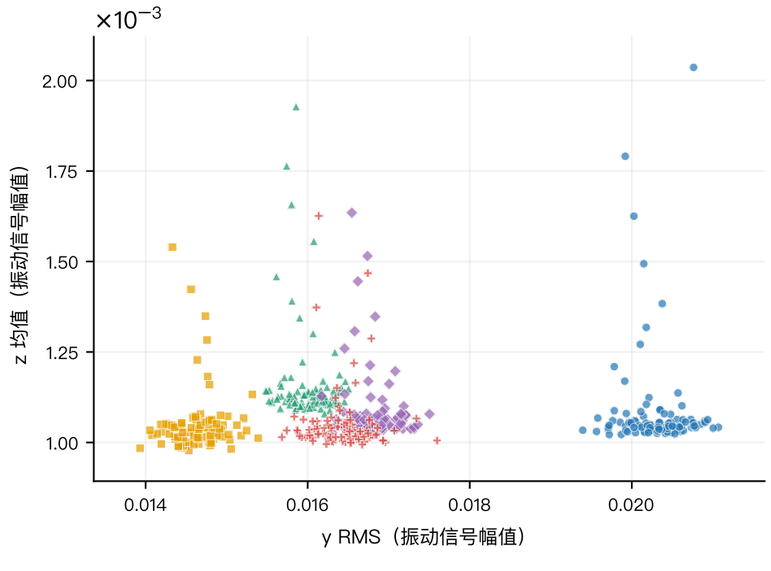
189 y RMS × y 峭度



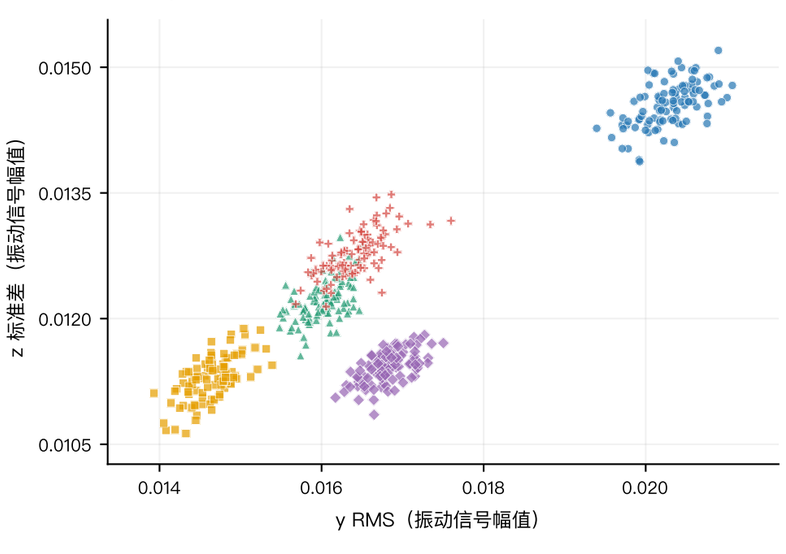
190 y RMS × y 峰值因子



191 y RMS × z 均值



192 y RMS × z 标准差



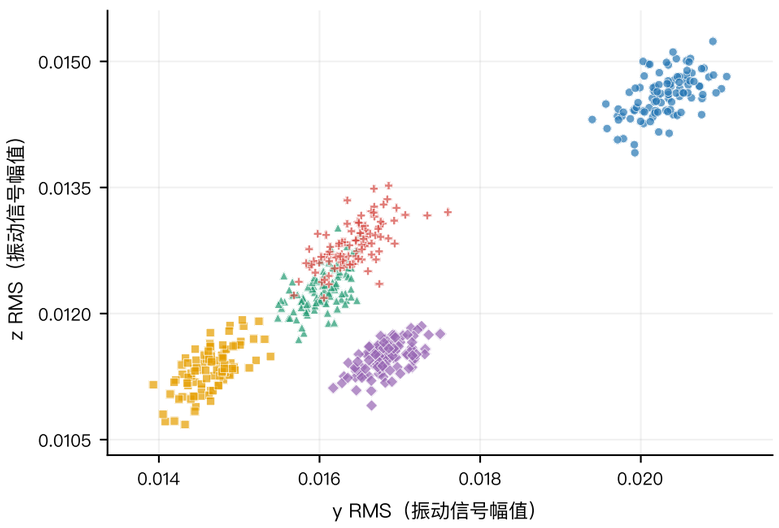
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

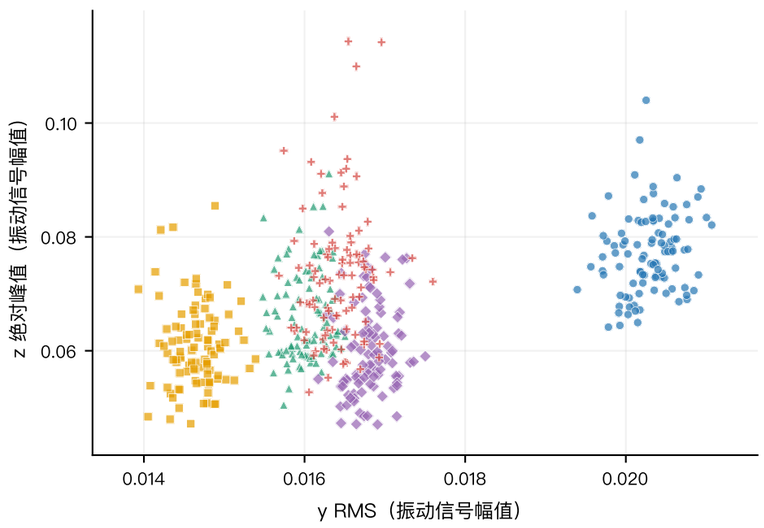
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

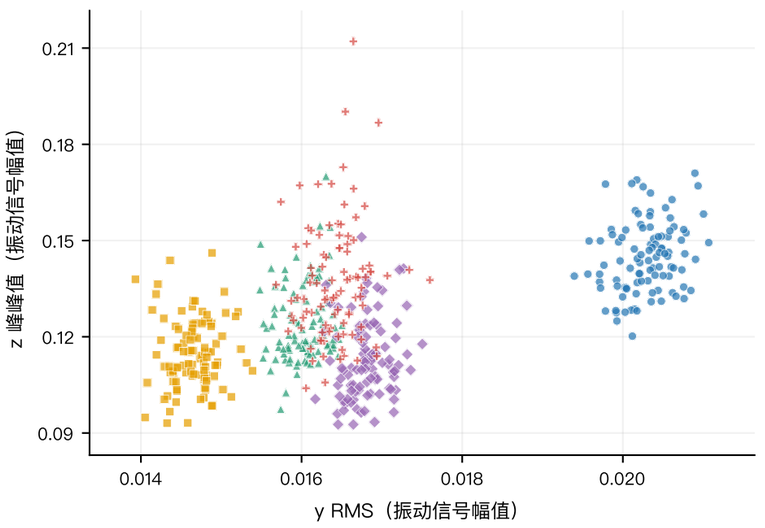
193　y RMS × z RMS



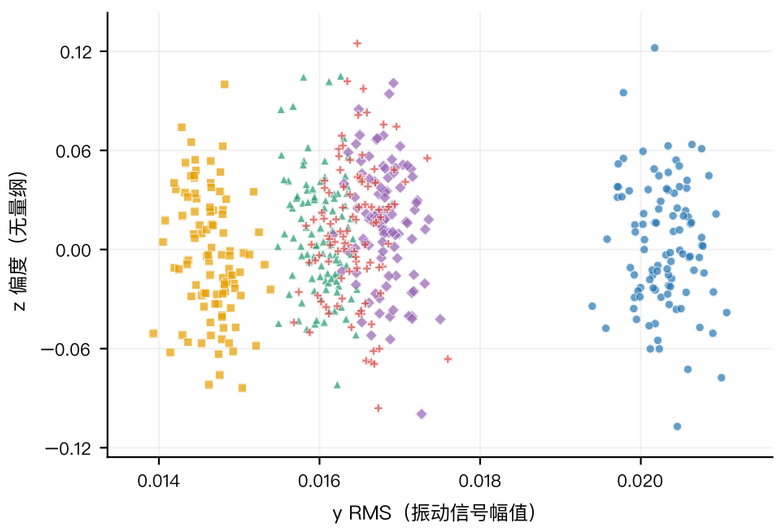
194　y RMS × z 绝对峰值



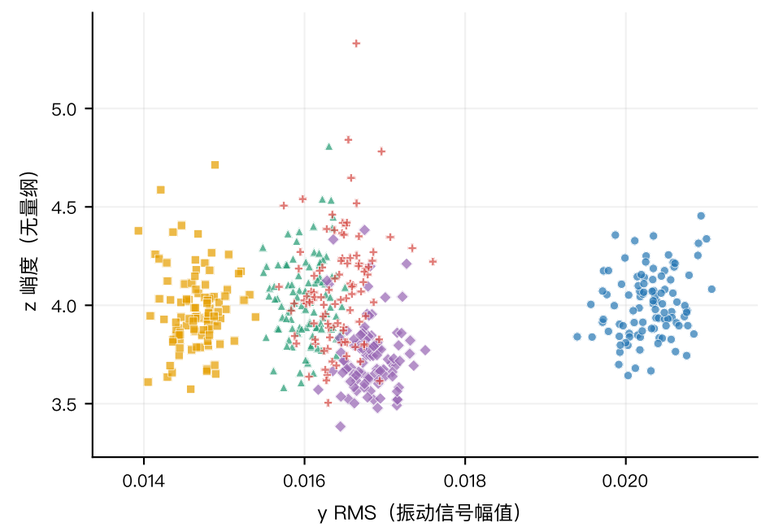
195　y RMS × z 峰峰值



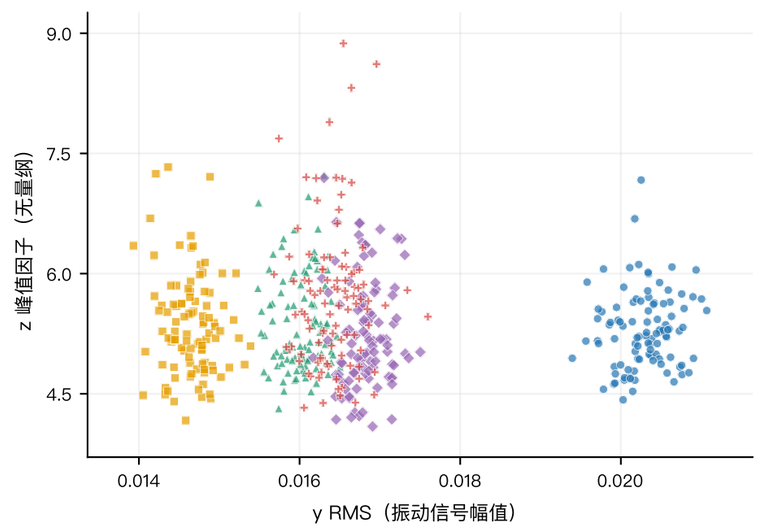
196　y RMS × z 偏度



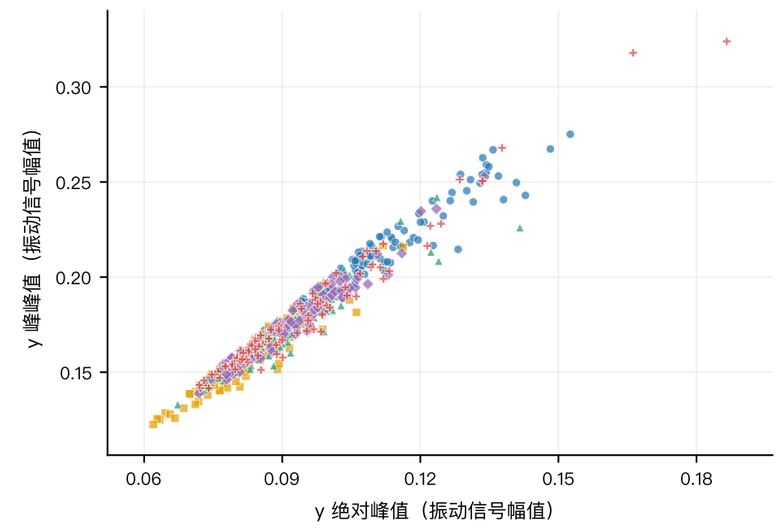
197　y RMS × z 峭度



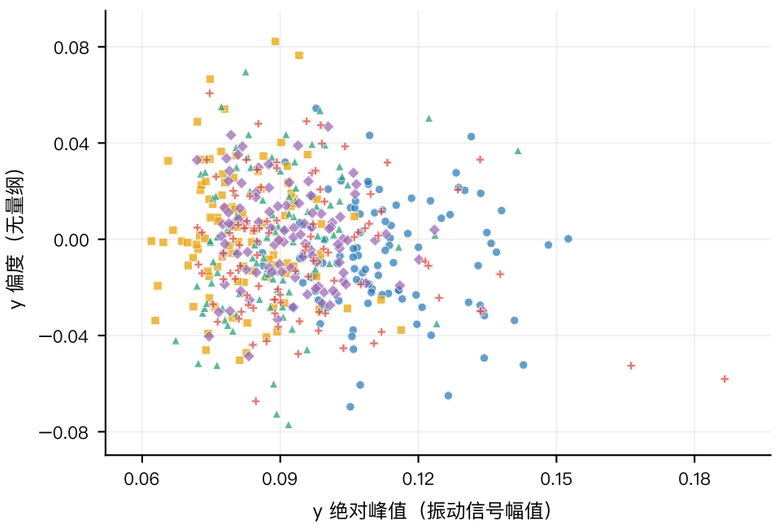
198　y RMS × z 峰值因子



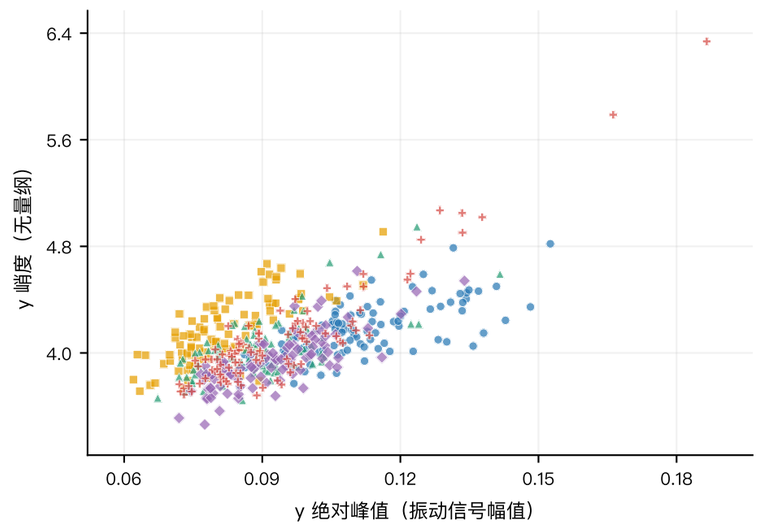
199　y 绝对峰值 × y 峰峰值



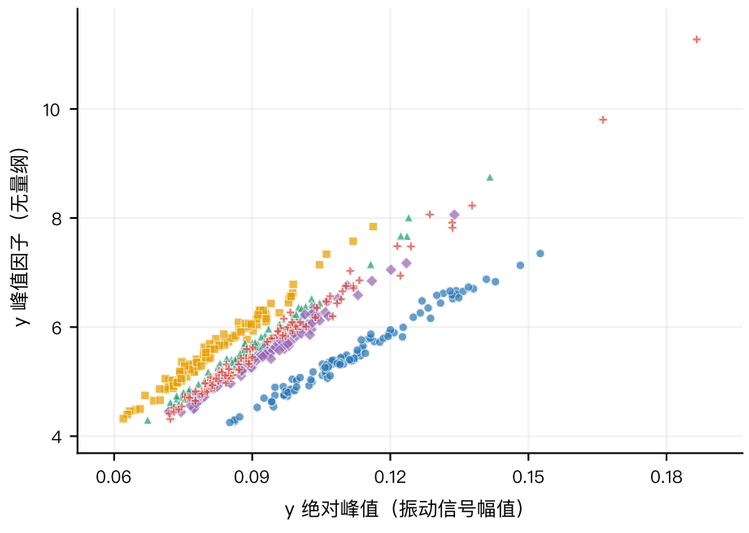
200　y 绝对峰值 × y 偏度



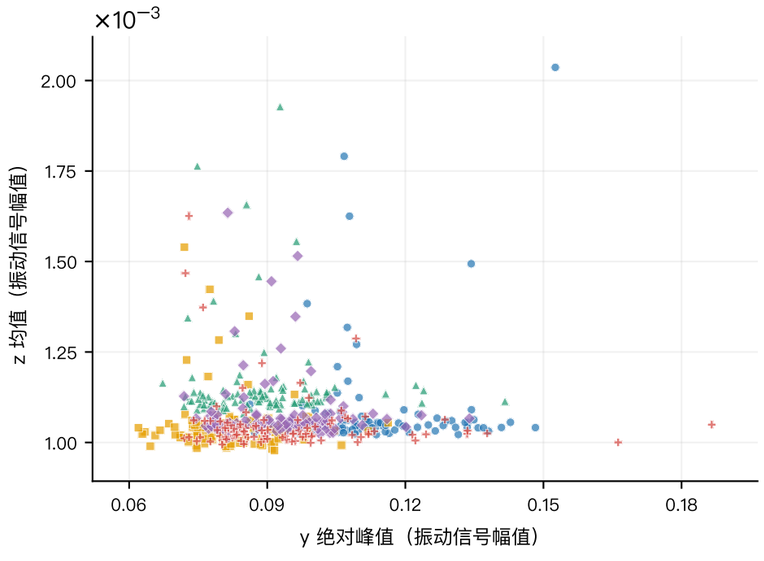
201　y 绝对峰值 × y 峭度



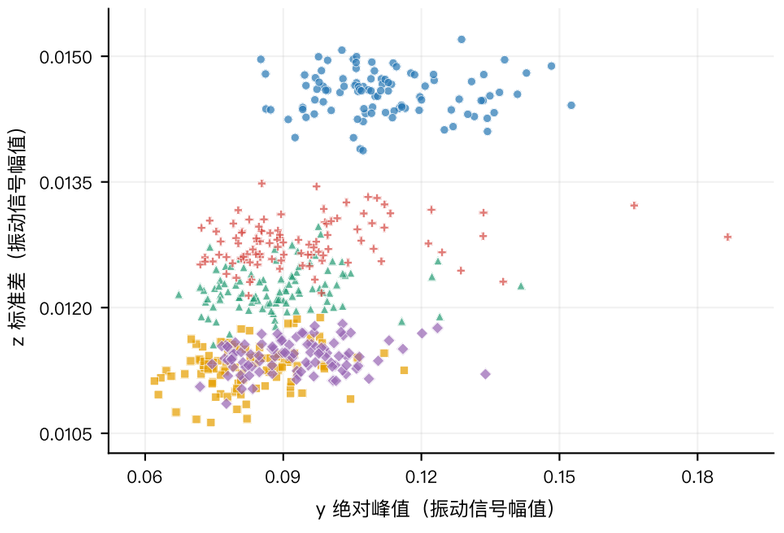
202　y 绝对峰值 × y 峰值因子



203　y 绝对峰值 × z 均值



204　y 绝对峰值 × z 标准差



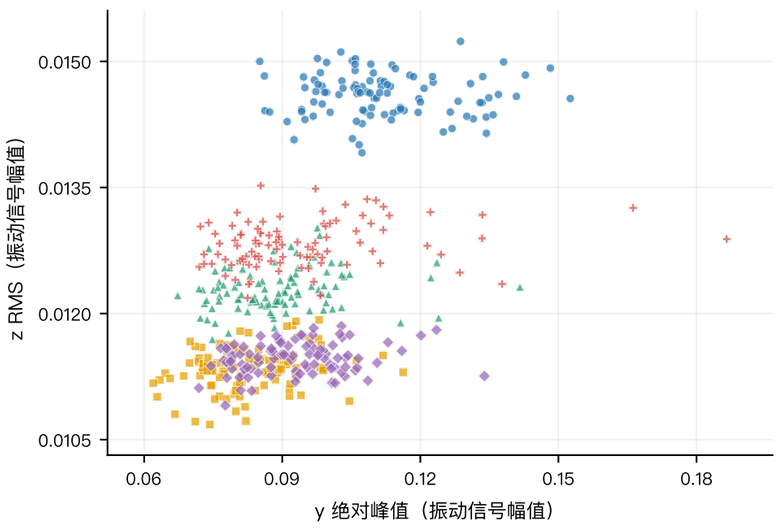
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

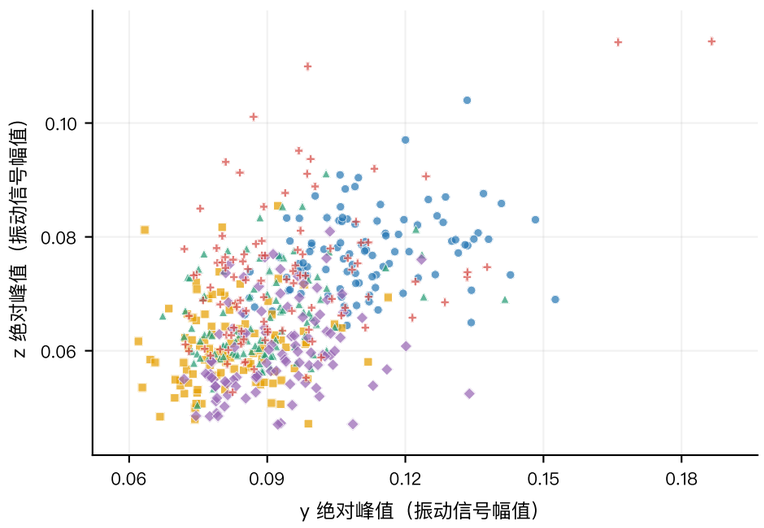
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

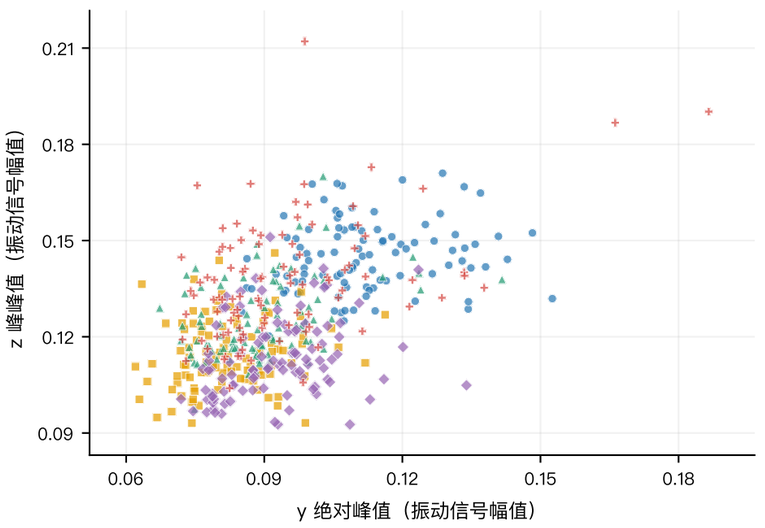
205 y 绝对峰值 × z RMS



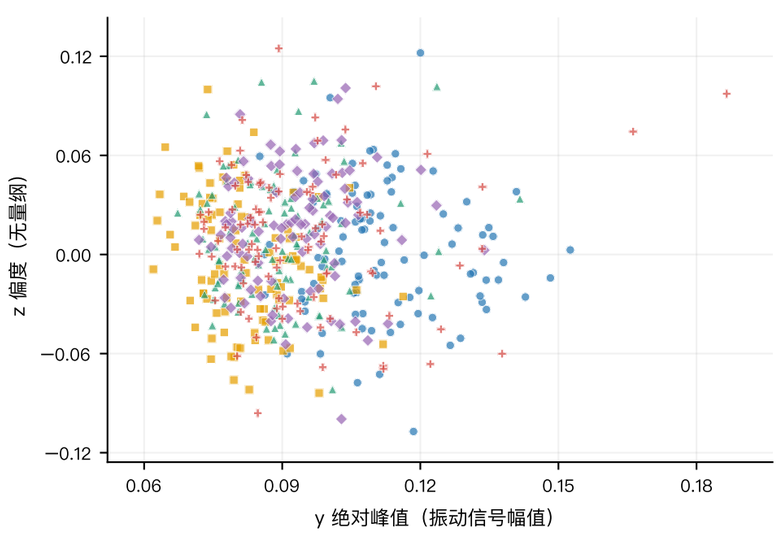
206 y 绝对峰值 × z 绝对峰值



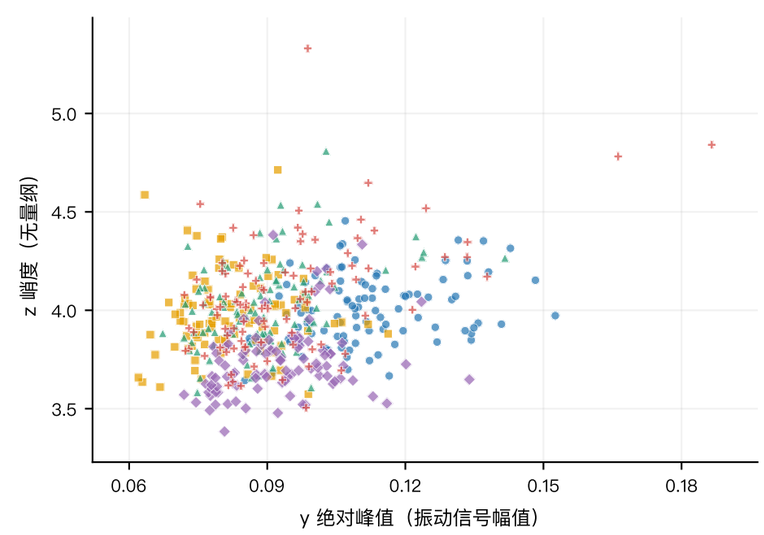
207 y 绝对峰值 × z 峰峰值



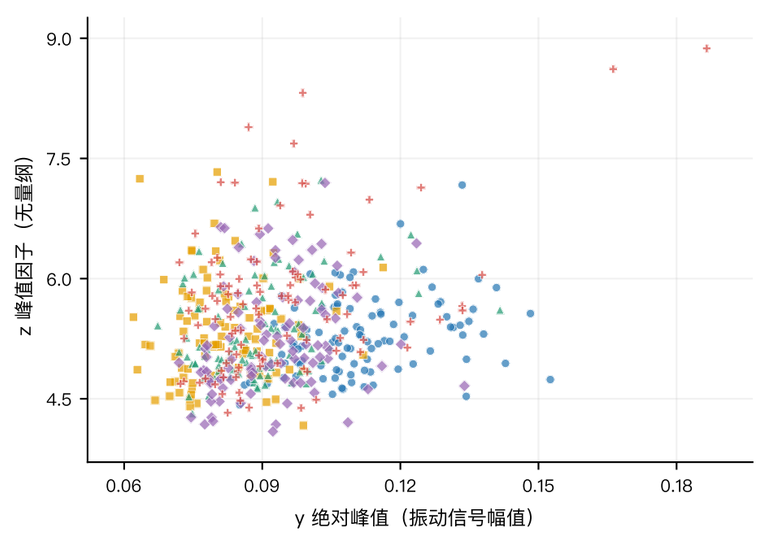
208 y 绝对峰值 × z 偏度



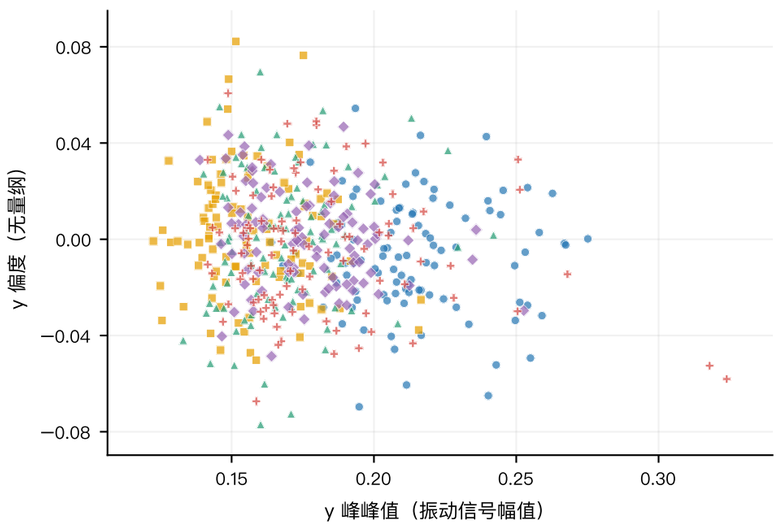
209 y 绝对峰值 × z 峭度



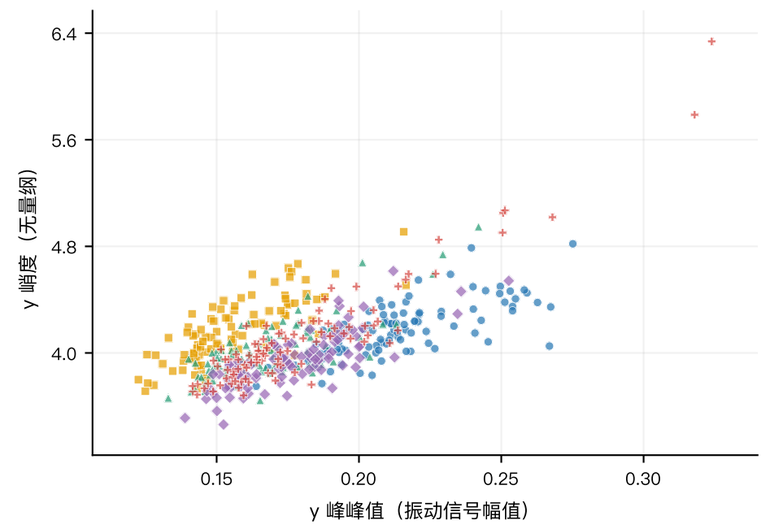
210 y 绝对峰值 × z 峰值因子



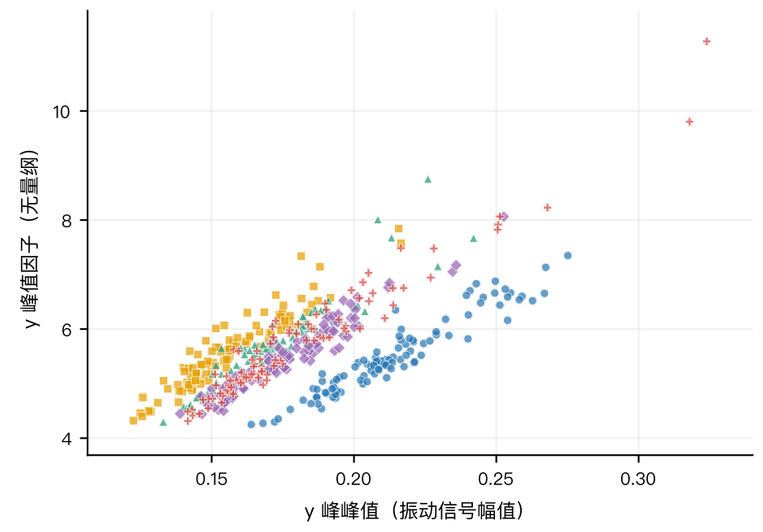
211 y 峰峰值 × y 偏度



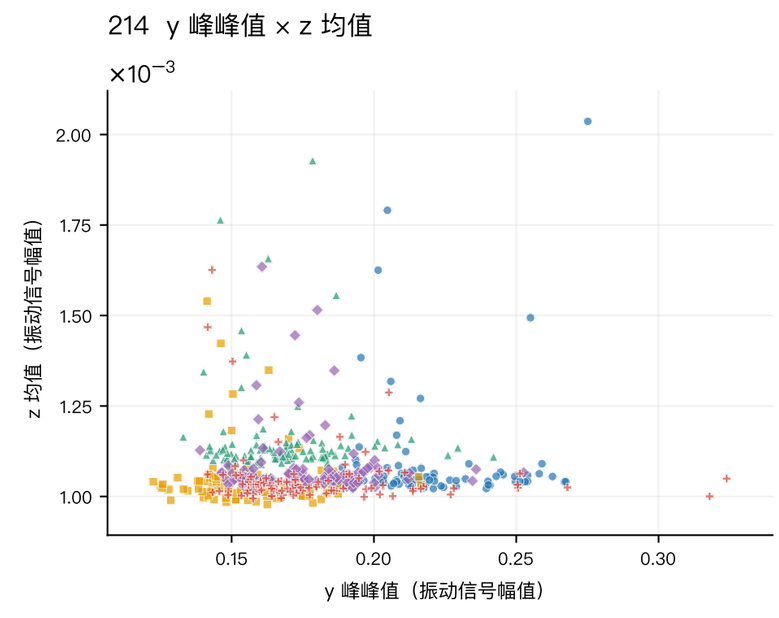
212 y 峰峰值 × y 峭度



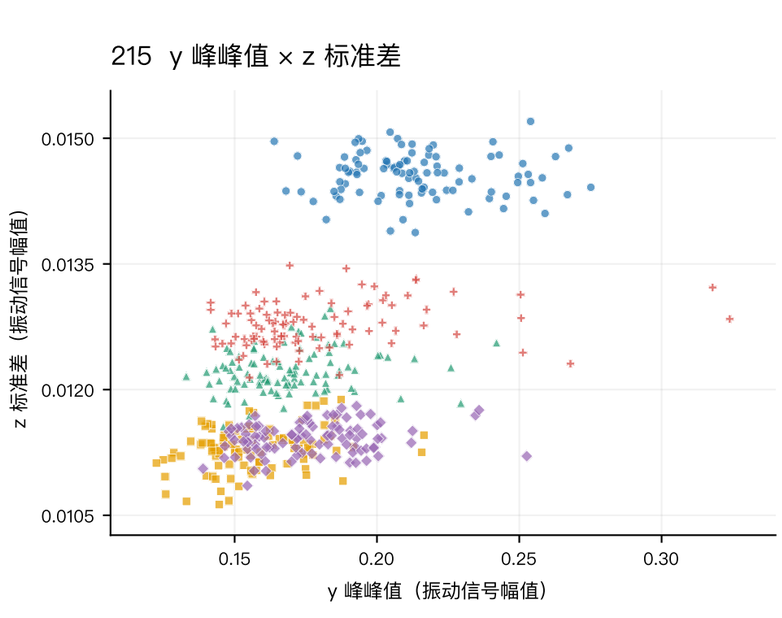
213 y 峰峰值 × y 峰值因子



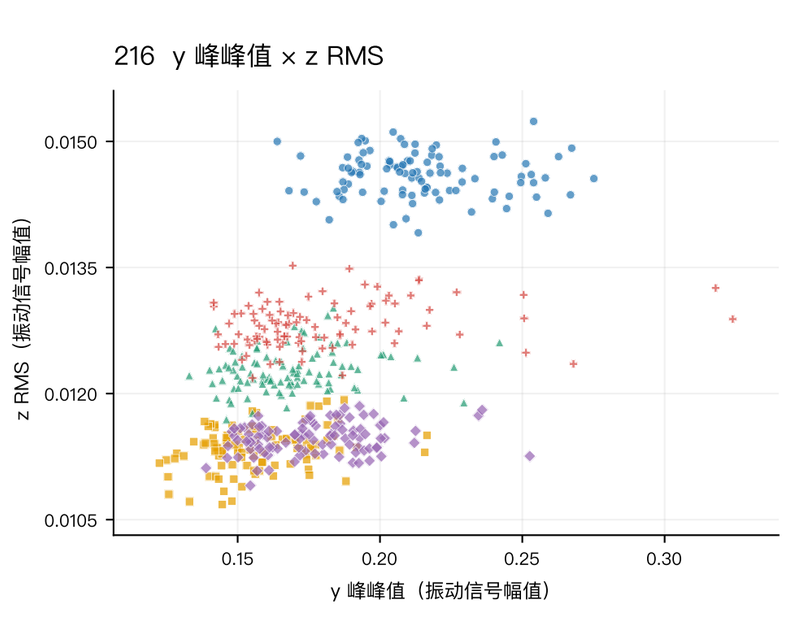
214 y 峰峰值 × z 均值



215 y 峰峰值 × z 标准差



216 y 峰峰值 × z RMS



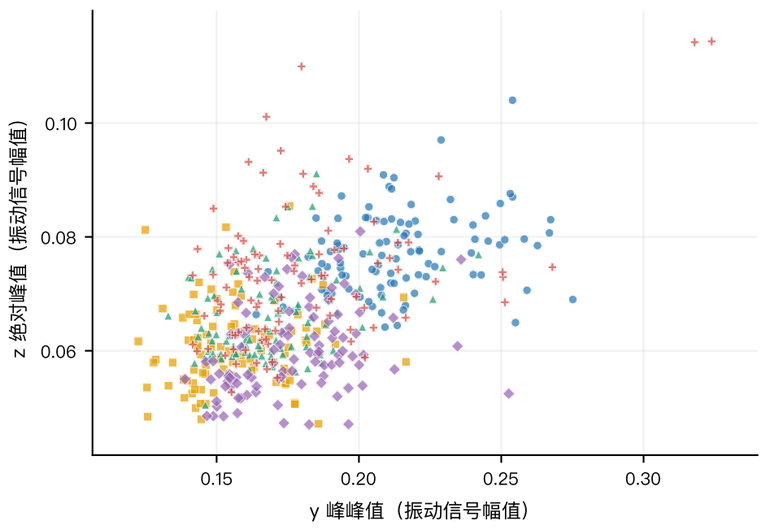
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

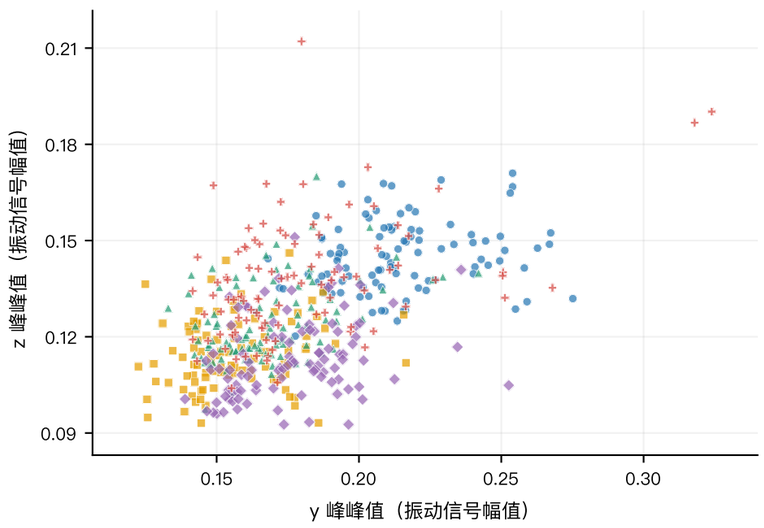
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

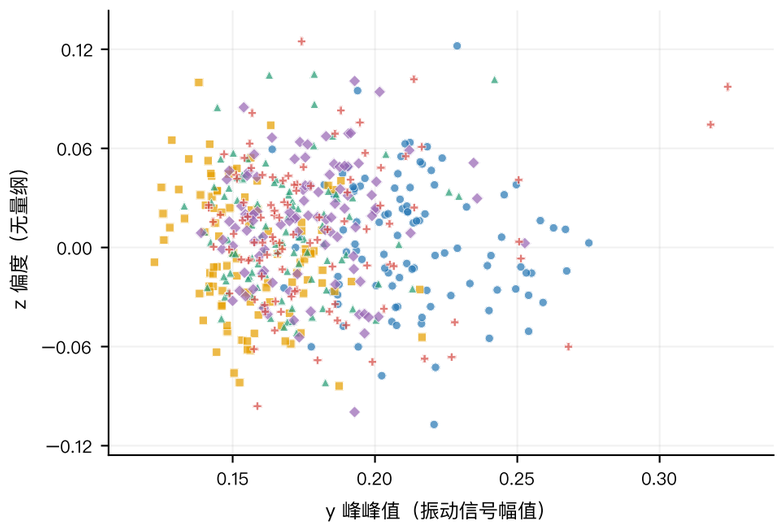
217 y 峰峰值 × z 绝对峰值



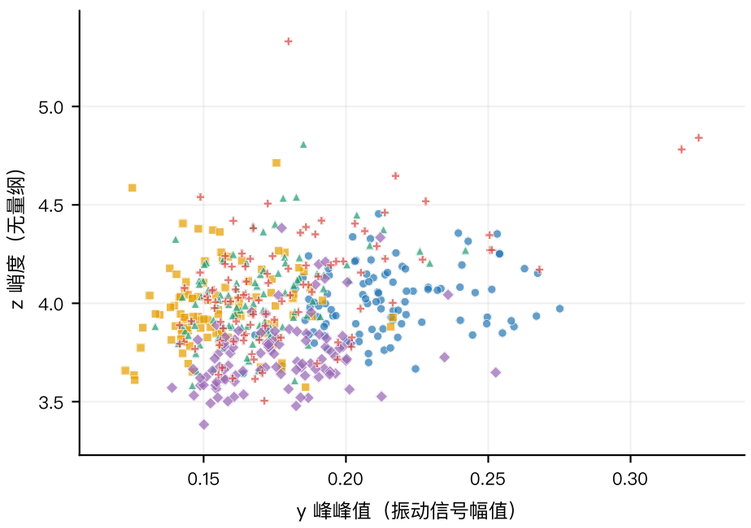
218 y 峰峰值 × z 峰峰值



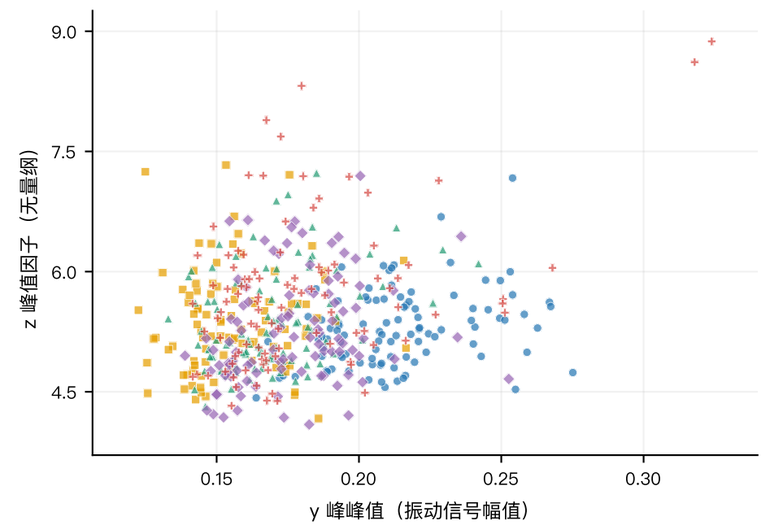
219 y 峰峰值 × z 偏度



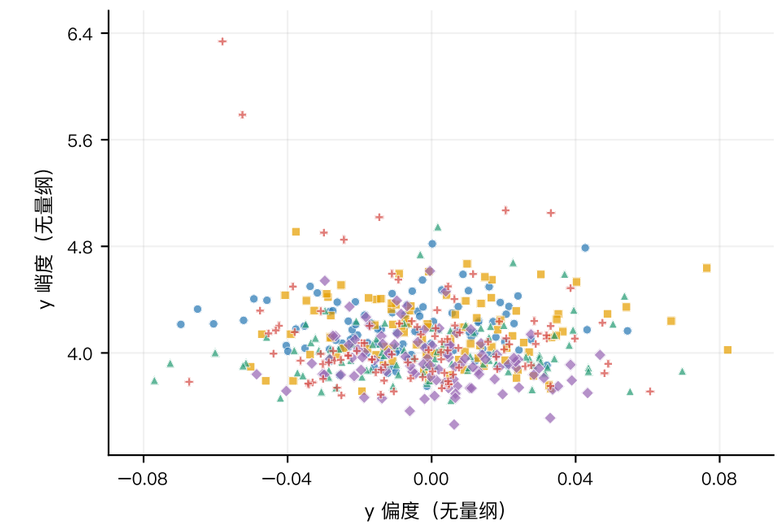
220 y 峰峰值 × z 峭度



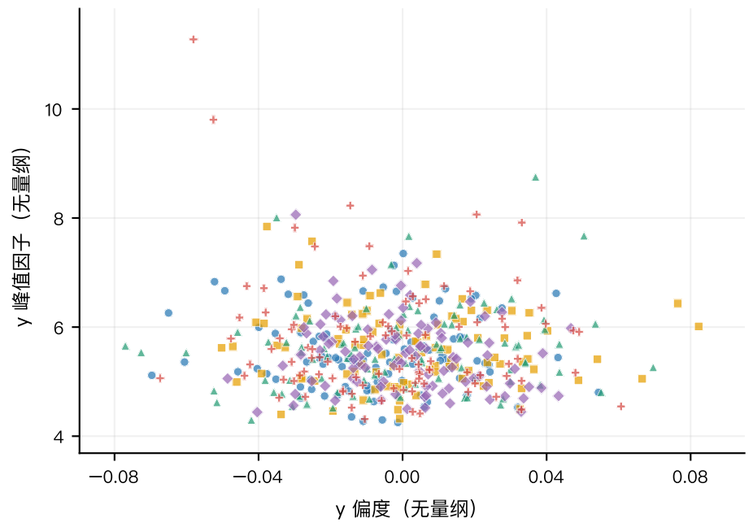
221 y 峰峰值 × z 峰值因子



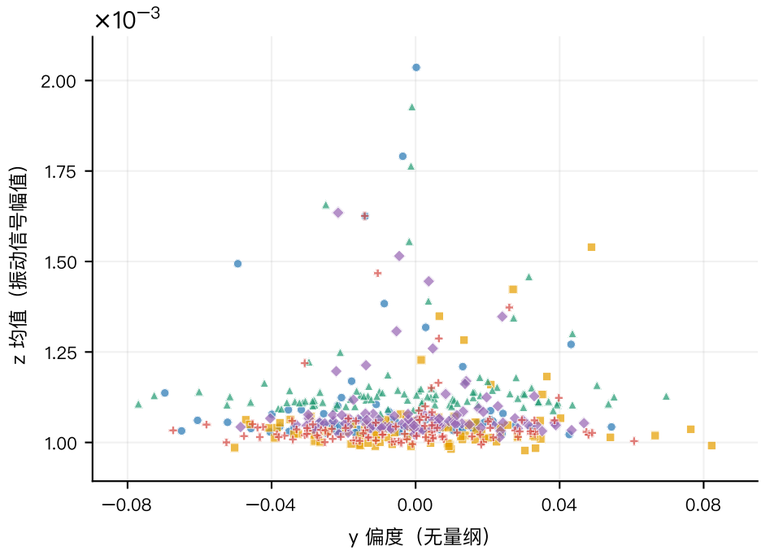
222 y 偏度 × y 峭度



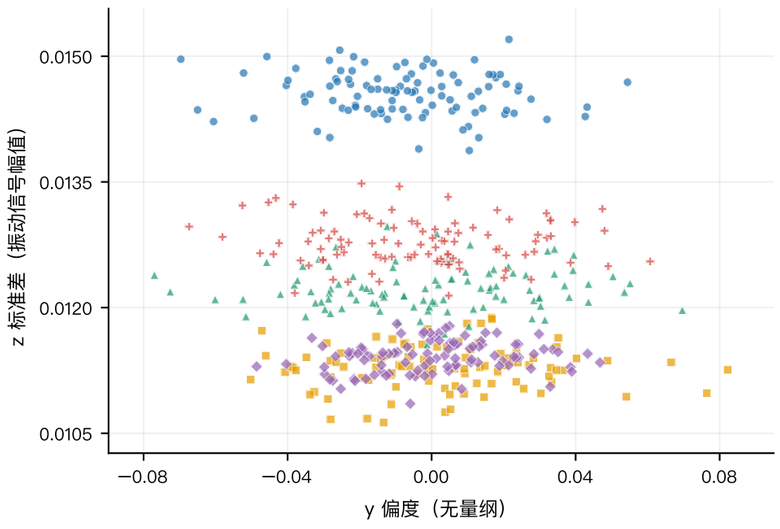
223 y 偏度 × y 峰值因子



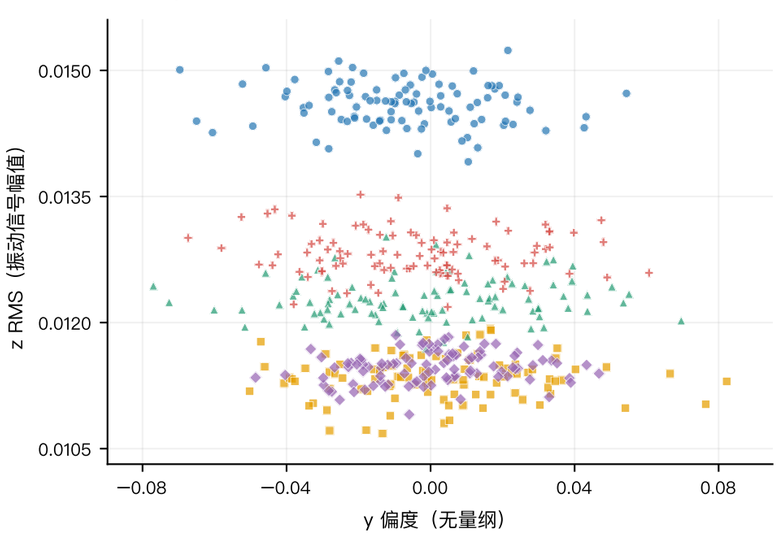
224 y 偏度 × z 均值



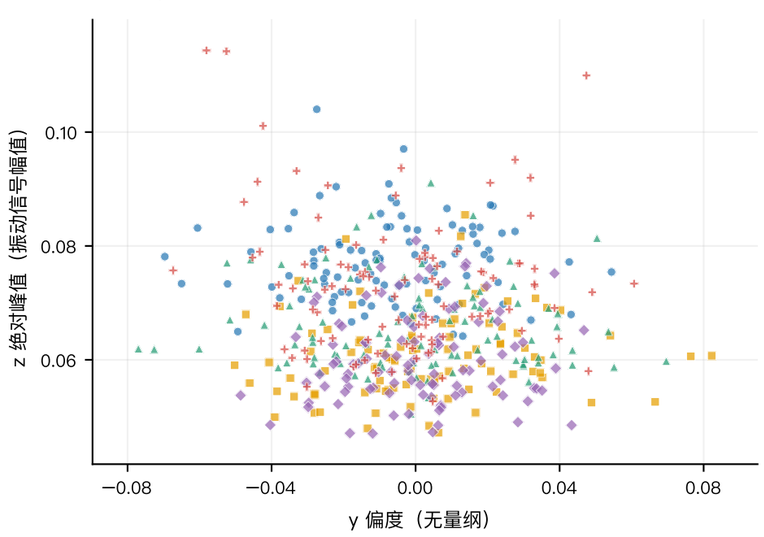
225 y 偏度 × z 标准差



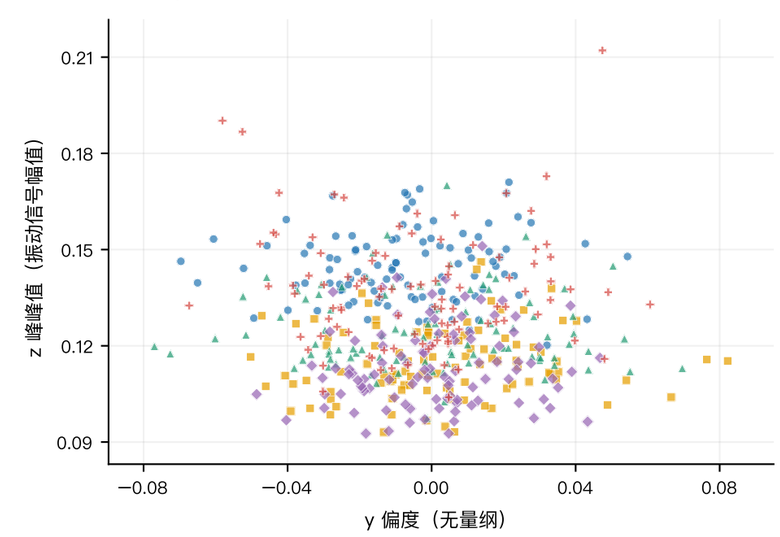
226 y 偏度 × z RMS



227 y 偏度 × z 绝对峰值



228 y 偏度 × z 峰峰值



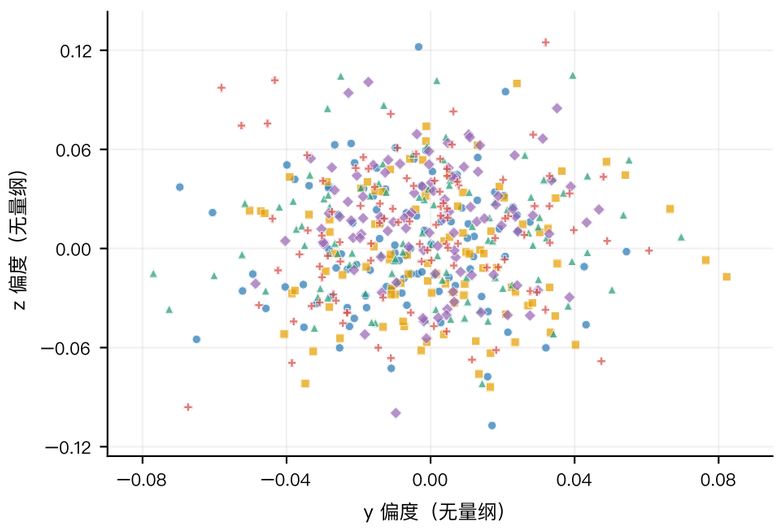
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

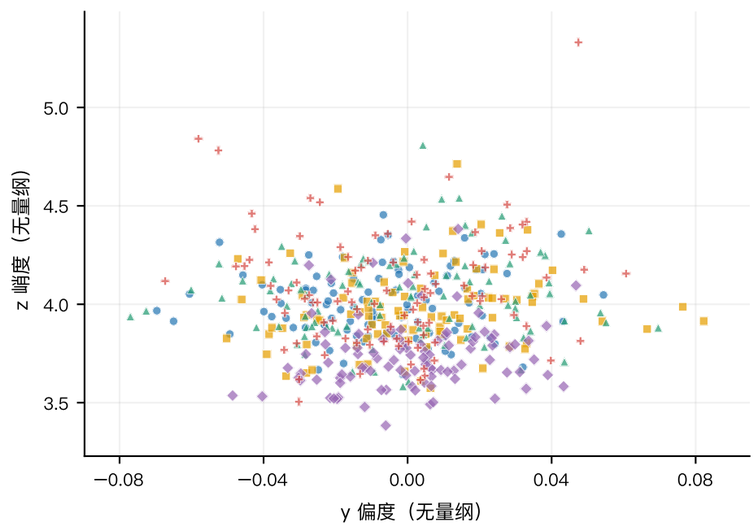
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

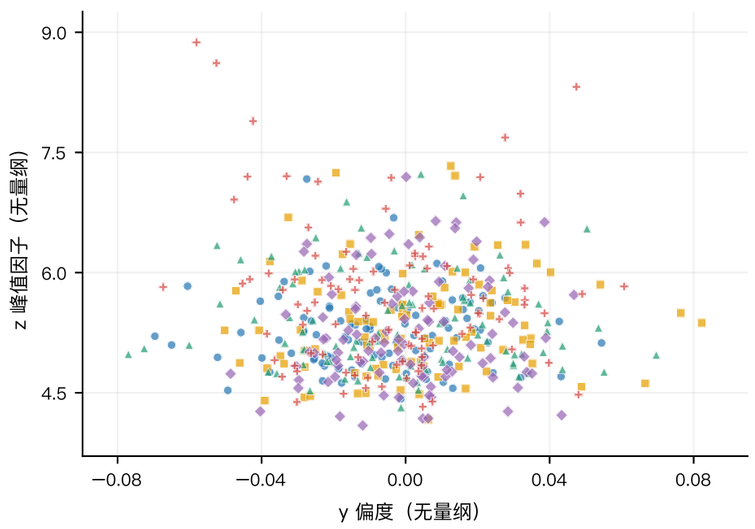
229 y 偏度 × z 偏度



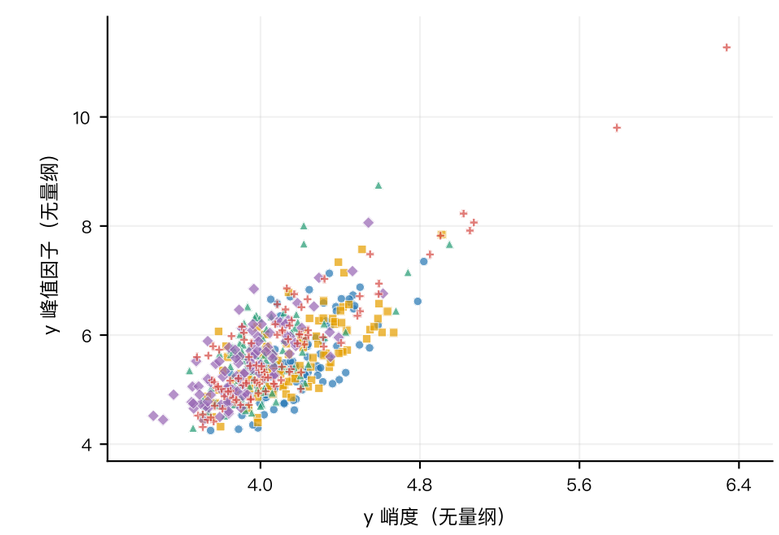
230 y 偏度 × z 峭度



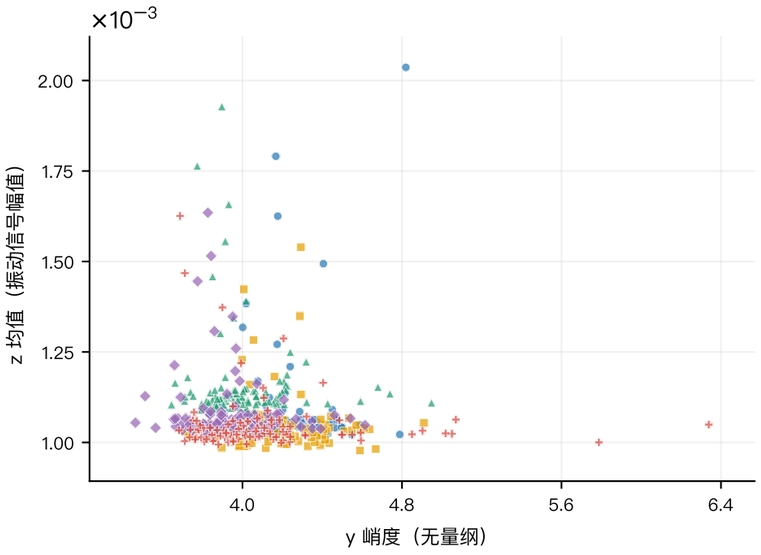
231 y 偏度 × z 峰值因子



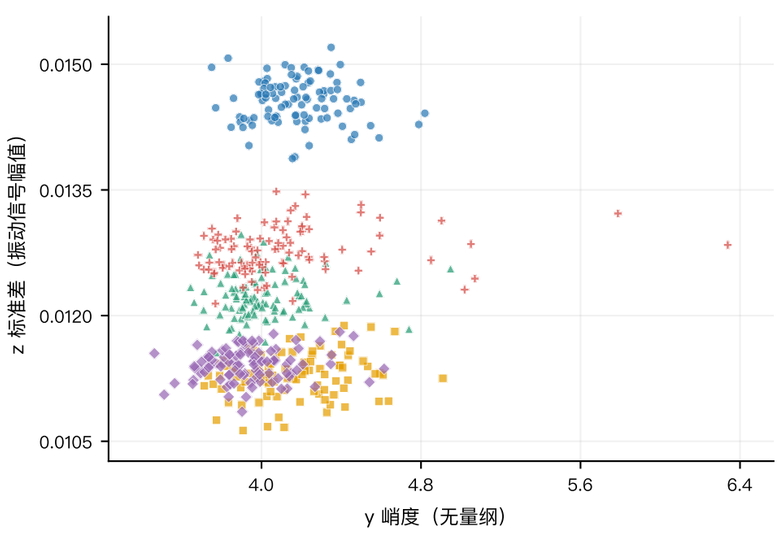
232 y 峭度 × y 峰值因子



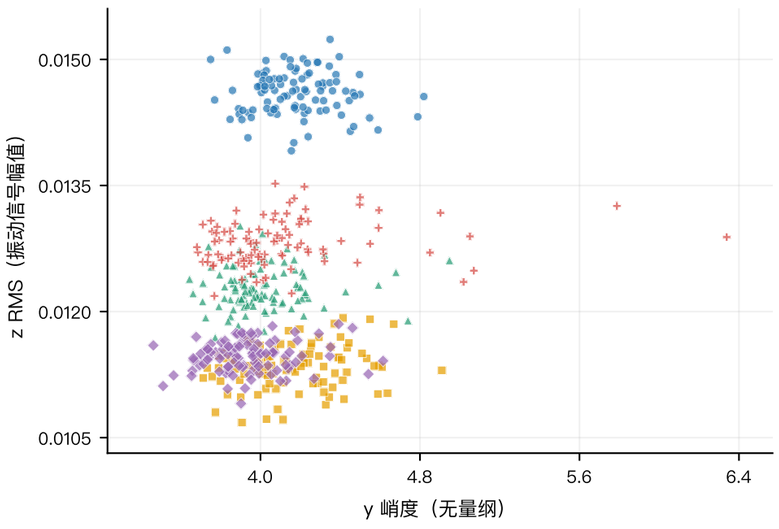
233 y 峭度 × z 均值



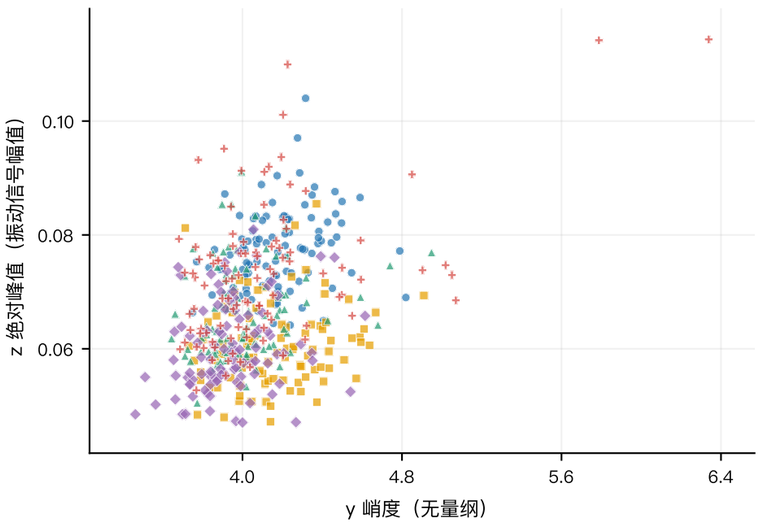
234 y 峭度 × z 标准差



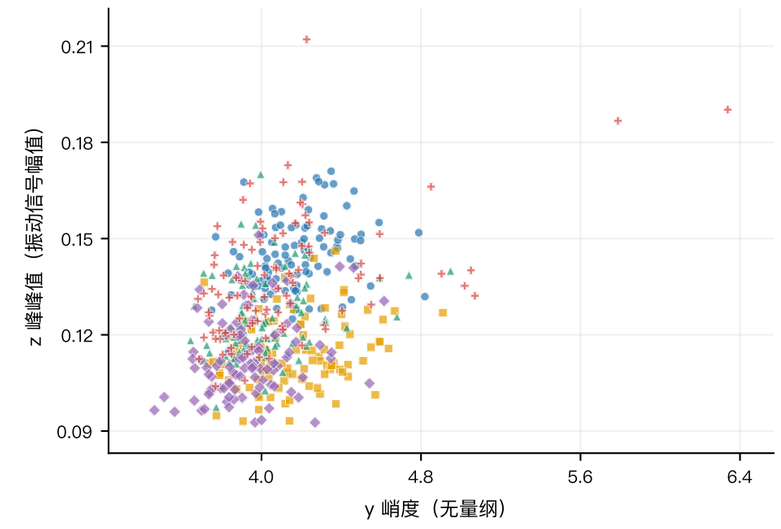
235 y 峭度 × z RMS



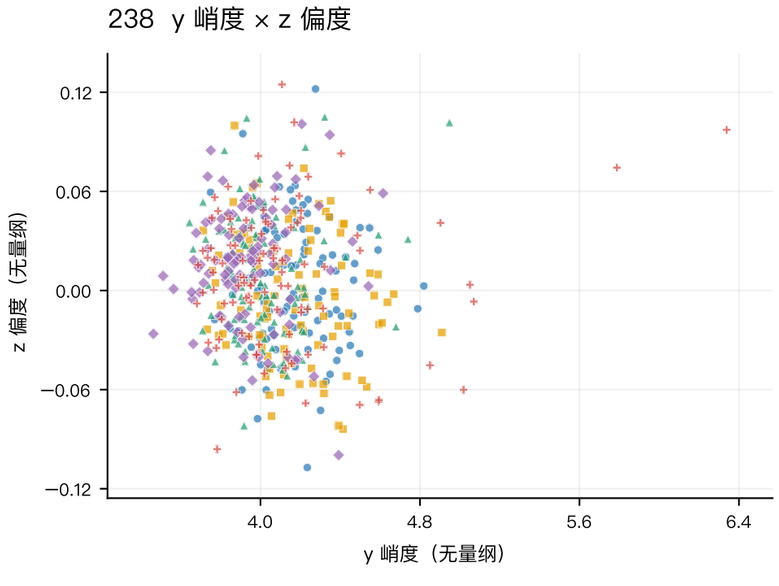
236 y 峭度 × z 绝对峰值



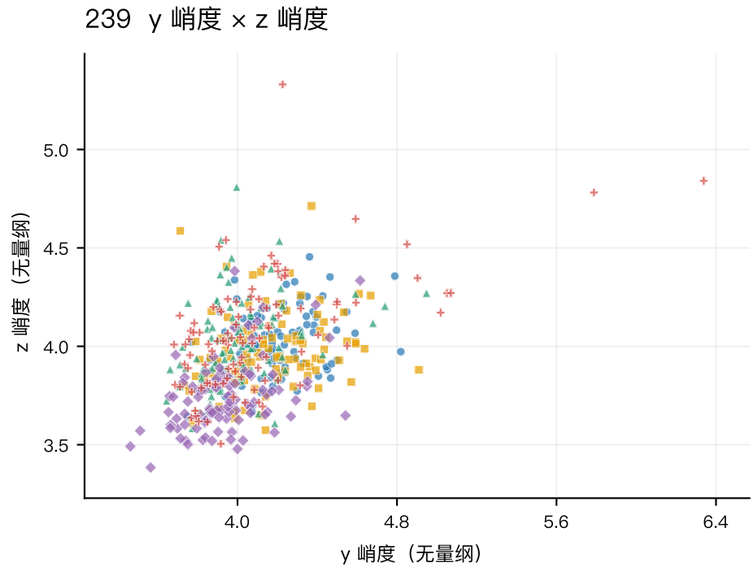
237 y 峭度 × z 峰峰值



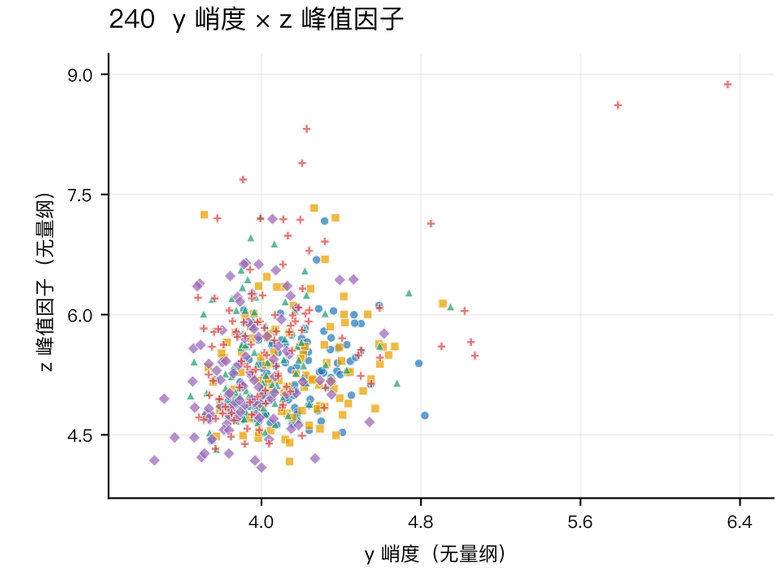
238 y 峭度 × z 偏度



239 y 峭度 × z 峭度



240 y 峭度 × z 峰值因子



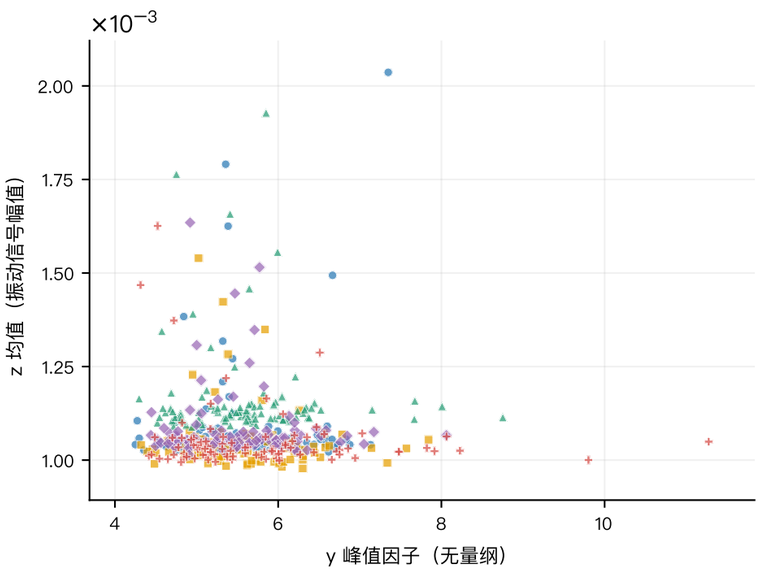
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

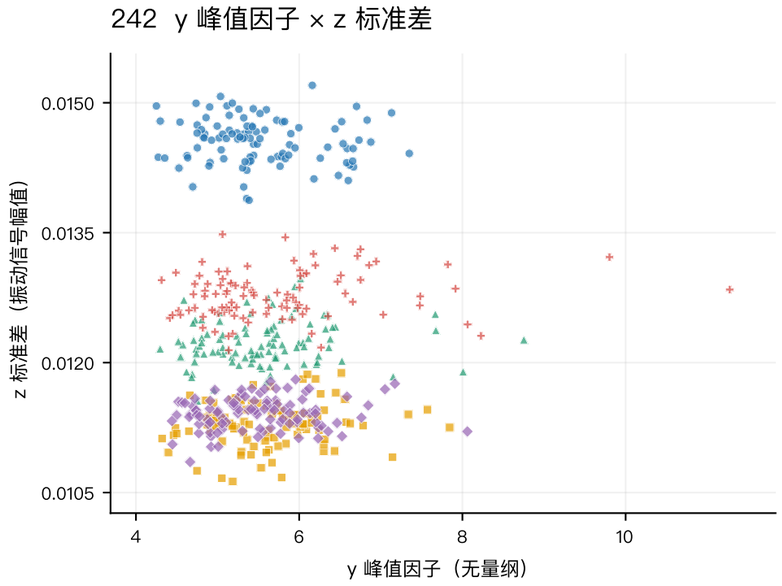
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

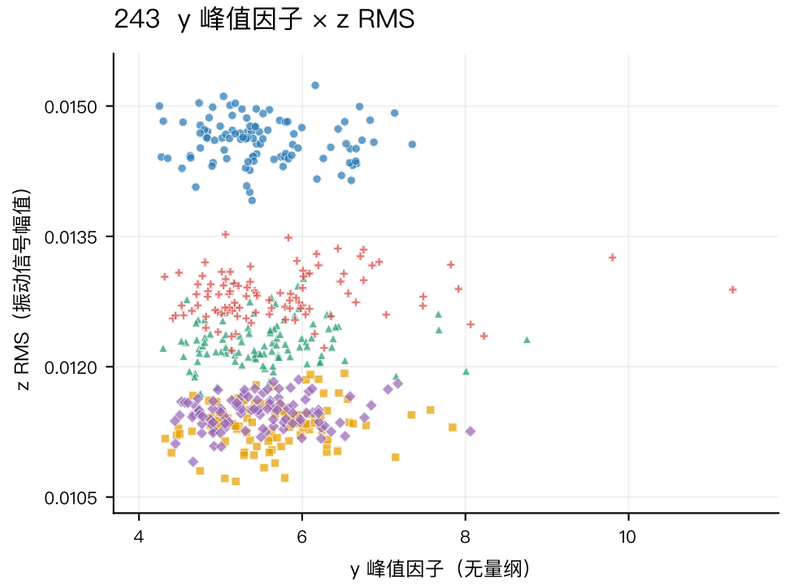
241 y 峰值因子 × z 均值



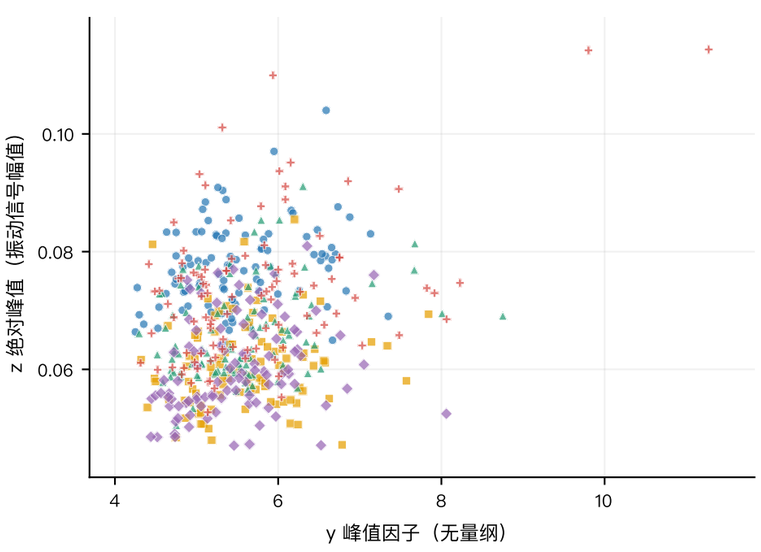
242 y 峰值因子 × z 标准差



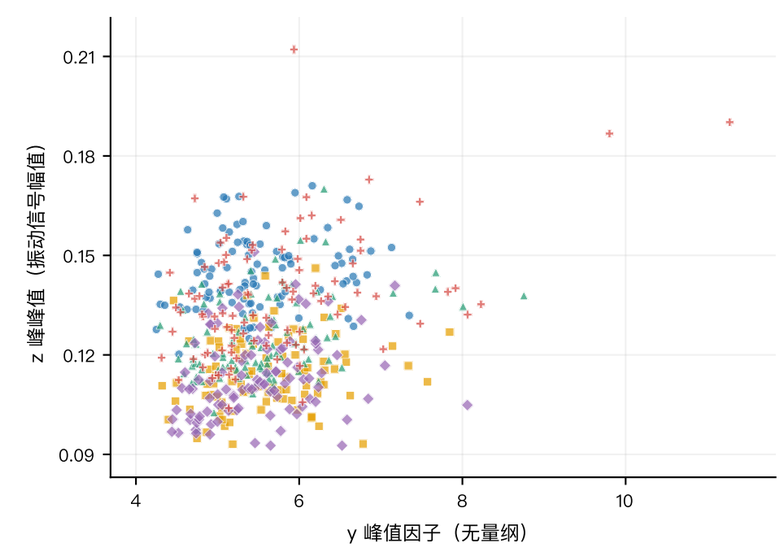
243 y 峰值因子 × z RMS



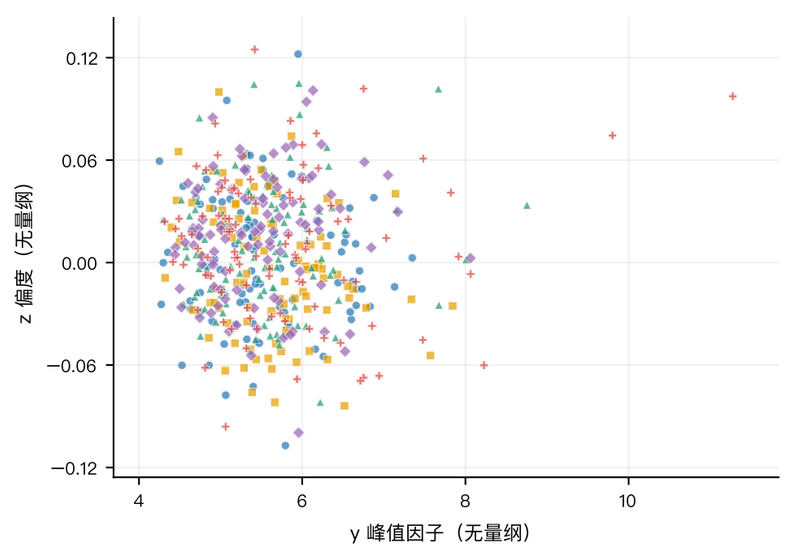
244 y 峰值因子 × z 绝对峰值



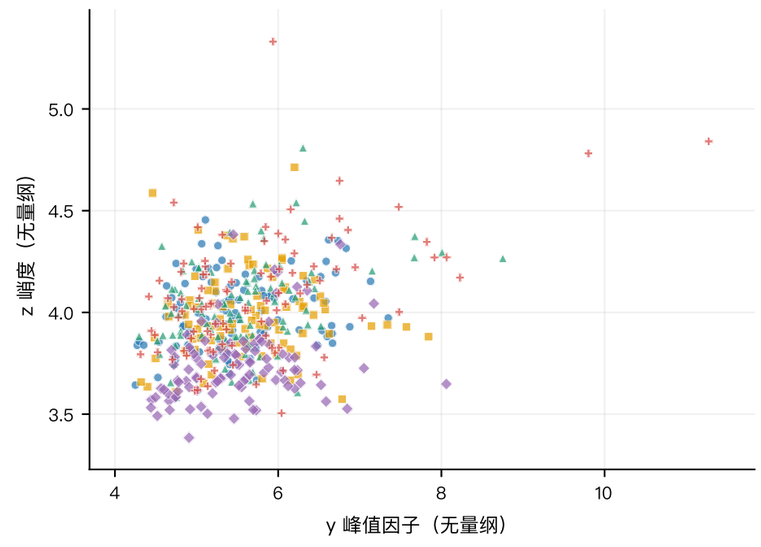
245 y 峰值因子 × z 峰峰值



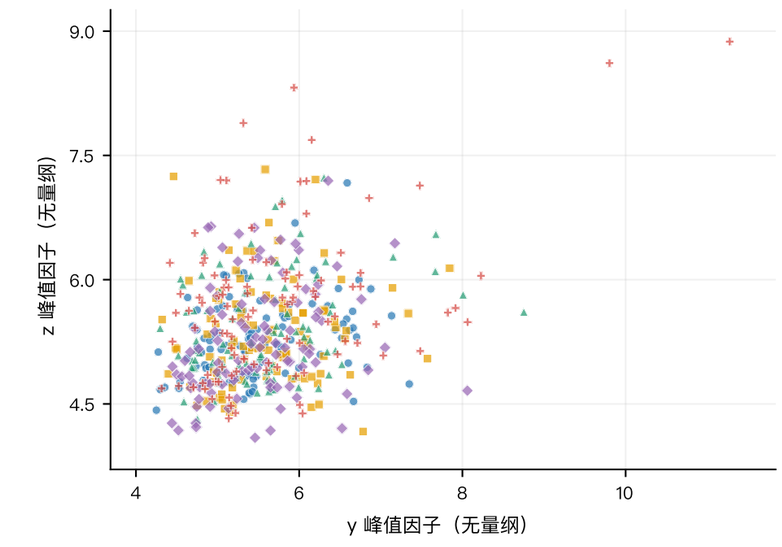
246 y 峰值因子 × z 偏度



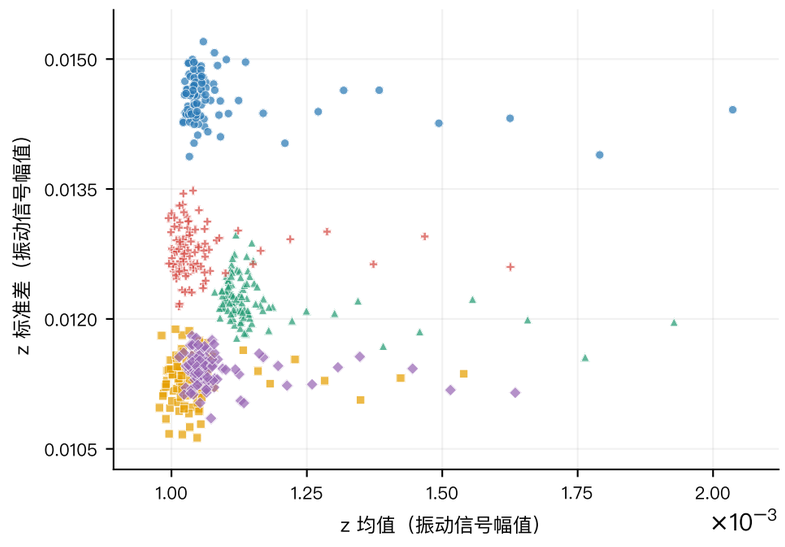
247 y 峰值因子 × z 峭度



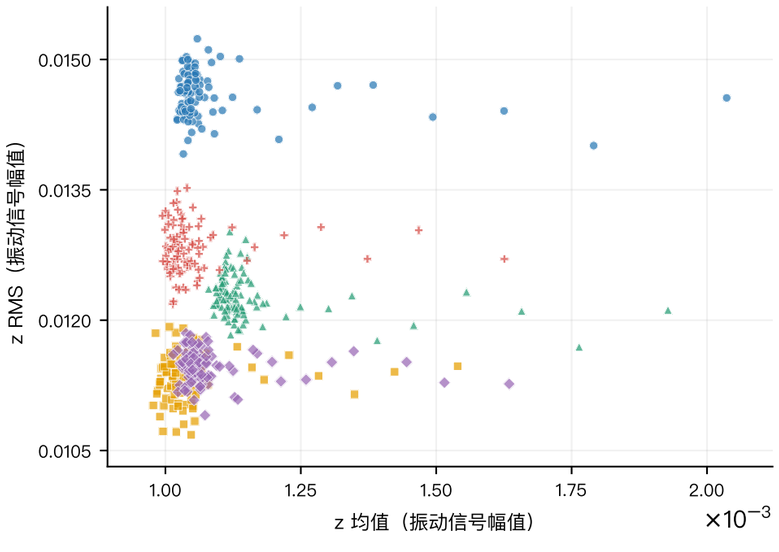
248 y 峰值因子 × z 峰值因子



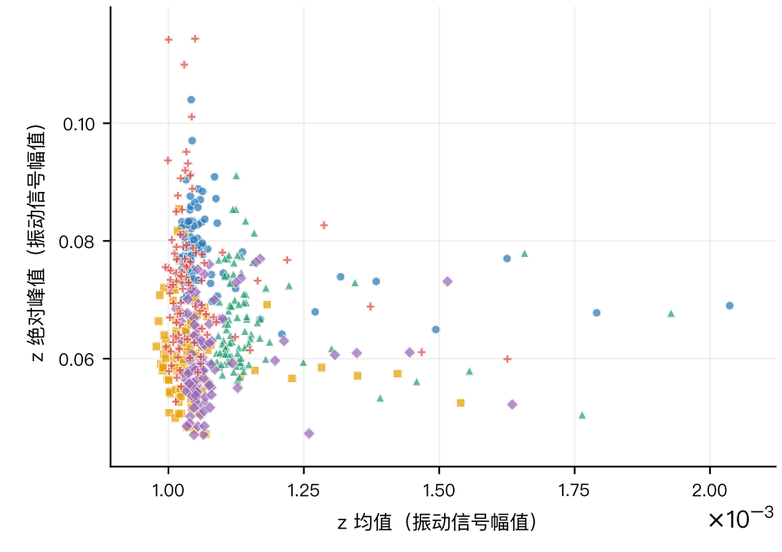
249 z 均值 × z 标准差



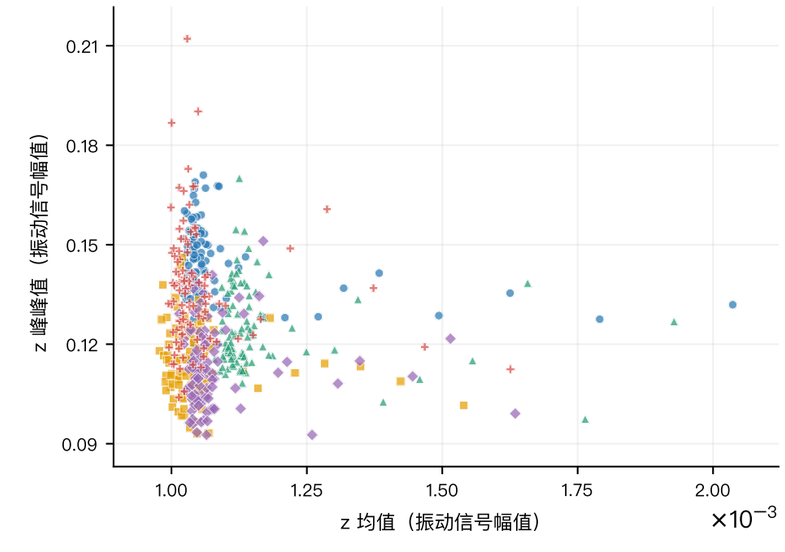
250 z 均值 × z RMS



251 z 均值 × z 绝对峰值



252 z 均值 × z 峰峰值



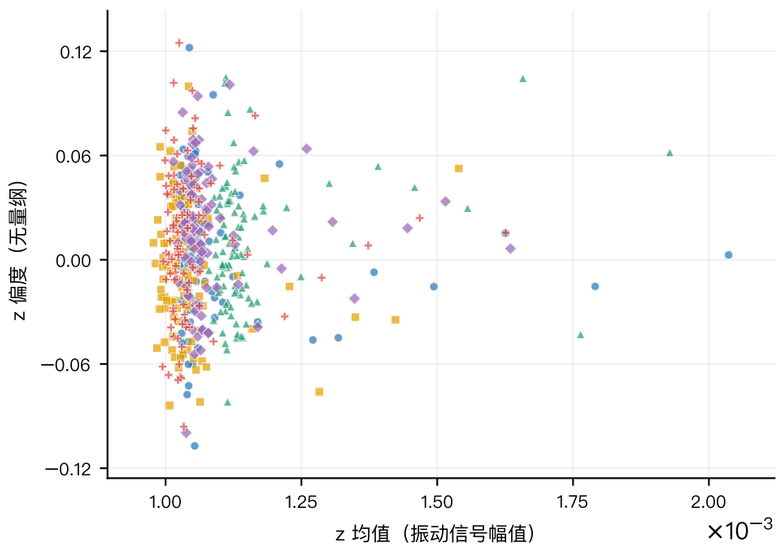
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

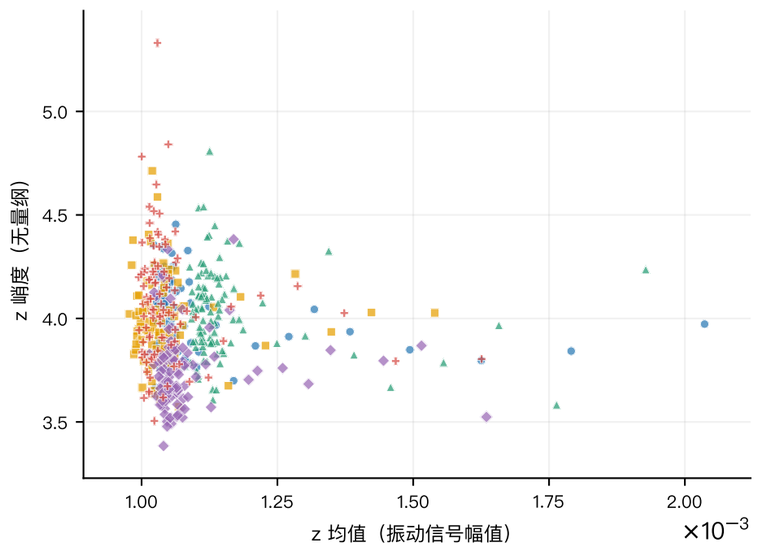
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

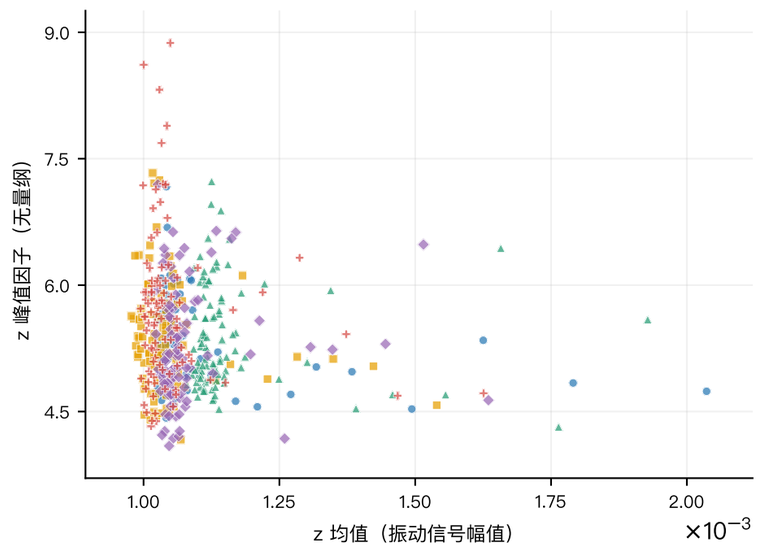
253 z 均值 × z 偏度



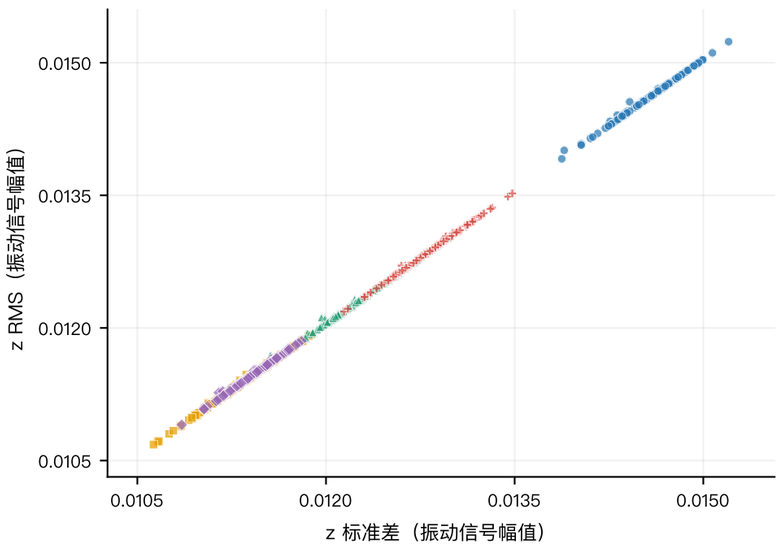
254 z 均值 × z 峭度



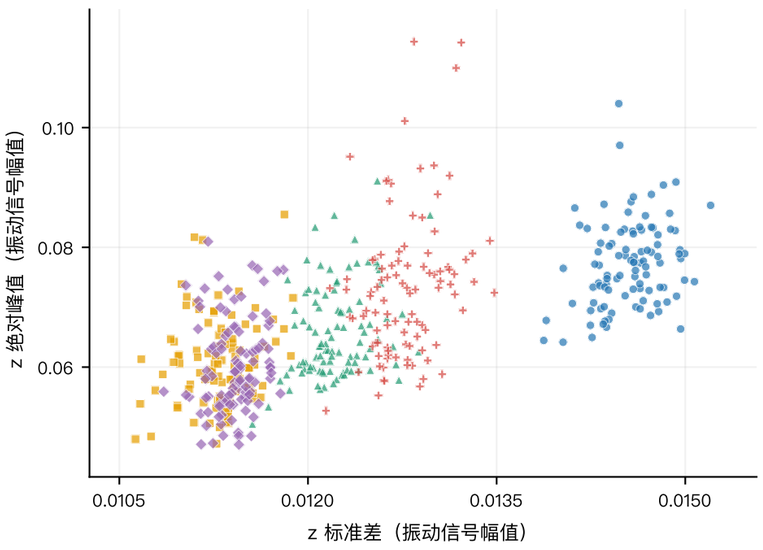
255 z 均值 × z 峰值因子



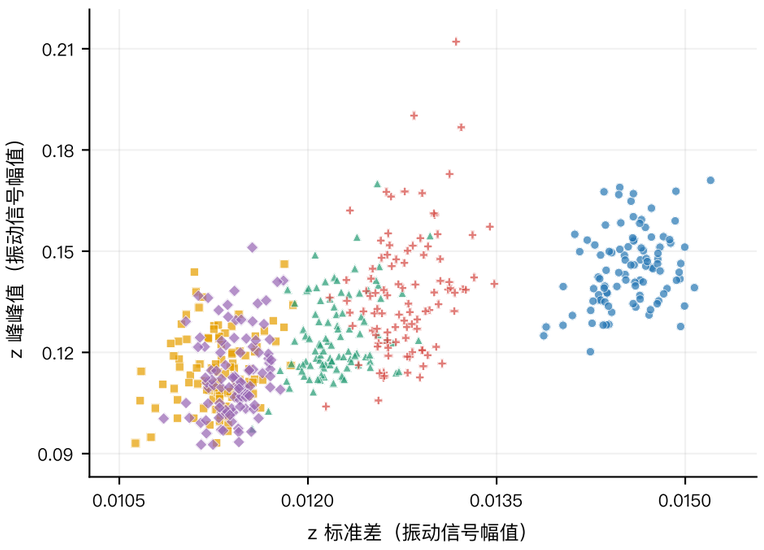
256 z 标准差 × z RMS



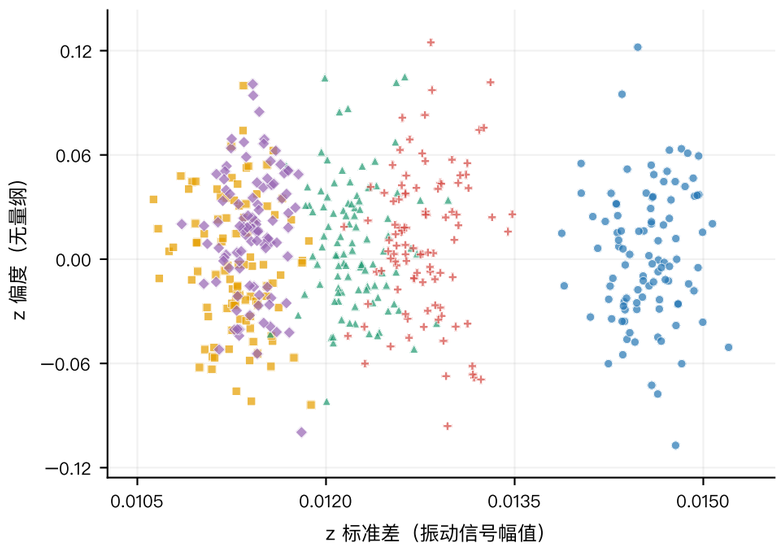
257 z 标准差 × z 绝对峰值



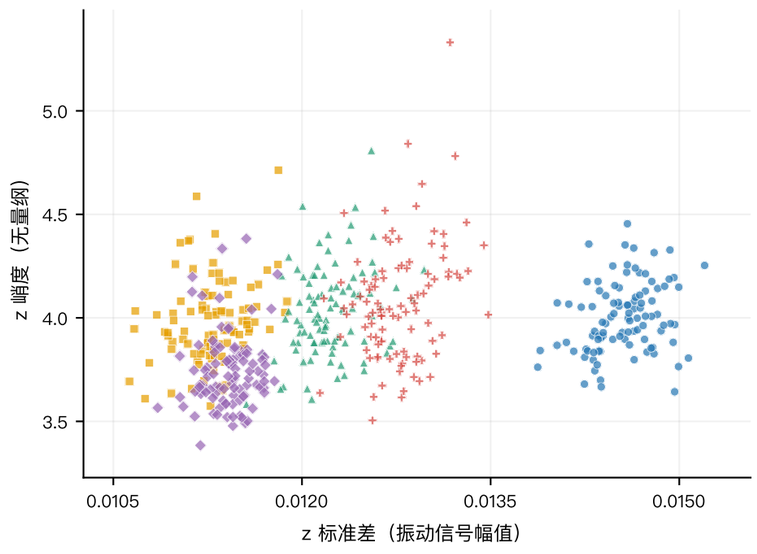
258 z 标准差 × z 峰峰值



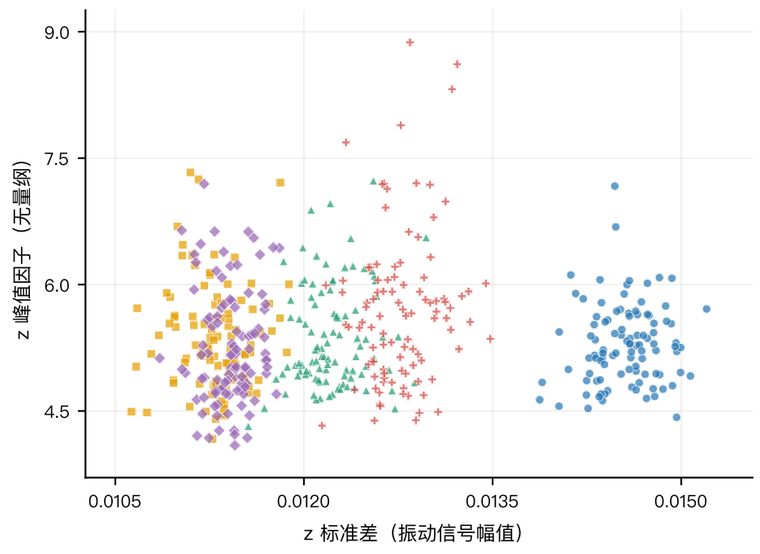
259 z 标准差 × z 偏度



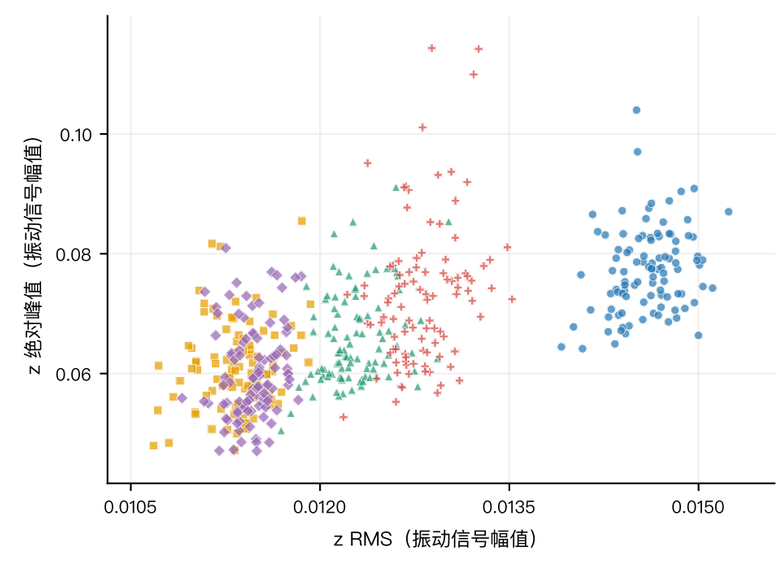
260 z 标准差 × z 峭度



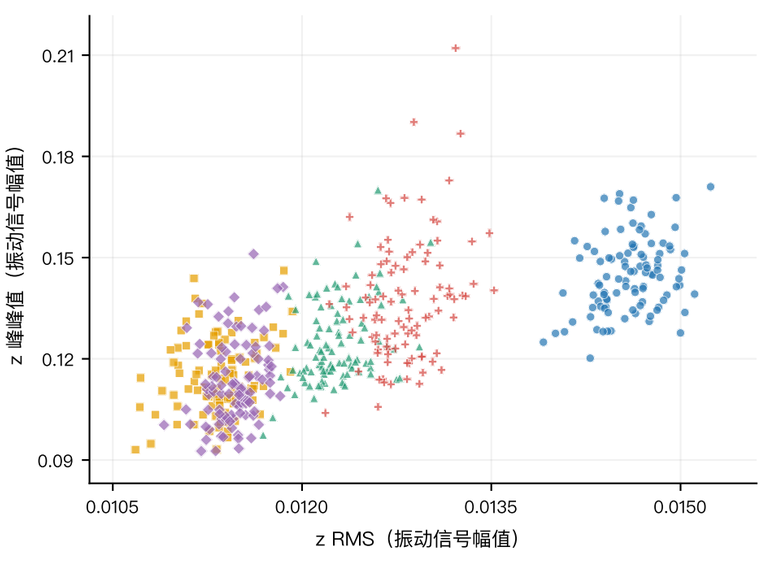
261 z 标准差 × z 峰值因子



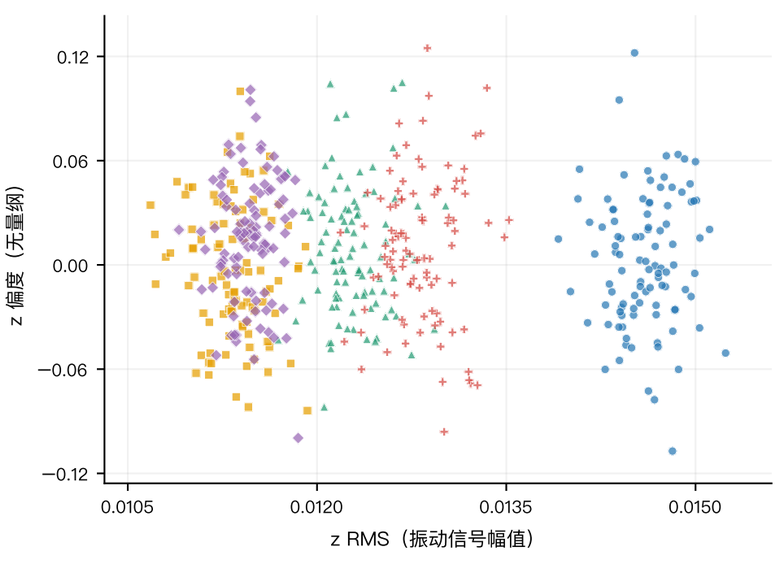
262 z RMS × z 绝对峰值



263 z RMS × z 峰峰值



264 z RMS × z 偏度



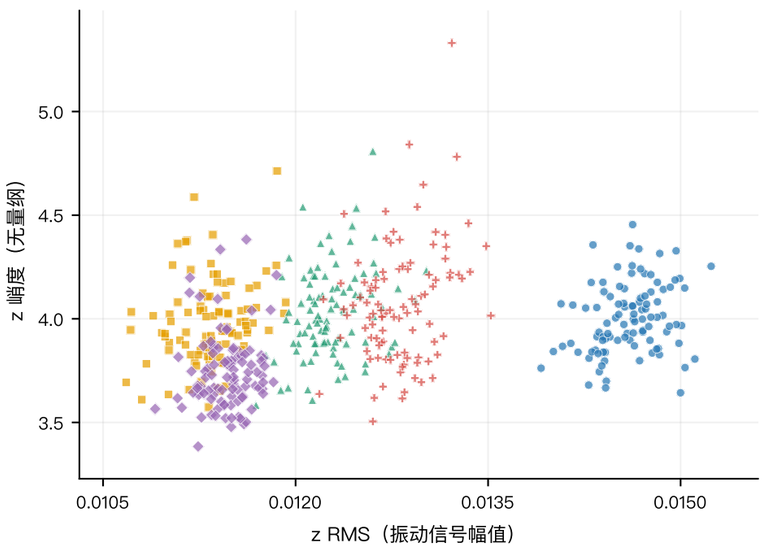
每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。

30_2 | 全部双特征组合

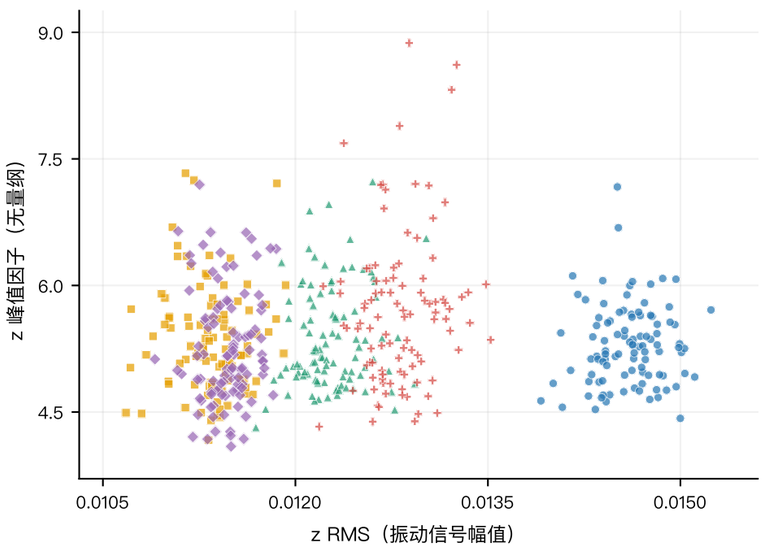
24 项特征 · 276 种配对 · 每图 500 点 · 每种状态 100 个窗口

健康　齿部故障　缺齿　齿根故障　齿面故障

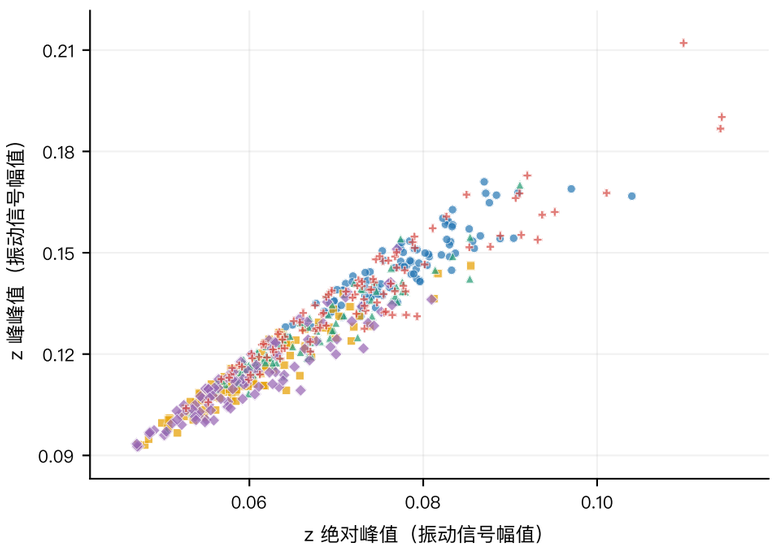
265 z RMS × z 峭度



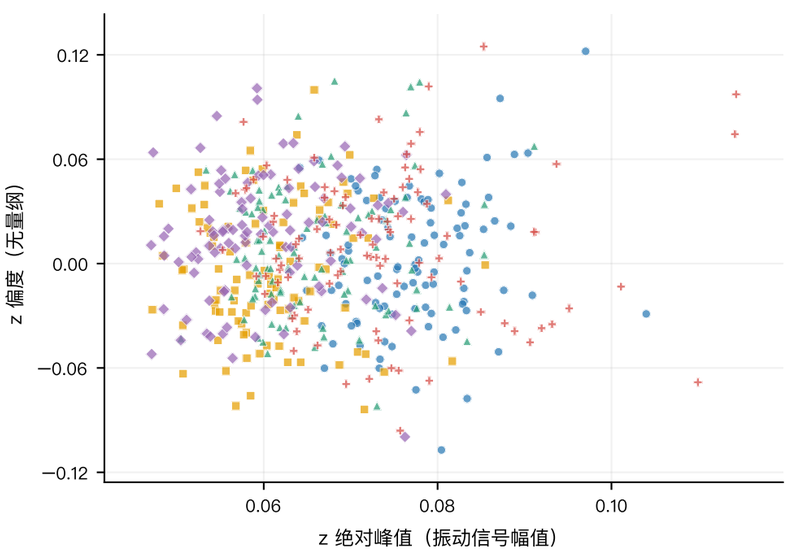
266 z RMS × z 峰值因子



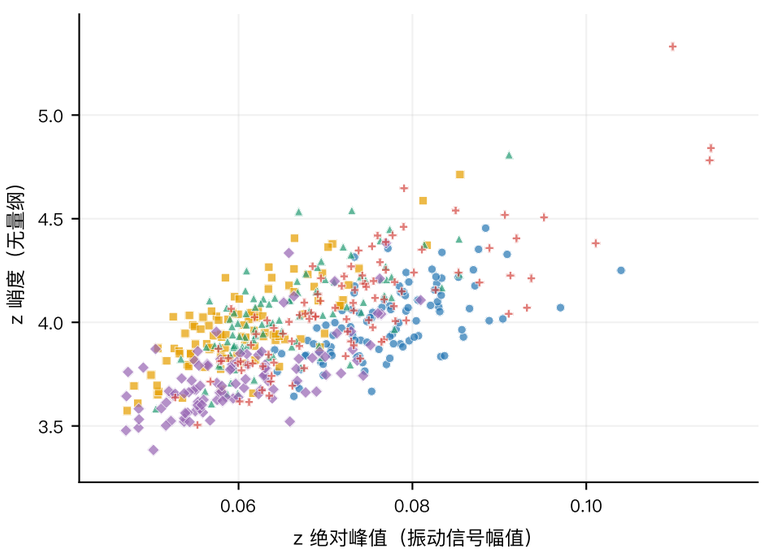
267 z 绝对峰值 × z 峰峰值



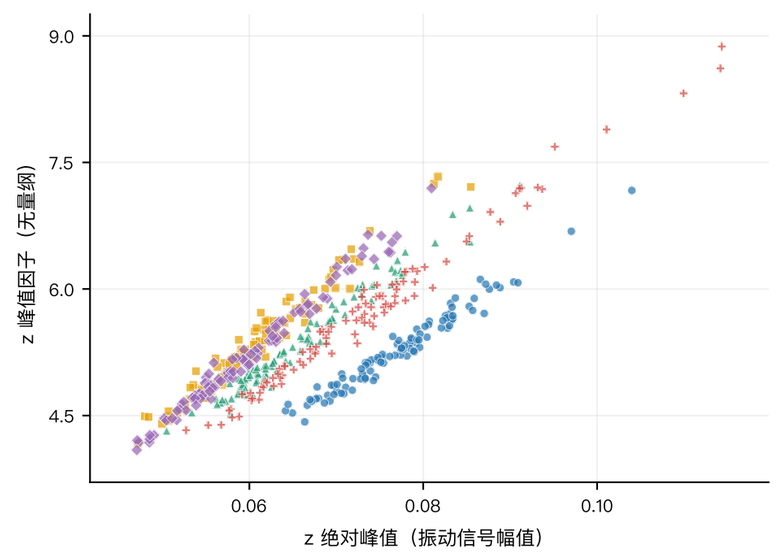
268 z 绝对峰值 × z 偏度



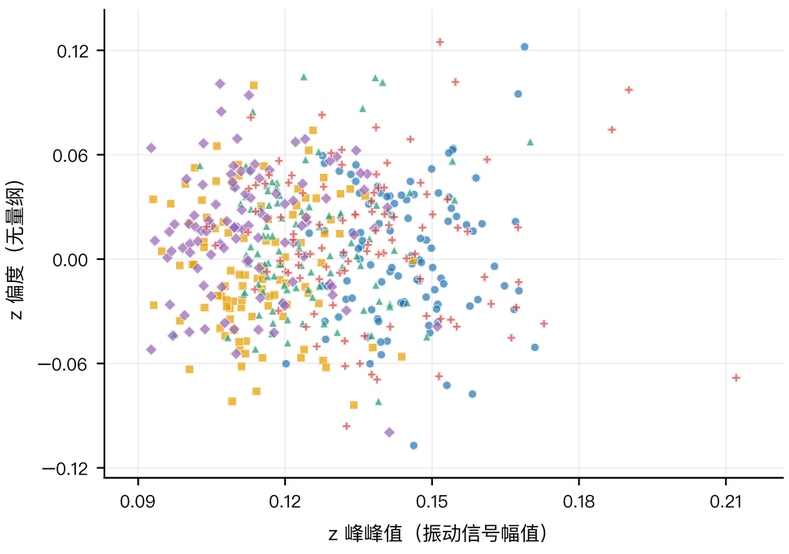
269 z 绝对峰值 × z 峭度



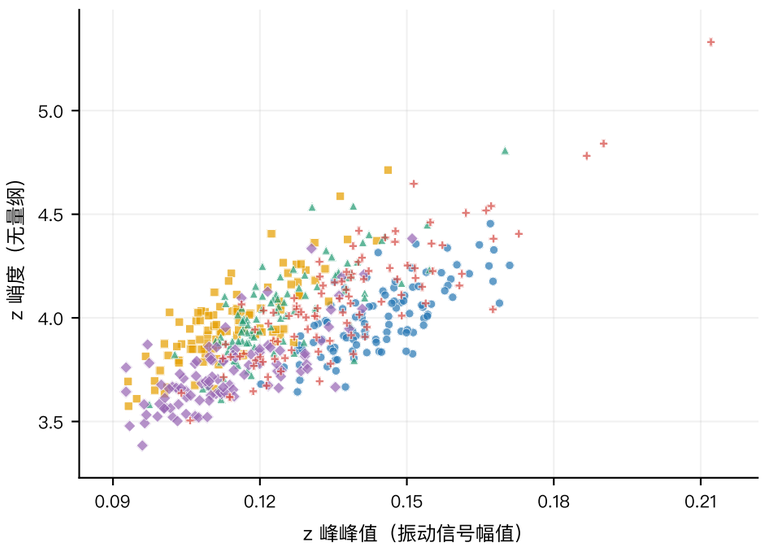
270 z 绝对峰值 × z 峰值因子



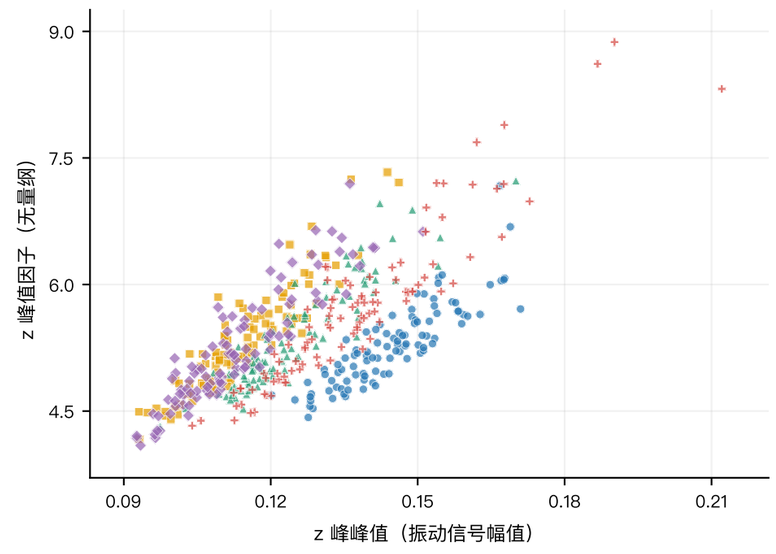
271 z 峰峰值 × z 偏度



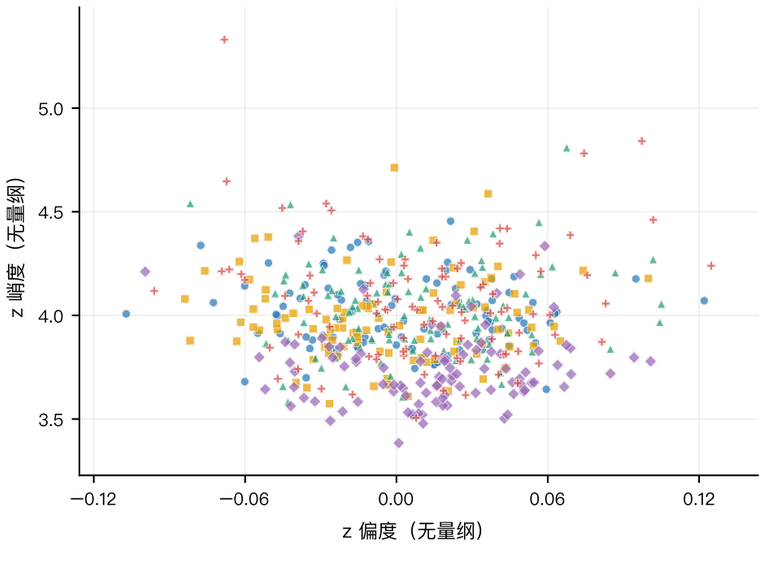
272 z 峰峰值 × z 峭度



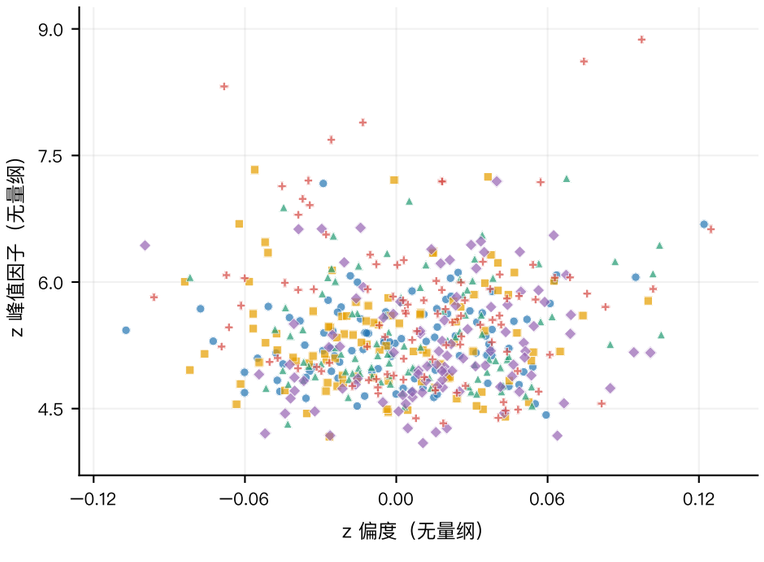
273 z 峰峰值 × z 峰值因子



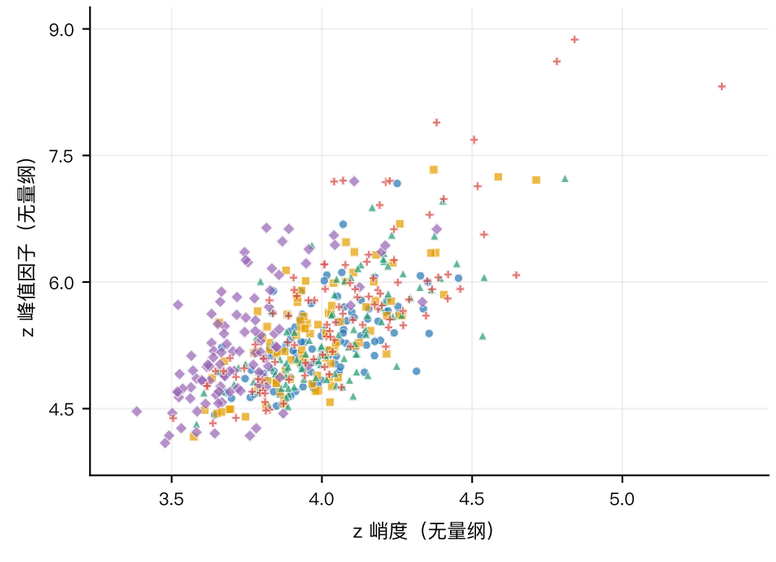
274 z 偏度 × z 峭度



275 z 偏度 × z 峰值因子



276 z 峭度 × z 峰值因子



每点来自同一窗口；图内五类共用坐标。各图坐标范围分别设置，展示原始特征值。